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UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, DC 20549

Form 10-K

☑ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2019

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to

Commission file number 1-33579

INTERDIGITAL, INC.

(Exact name of registrant as specified in its charter)

Pennsylvania	**82-4936666**
(State or other jurisdiction of incorporation or organization)	*(IRS Employer Identification No.)*

200 Bellevue Parkway, Suite 300, Wilmington, DE 19809-3727
(Address of Principal Executive Offices and Zip Code)
Registrant's telephone number, including area code (302) 281-3600

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock (par value $0.01 per share)	IDCC	NASDAQ Stock Market LLC

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☑ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☑

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☑ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☑ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☑	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐

Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☑

The aggregate market value of the voting and non-voting common equity held by non-affiliates computed by reference to the price at which the common equity was last sold, or the average bid and asked price of such common equity, as of the last business day of the registrant's most recently completed second fiscal quarter: $1,989,033,542 as of June 28, 2019.

The number of shares outstanding of the registrant's common stock was 30,722,894 as of February 18, 2020.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's definitive proxy statement to be filed pursuant to Regulation 14A in connection with the registrant's 2020 annual meeting of shareholders are incorporated by reference into Items 10, 11, 12, 13 and 14 of Part III of this Form 10-K.

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In this Form 10-K, the words "we," "our," "us," "the Company" and "InterDigital" refer to InterDigital, Inc. and/or its subsidiaries, individually and/or collectively, unless otherwise indicated or the context otherwise requires. InterDigital® is a registered trademark of InterDigital, Inc. All other trademarks, service marks and/or trade names appearing in this Form 10-K are the property of their respective holders.

EXPLANATORY NOTE ABOUT INTERDIGITAL, INC.

On April 3, 2018, for the purpose of reorganizing its holding company structure, InterDigital, Inc., a Pennsylvania corporation and then-existing NASDAQ-listed registrant (the "Predecessor Company"), executed an Agreement and Plan of Merger ("Merger Agreement") with InterDigital Parent, Inc., a Pennsylvania corporation (the "Successor Company") 100% owned by the Predecessor Company, and another newly formed Pennsylvania corporation owned 100% by the Successor Company ("Merger Sub"). Pursuant to the Merger Agreement, on April 3, 2018, Merger Sub merged (the "Merger" or "Reorganization") with and into the Predecessor Company, with the Predecessor Company surviving. As a result of the Merger, the Predecessor Company is now a wholly owned subsidiary of the Successor Company. Neither the business conducted by the Successor Company and the Predecessor Company in the aggregate, nor the consolidated assets and liabilities of the Successor Company and the Predecessor Company in the aggregate, changed as a result of the Reorganization. By virtue of the Merger, each share of the Predecessor Company's outstanding common stock was converted, on a share-for-share basis,

into a share of common stock of the Successor Company. As a result, each shareholder of the Predecessor Company became the owner of an identical number of shares of common stock of the Successor Company. Immediately following the Reorganization, the Successor Company was renamed as “InterDigital, Inc.,” identical to the Predecessor Company’s name prior to the Merger. The Successor Company’s common stock continues to be traded under the name “InterDigital, Inc.” and continues to be listed on the NASDAQ Global Select Market under the ticker symbol “IDCC.” In addition, immediately following the Merger the directors and executive officers of the Successor Company were the same individuals who were directors and executive officers, respectively, of the Predecessor Company immediately prior to the Merger.

For the purpose of this Annual Report on Form 10-K, references to the Company, our Board of Directors or any committee thereof, or our management, employees, business or financial results at or for any period prior to the Merger refer to those of the Predecessor Company and thereafter to those of the Successor Company.

PART I

Item 1. *BUSINESS.*

Overview

InterDigital, Inc. ("InterDigital") is a research and development company that licenses its innovations to the global wireless and consumer electronics industries. We design and develop advanced technologies that enable connected, immersive experiences in a broad range of communications and entertainment products and services. Since our founding in 1972, our engineers have designed and developed a wide range of innovations that are used in wireless products and networks, from the earliest digital cellular systems to 5G and, today, solutions that we believe will shape the world beyond 5G. With the acquisition of the patent licensing business of visual technology industry leader Technicolor SA ("Technicolor") in 2018 (the "Technicolor Patent Acquisition"), followed by the acquisition of their Research & Innovation unit in 2019 (the "R&I Acquisition" and, together with the Technicolor Patent Acquisition, the "Technicolor Acquisitions"), we are now a leader in video processing, encoding/decoding, and display technology, with a significant Artificial Intelligence ("AI") research effort that intersects with both wireless and visual technologies.

InterDigital is one of the largest pure research & development and licensing companies in the world, with one of the most significant patent portfolios in the technology industry. As of December 31, 2019, InterDigital's wholly owned subsidiaries held a portfolio of approximately 32,000 patents and patent applications related to wireless communications, video coding, display technology, and other areas relevant to the wireless and consumer electronics industries. Our portfolio includes numerous patents and patent applications that we believe are or may be essential or may become essential to standards established by many Standards Development Organizations ("SDOs"), including cellular and other wireless communications and video technology standards. Those wireless standards include 3G, 4G and the IEEE 802 suite of standards, as well as patents and patent applications that we believe are or may become essential to 5G standards that currently exist and as they continue to develop. Our video technology portfolio includes patents and applications relating to standards established by ISO/IEC Moving Picture Expert Group (MPEG), the ITU-T Video Coding Expert Group (VCEG), the Joint Collaborative Team on Video Coding (JCT-VC) and the Joint Video Expert Team (JVET), among others.

Our wireless portfolio has largely been built through internal development, supplemented by joint development projects with other companies, as well as select acquisitions of patents and companies. Products incorporating our patented inventions in wireless include: mobile devices, such as cellular phones, tablets, notebook computers and wireless personal digital assistants; wireless infrastructure equipment, such as base stations; components, dongles and modules for wireless devices; and Internet of Things ("IoT") devices and software platforms. Our video technology portfolio largely represents patents and applications that came to InterDigital as a result of the Technicolor Patent Acquisition, supplemented by internal development. Our patented inventions in video are incorporated in a range of products and services, including cellular phones, notebook computers, televisions, gaming consoles, set-top boxes, streaming devices and other consumer electronics.

InterDigital derives revenues primarily from patent licensing, with contributions from patent sales, product sales, technology solutions licensing and sales and engineering services. On January 1, 2018, we adopted the requirements of new revenue accounting guidance, ASU No. 2014-09 "*Revenue from Contracts with Customers (Topic 606)*" ("ASC 606"), using the modified retrospective method. Consistent with the modified retrospective adoption method, our results of operations for periods prior to our adoption of ASC 606 remain unchanged and are presented in accordance with ASC Topic 605, "*Revenue Recognition*" ("ASC 605").

In 2019, our total revenues were $318.9 million, including recurring revenues of $298.2 million, which consists of current patent royalties and current technology solutions revenue. In 2018, our total revenues were $307.4 million, which consisted of recurring revenues of $280.3 million. Additional information about our revenues, profits and assets, as well as additional financial data, is provided in the selected financial data in Part II, Item 6, and in the financial statements and accompanying Notes in Part II, Item 8, of this Form 10-K.

Our Strategy

Our objective is to continue to be a leading designer and developer of technology solutions and innovation for the wireless and consumer electronics industries and to

monetize those solutions and innovations primarily through licensing, combined with patent sales and other revenue opportunities.

To execute our strategy, we intend to:

- ***Continue to invest in advanced research and development related to wireless, video, IoT and AI.*** We intend to grow or maintain a leading position in advanced wireless technology, IoT, video coding, and other related technology areas by leveraging our expertise to guide internal research and development capabilities, direct our

efforts in partnering with leading inventors and industry players to source new technologies and pursue select acquisitions of technologies, businesses and/or companies.

- ***Grow our patent-based revenue.*** We intend to grow our licensing revenue base by adding licensees and leveraging the size of the overall mobile technology market, expanding our licensing revenue in the consumer electronics market, and expanding into adjacent and new technology areas that align with our intellectual property position. These licensing efforts can be self-driven or executed in conjunction with licensing partnerships, trusts and other efforts, and may involve the vigorous enforcement and defense of our intellectual property through litigation and other means. We also believe that our ongoing research efforts and associated patenting activities may enable us to sell patent assets that are not vital to our core licensing programs and to execute patent swaps that can strengthen our overall portfolio.
- ***Maintain a collaborative relationship with key industry players and worldwide standards bodies.*** We intend to continue contributing to the ongoing process of defining wireless, video and other standards and other industry-wide efforts and incorporating our inventions into those technology areas. Those efforts, and the knowledge gained through them, support internal development efforts and help guide technology and intellectual property sourcing through partners and other external sources.
- ***Pursue commercial opportunities for our advanced platforms and solutions.*** As part of our ongoing research and development efforts, InterDigital often builds out entire functioning platforms in various technology areas. Moreover, we believe that our advanced capabilities in visual technologies will continue to result in solutions that can be implemented in adjacent industries, such as content production, gaming, and other areas. We seek to bring those technologies, as well as other technologies we may develop or acquire, to market through various methods including technology licensing, stand-alone commercial initiatives, joint ventures and partnerships.

Technology Research and Development

InterDigital R&I

InterDigital operates a diversified research and development operation, InterDigital Research & Innovation ("InterDigital R&I"). InterDigital R&I was created through the combination of InterDigital's research team with Technicolor's R&I team acquired in the R&I Acquisition.

As an early and ongoing participant in the digital wireless market, InterDigital developed pioneering solutions for the primary cellular air interface technologies in use today. That early involvement, and our continued development of advanced digital wireless technologies, have enabled us to create our significant worldwide portfolio of patents. In addition, InterDigital was among the first companies to participate in standardization and platform development efforts related to Machine-to-Machine ("M2M") communications and IoT technology. With the completion of the Technicolor Acquisitions, InterDigital R&I is a leader in key video technologies, including emerging technologies such as digital avatars, immersive video and AI. Our current research efforts are focused on a variety of areas related to future technology and devices, including cellular wireless technology, advanced video coding and transmission, and AI.

Our capabilities in the development of advanced technologies are based on the efforts of a highly specialized engineering team, leveraging leading-edge equipment and software platforms. As of December 31, 2019, InterDigital employed approximately 290 engineers, approximately 90% of whom hold advanced degrees (including 107 doctorate degrees). Over the last three years, investment in development has ranged from $69.7 million to $75.7 million, and the largest portion of this expense has been personnel costs. Additional information about our development expenses is provided in the results of operations, under the heading "*Operating Expenses*," in Part II, Item 7, of this Form 10-K.

Cellular Wireless Technology

We have a long history of developing cellular technologies, including those related to CDMA and TDMA and, more recently, OFDM/OFDMA and MIMO. Many of our inventions are being used in all 2G, 3G, 4G and 5G wireless networks and mobile terminal devices. We also continue to be engaged in development efforts to build and enhance our 3GPP technology portfolio in areas including 5G NR, LTE-Advanced, and cellular IoT. Further, we continue to develop additional technologies in response to existing or perceived challenges of connected devices in the expanding terminal markets. These include technologies for automobiles, wearables, smart homes, drones and other connected consumer

electronic products. We are developing solutions that enable connectivity in both licensed and unlicensed spectrum, and across a large range of frequencies up to the millimeter wave bands.

Our wireless research and development activities focus on solutions that apply to 3GPP and other wireless market segments. Segments outside of 3GPP primarily fall within the scope of the IEEE 802, IETF and ETSI standards. We continue to grow a portfolio of technology related to Wi-Fi, Internet Standards, and Edge Computing, that includes, for example,

improvements to the IEEE 802.11 PHY and MAC to increase peak data rates (802.11ax, 802.11ay), integrated access and backhaul, and terminal mobility for edge and fog computing services.

Advanced Video Coding and Transmission Technology

An important and growing segment of wireless traffic is devoted to video streaming. InterDigital has been active for a number of years in developing advanced technologies that address the challenges of video as it relates to mobile, and we further enhanced our capabilities in this area with the completion of the R&I Acquisition. Specifically, in the area of video research and standards, we have been actively engaged in video standards development work in the ISO/IEC Moving Picture Expert Group (MPEG), the ITU-T Video Coding Expert Group (VCEG), the Joint Collaborative Team on Video Coding (JCT-VC) and the Joint Video Expert Team (JVET). Those efforts have focused on H.265/HEVC versions 1 to 4 and MPEG DASH, as well as development of the FVC/H.266 and the MPEG Immersive (MPEG-I) standards suite for the future. Beyond standards, InterDigital R&I is conducting research in groundbreaking areas such as digital doubles and digital twins, immersive video, augmented and mixed reality, and other emerging technologies.

Artificial Intelligence

In addition to our historical work in major wireless standards that integrate some AI capabilities, the R&I Acquisition brought an advanced AI lab to InterDigital that is researching a variety of aspects of AI that intersect with video and wireless technology. Those areas of research include: energy-efficient deep learning, aimed at reducing the energy-intensive rollout of AI into specific service areas; deep video and M2M compression, seeking to design disruptive video codecs based on deep learning techniques; AI for dynamic wireless environments, focused on learning and optimizing wireless systems, particularly when channel dynamics are highly dynamic; and explainable or interpretable AI, addressing weaknesses in neural networks in providing transparency and generating trust.

Patent Portfolio; R&D Facilities

As of December 31, 2019, our patent portfolio consisted of approximately 32,000 patents and patent applications worldwide. The patents and applications comprising our portfolio relate predominantly to cellular wireless standards, including 3G, 4G and 5G technologies (sometimes referred to as "3GPP"), other wireless standards, including 802.11 (Wi-Fi) technology, and a variety of video technologies and standards, such as HEVC. Issued patents expire at differing times ranging from 2020 through 2038. We currently operate eight research and development facilities in five countries: Berlin, Germany; Conshohocken, Pennsylvania, USA; London, United Kingdom; Montreal, Canada; New York, New York, USA; Palo Alto, California, USA; Rennes, France; and San Diego, California, USA.

Our Revenue Sources

Patent-Based Revenue

Overview of Patent Licenses

We believe that companies making, importing, using or selling products compliant with the standards covered by our patent portfolio, including all manufacturers of mobile handsets, tablets and other devices, require a license under our patents and will require licenses under patents that may issue from our pending patent applications. We have successfully entered into license agreements with many of the leading mobile communications companies globally, including Apple Inc. ("Apple"), Google LLC ("Google"), LG Electronics, Inc. ("LG"), Samsung Electronics Co., Ltd. ("Samsung"), Sony Corporation of America ("Sony"), and ZTE Corporation ("ZTE"), among others.

We have striven to be recognized within the licensing industry for the transparency of our business, fairness and flexibility of our approach, and our willingness to work with licensees. In furtherance of this objective, in January 2020, we made publicly available our rates, portfolio data and licensing policies with regard to mobile handsets, potentially setting a new industry standard for transparency in licensing.

Most of our patent license agreements are structured on a variable royalty basis, while others are structured on a fixed-fee basis or a combination thereof. Upon entering into a new patent license agreement, the licensee typically agrees to pay consideration for sales made prior to the effective date of the license agreement (i.e., past patent royalties) and also agrees to pay royalties or license fees on licensed products sold during the term of the agreement. We expect that, for the most part, new license agreements will follow this model. Almost all of our patent license agreements provide for the payment of royalties based on sales of licensed products designed to operate in accordance with particular standards (convenience-based licenses), as opposed to the payment of royalties if the manufacture, sale or use of the licensed product infringes one of our patents (infringement-based licenses).

Some of our patent licenses are fixed-fee agreements, requiring no additional payments relating to designated sales under agreed upon conditions. Those conditions can include paid-up licenses for a period of time, for a class of products, for a

number of products sold, under certain patents or patent claims, for sales in certain countries or a combination thereof. Licenses become paid-up based on the payment of fixed amounts or after the payment of royalties for a term.

Some of our patent license agreements provide for the non-refundable prepayment of royalties that are usually made in exchange for prepayment discounts. As the licensee reports sales of covered products, the royalties are calculated and either applied against any prepayment or become payable in cash or other consideration. Additionally, royalties on sales of licensed products under the license agreement become payable or applied against prepayments based on the royalty formula applicable to the particular license agreement. These formulas include flat dollar rates per unit, a percentage of sales, a percentage of sales with a per-unit cap and other similar measures. The formulas can also vary by other factors, including territory, covered standards, quantity and dates sold. Our license agreements typically contain provisions that give us the right to audit our licensees' books and records to ensure compliance with the licensees' reporting and payment obligations under those agreements. From time to time, these audits reveal underreporting or underpayments under the applicable agreements. In such cases, we seek payment for the amount owed and enter into negotiations with the licensee to resolve the discrepancy.

For a discussion of our revenue recognition policies with respect to patent license agreements, see “Item 7. *Management’s Discussion and Analysis of Financial Condition and Results of Operations - Overview - Critical Accounting Policies and Estimates - Revenue Recognition - Patent License Agreements*.”

Licensing Through Platforms

As part of the Technicolor Patent Acquisition, we assumed Technicolor's rights and obligations under a joint licensing program with Sony relating to digital televisions ("DTVs") and standalone computer display monitors ("CDMs") (such program, the "Madison Arrangement"), including Technicolor's role as sole licensing agent. Under the Madison Arrangement, Technicolor and Sony combined portions of their respective DTV and CDM patent portfolios and created a combined licensing opportunity to DTV and CDM manufacturers. As licensing agent for the Madison Arrangement, we are responsible for making decisions regarding the prosecution and maintenance of the combined patent portfolio and the licensing and enforcement of the combined patent portfolio in the field of use of DTVs and CDMs. Refer to Note 5, "*Business Combinations and Other Transactions*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information about the Madison Arrangement.

In third quarter 2016, InterDigital joined Avanci, the industry’s first marketplace for the licensing of cellular standards-essential technology for the IoT. The licensing platform brings together some of InterDigital’s peers in standards-essential technology leadership, and makes 2G, 3G, 4G and 5G standards-essential patents available to IoT players in specific product segments with one flat-rate license. The Avanci licensing programs in specific product segments for the IoT industry will provide access to the entire applicable standards-essential wireless patent portfolios held by all of the platform participants, as well as any additions to their portfolios during the term of the license. Since December 2017, Avanci has announced signed patent license agreements with BMW Group, Audi and Porsche, the Volkswagen Group Companies and Volvo Cars.

In 2013, InterDigital formed the Signal Trust for Wireless Innovation (the "Signal Trust"). The goal of the Signal Trust is to monetize a large patent portfolio related to cellular infrastructure. More than 500 patents and patent applications were transferred from InterDigital to the Signal Trust, focusing primarily on 3G and LTE technologies and developed by InterDigital's engineers and researchers over more than a decade. A number of these innovations have been contributed to the worldwide standards process, resulting in a portfolio that includes patents for pioneering inventions that we believe are used pervasively in the cellular wireless industry. InterDigital is the primary beneficiary of the Signal Trust. The distributions from the Signal Trust will support continued research related to cellular wireless technologies. A small portion of the proceeds from the Signal Trust are used to fund, through the Signal Foundation for Wireless Innovation, scholarly analysis of intellectual property rights and the technological, commercial and creative innovations they facilitate.

Patent Sales

We also pursue, on occasion, targeted sales of portions of our patent portfolio. This strategy is based on the expectation that our portfolio and continued research efforts extend well beyond the requirements for a successful licensing program. In addition, the strategy leverages the desire from new entrants in the mobile technology space to build strong intellectual property positions to support their businesses.

Other Potential Revenue Opportunities

Our strong technology expertise and research and development team also form the basis for other potential revenue opportunities, focused around areas such as engineering services, research joint ventures and the continued development, commercialization and licensing of research and development projects that have progressed to a pre-commercial or commercial phase. We also currently recognize revenue from the licensing of technology that has been developed by our engineering teams and is integrated into other companies' technology products.

In all of its technology areas, InterDigital works to incubate and commercialize market-ready technologies. These include technologies that were developed as part of our standards development efforts, as well as technologies developed outside the scope of those efforts. Those commercial efforts sometimes include the establishment of a separate commercial initiative focused on the specific opportunity. Although these initiatives are in their early stages, they are potential revenue opportunities for the Company.

In 2012, we formed a joint venture with Sony called Convida Wireless. The joint venture combined InterDigital's advanced M2M research capabilities with Sony's consumer electronics expertise with the purpose of driving new research in IoT communications and other connectivity areas. This joint venture was renewed in 2015 with its focus expanded to include advanced research and development into 5G and future wireless technologies, and further renewed in 2018.

Overview of Wireless Communications and Consumer Electronics Industries

The wireless communications industry continues to be significant worldwide, and the number of device types entering the market is growing. For example, the introduction of 5G wireless networks is expected to drive a significant upgrade cycle for mobile phones, and 5G technology is expected to be implemented in an expanding range of products. In particular, IoT is an important new market that is expected to result in a significant increase in the number of connected devices worldwide and unlock new business capabilities. IoT is still in its early stages, and estimates vary broadly as far as how many connections it will yield, but by some estimates there could be as many as 120 billion connected devices by 2030, a significant portion of which is expected to comprise cellular IoT devices. According to data from IHS Markit, more than 2 billion devices in the video, audio and IoT/other technology areas were shipped in 2019. Those devices include TV displays, computer displays, set-top boxes, gaming consoles, wireless assistants and headphones, wearables, smart home devices, and other types of consumer electronic devices that implement video or wireless technology, or a combination of both. The consumer electronics industry is also experiencing significant change, as technology-enabled services such as video streaming and 4K UHD video are being adopted globally.

To achieve economies of scale and support interoperability among different participants, products for the wireless industry have typically been designed to operate in accordance with certain standards. Industry standards are formal guidelines for engineers, designers, manufacturers and service providers that regulate and define the use of the radio frequency spectrum in conjunction with providing detailed specifications for wireless communications products. A primary goal of the standards is to ensure interoperability of products marketed by multiple companies. A large number of international and regional wireless SDOs, including the ITU, ETSI, TIA (USA), IEEE, ATIS (USA), TTA (Korea), ARIB (Japan) and ANSI, have responsibility for the development and administration of wireless communications standards. New standards are typically adopted with each new generation of products, are often compatible with previous generations and are defined to ensure equipment interoperability and regulatory compliance. The consumer electronics industry also implements many of the same standards, including standards related to Wi-Fi and increasingly, cellular technologies, as well as a broad range of video coding standards that are governed by regional and global SDOs.

SDOs typically ask participating companies to declare formally whether they believe they hold patents or patent applications essential to a particular standard and whether they are willing to license those patents on either a royalty-bearing basis on fair, reasonable and nondiscriminatory terms or on a royalty-free basis. To manufacture, have made, sell, offer to sell or use such products on a non-infringing basis, a manufacturer or other entity doing so must first obtain a license from the holder of essential patent rights. The SDOs do not have enforcement authority against entities that fail to obtain required licenses, nor do they have the ability to protect the intellectual property rights of holders of essential patents.

InterDigital often publicly characterizes aspects of its business, including license agreements and development projects, as pertaining to mobile industry standards such as, for example, 3G, 4G, 5G, Wi-Fi and HEVC. In doing this, we generally rely on the positions of the applicable SDOs in defining the relevant standards. However, the definitions may evolve or change over time, including after we have characterized certain transactions.

Business Activities

2019 Patent Licensing Activity

Direct Licenses

During fourth quarter 2019, we entered into a multi-year, worldwide, non-exclusive, royalty-bearing patent license and settlement agreement with ZTE. The agreement covers the sale of ZTE's 3G, 4G and 5G handset and tablet products, as well as 802.11 and HEVC technologies incorporated into such products.

Also during fourth quarter 2019, we entered into a multi-year, worldwide, non-exclusive, royalty-bearing patent license agreement with u-blox AG ("u-blox"). The agreement covers the sale of u-blox's 3G and 4G machine to machine modules and certain consumer modules.

During second quarter 2019, we entered into a multi-year, worldwide, non-exclusive, royalty-bearing patent license agreement with Teltronic S.A.U. ("Teltronic"). The agreement covers the sale of Teltronic's 4G terminal units as well as 3G and 4G infrastructure equipment.

Also during second quarter 2019, we entered into a Settlement Agreement and First Amendment to the Patent License Agreement with Asustek Computer Incorporated ("Asus"). The agreement provides for, among other things, a multi-year amendment to our 2008 patent license agreement with Asus (the "2008 Asus PLA") that adds coverage for 4G technologies and amends certain other terms of the 2008 Asus PLA.

Licenses Through Platforms

During fourth quarter 2019, Google was granted a multi-year, worldwide, non-exclusive, royalty-bearing patent license covering the sale of certain of its 3G and 4G mobile communication devices. We entered into this agreement through a licensing platform.

Also during fourth quarter 2019, as part of the Madison Arrangement, we entered into a multi-year, non-exclusive, royalty-bearing patent license agreement with Funai Electric Co., Ltd. ("Funai"). The agreement covers the U.S. sales of Funai's DTVs.

During 2019, Avanci announced that it entered into several patent license agreements with new licensees, including Audi and Porsche, the Volkswagen Group Companies and Volvo Cars.

Customers Generating Revenues Exceeding 10% of Total 2019 Revenues

Apple, Samsung and LG Electronics comprised approximately 35%, 25% and 10% of our total 2019 revenues, respectively.

In 2016, we entered into a multi-year, royalty-bearing, worldwide and non-exclusive patent license agreement with Apple (the “Apple PLA”). The agreement sets forth terms covering the sale by Apple of its products and services, including, but not limited to, its 3G, 4G and future generation cellular and wireless-enabled products. The Apple PLA gives Apple the right to terminate certain rights and obligations under the license for the period after September 30, 2021, but has the potential to provide a license to Apple for a total of up to six years. During 2019, we recognized a total of $111.7 million of revenue associated with the Apple PLA under ASC 606.

In 2014, we entered into a patent license agreement with Samsung (the “Samsung PLA”). The royalty-bearing license agreement sets forth terms covering the sale by Samsung of 3G, 4G and certain future generation wireless products. The Samsung PLA provided Samsung the right to terminate certain rights and obligations under the license for the period after 2017 but had the potential to provide a license to Samsung for a total of ten years, including 2013. Samsung did not elect to terminate such rights and obligations, and the period for such election has expired. Accordingly, the term of the Samsung PLA ends on December 31, 2022. During 2019, we recognized a total of $78.3 million of revenue associated with the Samsung PLA under ASC 606.

In 2017, we entered into a multi-year, worldwide, non-exclusive patent license agreement with LG (the “LG PLA”), a global leader and technology innovator in consumer electronics, mobile communications and home appliances. The LG PLA covers the 3G, 4G and 5G terminal unit products of LG and its affiliates and sets forth a royalty of cash payments to InterDigital as well as a process for the transfer of patents from LG to InterDigital. The term of the LG PLA expires on December 31, 2020. During 2019, we recognized a total of $31.8 million of revenue associated with the LG PLA under ASC 606.

Patent Infringement and Declaratory Judgment Proceedings

From time to time, if we believe a party is required to license our patents in order to manufacture, use and/or sell certain products and such party refuses to do so, we may agree with such party to have royalty rates, or other terms, set by third party adjudicators (such as arbitrators) or, in certain circumstances, we may institute legal action against them. This legal action has typically taken the form of a patent infringement lawsuit or an administrative proceeding such as a Section 337 proceeding before the United States International Trade Commission ("USITC" or the "Commission"). In a patent infringement lawsuit, we would typically seek damages for past infringement and an injunction against future infringement. In a USITC proceeding, we would seek an exclusion order to bar infringing goods from entry into the United States, as well as a cease and desist order to bar further sales of infringing goods that have already been imported into the United States. Parties

may bring administrative and/or judicial challenges to the validity, enforceability, essentiality and/or applicability of our patents to their products. Parties may also allege that our efforts to enter into a license with that party do not comply with any obligations we may have in connection with our participation in standards-setting organizations, and therefore that we are not entitled to the relief that we seek. For example, a party may allege that we have not complied with an obligation to offer (or be prepared to offer) a license to that party for patents that are or may become standards-essential patents ("SEPs") on fair, reasonable and non-discriminatory ("FRAND") terms and conditions, and may also file antitrust claims or regulatory complaints on that or other bases, and may seek damages or other relief based on such claims. In addition, a party might file a declaratory judgment action to seek a court's declaration that our patents are invalid, unenforceable, not infringed by the other party's products or are not SEPs. Our response to such a declaratory judgment action may include claims of infringement. When we include claims of infringement in a patent infringement lawsuit, a favorable ruling for the Company can result in the payment of damages for past patent royalties, the setting of a royalty for future sales or issuance by the court of an injunction enjoining the infringer from manufacturing, using and/or selling the infringing product.

Contractual Arbitration Proceedings

We and our licensees, in the normal course of business, may have disagreements as to the rights and obligations of the parties under applicable agreements. For example, we could have a disagreement with a licensee as to the amount of reported sales and royalties. Our patent license agreements typically provide for audit rights as well as private confidential arbitration as the mechanism for resolving disputes, and we may attempt to resolve such disputes in arbitration. In arbitration, licensees may seek to assert various claims, defenses, or counterclaims, such as claims based on waiver, promissory estoppel, breach of contract, fraudulent inducement to contract, antitrust, and unfair competition. Arbitration proceedings can be resolved through an award rendered by the arbitrators or by settlement between the parties. Parties to arbitration might have the right to have the award reviewed in a court of competent jurisdiction. However, based on public policy favoring the use of arbitration, it is generally difficult to have arbitration awards vacated or modified. The party securing an arbitration award may seek to have that award confirmed as a judgment through an enforcement proceeding. The purpose of such a proceeding is to secure a judgment that can be used for, if need be, seizing assets of the other party.

In addition, arbitration may be a particularly effective means for resolving disputes with prospective licensees concerning the appropriate FRAND terms and conditions for license agreements that include SEPs, particularly where negotiations have otherwise reached an impasse. Binding arbitration to resolve the terms and conditions of a worldwide FRAND license to our relevant portfolio of SEPs is an efficient and cost-effective mechanism, as it allows the parties to avoid piecemeal litigation in multiple jurisdictions and ensures that an enforceable patent license agreement that is consistent with FRAND commitments will be in place at the end of the arbitration process.

Competition

With respect to our technology development activities and resulting commercialization efforts, we face competition from companies, including in-house development teams at other wireless and consumer electronics device companies, semiconductor companies and wireless operators, developing other and similar technologies that are competitive with our products and solutions that we may market or set forth into the standards-setting arena.

Due to the exclusionary nature of patent rights, we do not compete, in a traditional sense, with other patent holders for patent licensing relationships or sale transactions. Other patent holders do not have the same rights to the inventions and technologies encompassed by our patent portfolio. In any device or piece of equipment that contains intellectual property, the manufacturer may need to obtain licenses from multiple holders of intellectual property. In licensing our patent portfolio, we compete with other patent holders for a share of the royalties that certain licensees may argue to be the total royalty that is supported by a certain product or products, which may face practical limitations. We believe that licenses under a number of our patents are required to manufacture and sell 3G, 4G, 5G and other wireless products, as well as other consumer electronics devices. However, numerous companies also claim that they hold patents that are or may be essential or may become essential to standards-based technology deployed on wireless products and other consumer electronics devices. To the extent that multiple parties all seek royalties on the same product, the manufacturers could claim to have difficulty in meeting the financial requirements of each patent holder. In the past, certain manufacturers have sought antitrust exemptions to act collectively on a voluntary basis. In addition, certain manufacturers have sought to limit aggregate licensing fees or rates for SEPs. Similarly, potential purchasers of our patents often amass patent portfolios for defensive and/or cross-licensing purposes and could choose to acquire patent assets within the same general technology space from other patent holders.

Employees

As of December 31, 2019, we had approximately 487 employees, including approximately 206 employees in France who were subject to collective bargaining arrangements. We consider our employee relations to be good.

Geographic Concentrations

See Note 4, "*Geographic/Customer Concentration*," in the Notes to Condensed Consolidated Financial Statements included in Part II, Item 8, of this Form 10-K for financial information about geographic areas for the last three years.

Corporate Information

The ultimate predecessor company of InterDigital, Inc. was incorporated in 1972 under the laws of the Commonwealth of Pennsylvania and conducted its initial public offering in November 1981. Our headquarters are located in Wilmington, Delaware, USA. Our research and development activities are conducted primarily in facilities located in Berlin, Germany; Conshohocken, Pennsylvania, USA; London, United Kingdom; Montreal, Canada; New York, New York, USA; Palo Alto, California, USA; Rennes, France; and San Diego, California, USA. We are also a party to leases for several smaller research and/or office spaces, including in Brussels, Belgium; Buffalo, New York, USA; Indianapolis, Indiana, USA; Paris, France; San Francisco, California, USA; and Shanghai, China. In addition, we own a building in Washington, District of Columbia, USA, that houses administrative office space.

Our Internet address is *www.interdigital.com*, where, in the "Investors" section, we make available, free of charge, our Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, certain other reports and filings required to be filed under the Securities Exchange Act of 1934, as amended (the "Exchange Act"), and all amendments to those reports or filings as soon as reasonably practicable after such material is electronically filed with or furnished to the United States Securities and Exchange Commission. The information contained on or connected to our website is not incorporated by reference into this Form 10-K.

Item 1A. *RISK FACTORS.*

We face a variety of risks that may affect our business, financial condition, operating results, the trading price of our common stock, or any combination thereof. You should carefully consider the following information and the other information in this Form 10-K in evaluating our business and prospects and before making an investment decision with respect to our common stock. If any of these risks were to occur, our business, financial condition, results of operations or prospects could be materially and adversely affected. In such an event, the market price of our common stock could decline and you could lose all or part of your investment. The risks and uncertainties we describe below are not the only ones facing us. Additional risks not presently known to us or that we currently deem immaterial may also affect our business.

Risks Related to Our Business

Our plans to license handset manufacturers in China may be adversely affected by a deterioration in United States-China trade and geopolitical relations, our customers facing economic uncertainty there or our failure to establish a positive reputation in China, which could materially adversely affect our long-term business, financial condition and operating results.

Companies headquartered in China currently comprise a substantial portion of the handset manufacturers that remain unlicensed to our patent portfolio. Our ability to license such manufacturers is, among other things, affected by the macroeconomic and geopolitical climate, as well as our business relationships and perceived reputation in China. The U.S. and Chinese governments are currently engaged in trade negotiations, and the U.S. State Department issued a travel advisory in January 2019 that advised U.S. citizens to exercise increased caution in China due to arbitrary enforcement of local laws. This travel advisory and other security concerns have restricted our ability to conduct in-person negotiations with prospective Chinese licensees in the past, and they, along with certain public health concerns (e.g., the current coronavirus outbreak), could do so in the future. In January 2020, the U.S. and China entered into Phase One of the Economic and Trade Agreement Between the United States of America and the People's Republic of China (the "Phase One Trade Agreement"). The Phase One Trade Agreement takes steps to ease certain trade tensions between the U.S. and China, including tensions involving intellectual property theft and forced intellectual property transfers by China. Although the Phase One Trade Agreement is an encouraging sign of progress in the trade negotiations between the U.S. and China, questions still remain as to the enforcement of its terms, the resolution of a number of other points of dispute between the parties, and the prevention of further tensions. If the U.S.-China trade dispute re-escalates or relations between the United States and China deteriorate, these conditions could adversely affect our ability to license our patent portfolio to Chinese handset manufacturers. Our ability to license such manufacturers could also be affected by economic uncertainty, particularly in the handset market, in China or by our failure to establish a positive reputation and relationships in China. The occurrence of any of these events could have an adverse effect on our ability to enter into

license agreements with Chinese handset manufacturers, which, in turn, could cause our long-term business, financial condition and operating results to be materially adversely affected.

Royalty rates, or other terms, under our patent license agreements could be subject to determination through arbitration or other third-party adjudications or regulatory or court proceedings, and arbitrators, judges or other third-party adjudicators

or regulators could determine that our patent royalty rates should be at levels lower than our agreed or historical rates or otherwise make determinations resulting in less favorable terms and conditions under our patent license agreements.

Historically, we strive for the terms of our patent license agreements, including our royalty rates, to be reached through arms-length bilateral negotiations with our licensees. We could agree, as we did with Huawei pursuant to our December 2013 settlement agreement, to have royalty rates, or other terms, set by third party adjudicators (such as arbitrators) and it is also possible that courts or regulators could decide to set or otherwise determine the FRAND consistency of such terms or the manner in which such terms are determined, including by determining a worldwide royalty rate for our SEPs. Changes to or clarifications of our obligations to be prepared to offer licenses to SEPs on FRAND terms and conditions could require such terms, including our royalty rates, to be determined through third party adjudications. Finally, we and certain of our current and prospective licensees have initiated, and we and others could in the future initiate, legal proceedings or regulatory proceedings requesting third party adjudicators or regulators to set FRAND terms and conditions for, or determine the FRAND-consistency of current terms and conditions in, our patent license agreements, and which could result in such third party adjudicators or regulators determining a worldwide royalty rate for our SEPs. To the extent that our patent royalty rates for our patent license agreements are determined through arbitration or other third party adjudications or regulatory or court proceedings rather than through bilateral negotiations, because such proceedings are inherently unpredictable and uncertain and there are currently few precedents for such determinations, it is possible that royalty rates may be lower than our historical rates, and this could also have a negative impact on royalties we are able to obtain from future licensees, which may have an adverse effect on our revenue and cash flow. In addition, to the extent that other terms and conditions for our patent license agreements are determined through such means, such terms and conditions could be less favorable than our historical terms and conditions, which may have an adverse effect on our licensing business.

Due to the nature of our business, we could continue to be involved in a number of costly litigation, arbitration and administrative proceedings to enforce or defend our intellectual property rights and to defend our licensing practices.

While some companies seek licenses before they commence manufacturing and/or selling devices that use our patented inventions, most do not. Consequently, we approach companies and seek to establish license agreements for using our inventions. We expend significant time and effort identifying users and potential users of our inventions and negotiating license agreements with companies that may be reluctant to take licenses. However, if we believe that a third party is required to take a license to our patents in order to manufacture, sell, offer for sale, import or use products, we have in the past commenced, and may in the future, commence legal or administrative action against the third party if they refuse to enter into a license agreement with us. In turn, we have faced, and could continue to face, counterclaims and other legal proceedings that challenge the essential nature of our patents, or that claim that our patents are invalid, unenforceable or not infringed. Litigation adversaries may allege that we have not complied with certain commitments to standards-setting organizations and therefore that we are not entitled to the relief that we seek. For example, a party may allege that we have not complied with an obligation to offer a license to a party on FRAND terms and conditions, and may also file antitrust claims, unfair competition claims or regulatory complaints on that or other bases, and may seek damages and other relief based on such claims. Litigation adversaries have also filed against us, and other third parties may in the future file, validity challenges such as inter partes proceedings in the USPTO, which can lead to delays of our patent infringement actions as well as potential findings of invalidity.

Litigation may be also required to enforce our intellectual property rights, protect our trade secrets, enforce patent license and confidentiality agreements or determine the validity, enforceability and scope of proprietary rights of others.

Third parties could commence litigation against us seeking to invalidate our patents or obtain a determination that our patents are not infringed, are not essential, are invalid or are unenforceable. In addition, current and prospective licensees have initiated proceedings against us claiming, and others in the future may claim, that we have not complied with our FRAND licensing commitments and/or engaged in anticompetitive or unfair licensing activities.

The cost of enforcing and defending our intellectual property and of defending our licensing practices has been and may continue to be significant. As a result, we could be subject to significant legal fees and costs, including in certain jurisdictions the costs and fees of opposing counsel if we are unsuccessful. In addition, litigation, arbitration and administrative proceedings require significant key employee involvement for significant periods of time, which could divert these employees from other business activities.

Potential patent and litigation reform legislation, potential USPTO and international patent rule changes, potential legislation affecting mechanisms for patent enforcement and available remedies, and potential changes to the intellectual property rights ("IPR") policies of worldwide standards bodies, as well as rulings in legal proceedings, may

affect our investments in research and development and our strategies for patent prosecution, licensing and enforcement and could have a material adverse effect on our licensing business as well as our business as a whole.

Potential changes to certain U.S. and international patent laws, rules and regulations may occur in the future, some or all of which may affect our research and development investments, patent prosecution costs, the scope of future patent coverage we secure, the number of forums in which we can seek to enforce our patents, the remedies that we may be entitled to in patent litigation, and attorneys' fees or other remedies that could be sought against us, and may require us to reevaluate and modify our research and development activities and patent prosecution, licensing and enforcement strategies. Similarly, legislation designed to reduce the jurisdiction and remedial authority of the USITC has periodically been introduced in Congress.

Any potential changes in the law, the IPR policies of standards bodies or other developments that reduce the number of forums available or the type of relief available in such forums (such as injunctive relief), restrict permissible licensing practices (such as our ability to license on a worldwide portfolio basis) or that otherwise cause us to seek alternative forums (such as arbitration or state court), would make it more difficult for us to enforce our patents, whether in adversarial proceedings or in negotiations. Because we have historically depended on the availability of certain forms of legal process to enforce our patents and obtain fair and adequate compensation for our investments in research and development and the unauthorized use of our intellectual property, developments that undermine our ability to do so could have a negative impact on future licensing efforts.

Rulings in our legal proceedings as well as those of third parties may affect our strategies for patent prosecution, licensing and enforcement. For example, in recent years, the USITC and U.S. courts, including the U.S. Supreme Court and the U.S. Court of Appeals for the Federal Circuit, have taken some actions that have been viewed as unfavorable to patentees, including us. Decisions that occur in the U.S. or in international forums may change the law applicable to various patent law issues, such as, for example, patentability, validity, claim construction, patent exhaustion, patent misuse, permissible licensing practices, available forums, and remedies such as damages and injunctive relief, in ways that are detrimental to the ability of patentees to enforce patents and obtain suitable relief.

We continue to monitor and evaluate our strategies for prosecution, licensing and enforcement with regard to these developments; however, any resulting change in such strategies may have an adverse impact on our business and financial condition.

Setbacks in defending our patent licensing practices could cause our cash flow and revenue to decline and could have an adverse effect on our licensing business.

Adverse decisions in litigation or regulatory actions relating to our licensing practices, including, but not limited to, findings that we have not complied with our FRAND commitments and/or engaged in anticompetitive or unfair licensing activities or that any of our license agreements are void or unenforceable, could have an adverse impact on our cash flow and revenue. Regulatory bodies may assess fines in the event of adverse findings, and as part of court or arbitration proceedings, a judgment could require us to pay damages (including the possibility of treble damages for antitrust claims). In addition, to the extent that legal decisions find patent license agreements to be void or unenforceable in whole or in part, that could lead to a decrease in the revenue associated with and cash flow generated by such agreements, and, depending on the damages requested, could lead to the refund of certain payments already made. Finally, adverse legal decisions related to our licensing practices could have an adverse effect on our ability to enter into license agreements, which, in turn, could cause our cash flow and revenue to decline.

Royalty rates could decrease for future license agreements due to downward product pricing pressures and competition over patent royalties.

Royalty payments to us under future license agreements could be lower than anticipated. Certain licensees and others in the wireless and consumer electronics industries, individually and collectively, are demanding that royalty rates for patents be lower than historic royalty rates and/or that such rates should be applied to royalty bases smaller than the selling price of an end product (such as the "smallest salable patent practicing unit"). There is also increasing downward pricing pressure on certain wireless products, including handsets, and other consumer electronics devices that we believe implement our patented inventions, and some of our royalty rates are tied to the pricing of these devices. In addition, a number of other companies also claim to hold patents that are essential with respect to products we aim to license. Demands by certain licensees to reduce royalties due to pricing pressure or the number of patent holders seeking royalties on these technologies, could result in a decrease in the royalty rates we receive for use of our patented inventions, thereby decreasing future revenue and cash flow.

Our plans to broaden our revenue opportunities through acquiring or developing technology in new or expanded areas, such as technologies in the consumer electronics and IoT spaces, and enhanced intellectual property sourcing and joint ventures, may not be successful and could materially adversely affect our long-term business, financial condition and operating results.

As part of our business strategy, we are seeking to broaden our revenue opportunities through targeted acquisitions, research partnerships, joint ventures and the continued development of new technologies. Increasingly, our future growth in

part depends on developing or acquiring technology in new or expanded areas that are used on cellular devices (such as video coding technologies) and adjacent industry segments outside of traditional cellular industries (such as other consumer electronics devices and the IoT, including the connected home and smart cities, automotive, mobile computing, mobile health and sensor technology), and on third parties incorporating our technology and solutions into device types used in these areas and industry segments. There is no guarantee that we will succeed in acquiring or developing technology and patents or partnering with inventors and research organizations to create new revenue opportunities and/or add new dimensions to our existing portfolio of intellectual property and potentially create new patent licensing programs. Also, our development activities may experience delays, which could reduce our opportunities for patent licensing or other avenues of revenue generation related to such development activities. In the event that any of these risks materialize, our long-term business, financial condition and operating results may be materially adversely affected.

Setbacks in defending and enforcing our patent rights could cause our revenue and cash flow to decline.

Some third parties have challenged, and we expect will continue to challenge, the infringement, validity and enforceability of certain of our patents. In some instances, certain of our patent claims could be substantially narrowed or declared invalid, unenforceable, not essential or not infringed. We cannot ensure that the validity and enforceability of our patents will be maintained or that our patents will be determined to be applicable to any particular product or standard. Moreover, third parties could attempt to circumvent certain of our patents through design changes. Any significant adverse finding as to the validity, infringement, enforceability or scope of our patents and/or any successful design-around of our patents could result in the loss of patent licensing revenue from existing licensees, through termination or modification of agreements or otherwise, and could substantially impair our ability to secure new patent licensing arrangements, either at all or on beneficial terms.

Our technologies may not become patented, adopted by wireless or video standards or widely deployed.

We invest significant resources in the development of advanced technology and related solutions. However, certain of our inventions that we believe will be employed in current and future products, including 4G, 5G, HEVC and others, are the subject of patent applications where no patent has been issued to us yet by the relevant patent issuing authorities. There is no assurance that these applications will issue as patents, either at all or with claims that would be required by products in the market currently or in the future. Our investments may not be recoverable or may not result in meaningful revenue if a sufficient number of our technologies are not patented and adopted by the relevant standards or if products based on the technologies in which we invest are not widely deployed. Competing technologies could reduce the opportunities for the adoption or deployment of technologies we develop. In addition, it is possible that in certain technology areas, such as in the IoT space, the adoption of proprietary systems could compete with or replace standards-based technology. It is also possible in certain technology areas, such as video coding and the IoT, that open source solutions such as AV1, VP-9 and OCF could compete with or replace proprietary standards-based technology. If the technologies in which we invest do not become patented or are not adopted by the relevant standards, or are not adopted by and deployed in the mainstream markets, at all or at the rate or within time periods that we expect, or in the case of open source solutions, do not infringe our technology, our business, financial condition and operating results could be adversely affected.

Delays in renewing or an inability to renew existing license agreements could cause our revenue and cash flow to decline.

Many of our license agreements have fixed terms. Although we endeavor to renew license agreements with fixed terms prior to the expiration of the license agreements, due to various factors, including the technology and business needs and competitive positions of our licensees and, at times, reluctance on the part of our licensees to participate in renewal discussions, we may not be able to renegotiate the license agreements on acceptable terms before or after the expiration of the license agreement, or at all. If there is a delay in renegotiating and renewing a license agreement prior to its expiration, there could be a gap in time during which we may be unable to recognize revenue from that licensee or we may be forced to renegotiate and renew the license agreement on terms that are more favorable to such licensee, and, as a result, our revenue and cash flow could be materially adversely affected. In addition, if we fail to renegotiate and renew our license agreements at all, our revenue and cash flow could be materially adversely affected.

Increased scrutiny by antitrust authorities may affect our strategies for patent prosecution, licensing and enforcement and may increase our costs of doing business and/or lead to monetary fines, penalties or other remedies or sanctions.

Domestic and foreign antitrust authorities have increased their scrutiny of the use of SEPs, including the enforcement of such patents against competitors and others. Such scrutiny has already resulted in enforcement actions against Qualcomm and could lead to additional investigations of, or enforcement actions against, us. Such inquiries and/or enforcement actions could impact the availability of injunctive and monetary relief, which may adversely affect our strategies for patent prosecution, licensing and enforcement and

increase our costs of operation. Such inquiries and/or enforcement actions could also result in monetary fines, penalties or other remedies or sanctions that could adversely affect our business and financial condition.

Our commercialization, licensing and/or mergers and acquisitions ("M&A") activities could lead to patent exhaustion or implied license issues that could materially adversely affect our business.

The legal doctrines of patent exhaustion and implied license may be subject to different judicial interpretations. Our commercialization or licensing of certain technologies and/or our M&A activities could potentially lead to patent exhaustion or implied license issues that could adversely affect our patent licensing program(s) and limit our ability to derive licensing revenue from certain patents under such program(s), whether through the assumption of license agreements that would result in our patents being captured by such agreements, the acquisition of a business that sells or licenses products that practice our patents, or otherwise. In the event of successful challenges by current or prospective licensees based on these doctrines that result in a material decrease to our patent licensing revenue, our financial condition and operating results may be materially adversely affected.

We may not be able to realize all of the anticipated benefits from the integration of the patent licensing business that we acquired from Technicolor in 2018 and the Research & Innovation unit of Technicolor (collectively, the "Technicolor business").

We may fail to realize the anticipated benefits from our integration of the Technicolor business on a timely basis, or at all, for a variety of reasons, including the following:

- failure of the acquisitions to materially increase the value of our core handset licensing business by not increasing the royalty amount we would otherwise derive on each handset, not accelerating the pace of licensing, or not allowing us to avoid litigation to protect our intellectual property;
- failure to continue to develop and expand our portfolio of video technology patent assets;
- failure to execute a successful business plan and licensing program related to consumer electronics;
- the risk that we could lose key employees of the Technicolor business;
- challenges associated with managing a geographically remote business;
- failure to forecast accurately the long-term value and costs of the Technicolor business or of certain assets acquired in the transactions;
- liabilities that are not covered by, or exceed the coverage under, the indemnification or other provisions of the acquisition-related agreements; and
- patent validity, infringement, exhaustion or enforcement issues not uncovered during our diligence process.

In the event that we fail to realize the anticipated benefits from the acquisition of the Technicolor business, our business and results of operations, and our stock price, may be adversely affected.

We have in the past and may in the future make acquisitions or engage in other strategic transactions that could result in significant changes, costs and/or management disruption and that may fail to enhance shareholder value or produce the anticipated benefits.

We have in the past and may in the future acquire companies, businesses, technology and/or intellectual property, enter into joint ventures or other strategic transactions. Acquisitions or other strategic transactions may increase our costs, including but not limited to accounting and legal fees, and may not generate financial returns or result in increased adoption or continued use of our technologies or of any technologies we may acquire.

Achieving the anticipated benefits of acquisitions depends in part upon our ability to integrate the acquired companies, businesses and/or assets in an efficient and effective manner. The integration of acquired companies or businesses may result in significant challenges, including, among others: successfully integrating new employees, technology and/or products; consolidating research and development operations; minimizing the diversion of management's attention from ongoing business matters; and consolidating corporate and administrative infrastructures. As a result, we may be unable to accomplish the integration smoothly or successfully.

In addition, we cannot be certain that the integration of acquired companies, businesses, technology and/or intellectual property with our business will result in the realization of the full benefits that we anticipate will be realized from such acquisitions. Our plans to integrate and/or expand upon research and development programs and technologies obtained through acquisitions may result in products or technologies that are not adopted by the market, or the market may adopt solutions competitive to our products or technologies. We may not derive any commercial value from the acquired technology or intellectual property or from future technologies or products based on the acquired

technology and/or intellectual property. In addition, to the extent we are separately seeking a patent license from a customer or customers of an acquired entity, the acquired entity may lose such customers. Following the completion of the acquisition, we may be subject to liabilities that are not covered by, or exceed the coverage under, the indemnification protection we may obtain, and we may encounter patent

validity, infringement or enforcement issues or unforeseen expenses not uncovered during our diligence process. Any acquired company or business would be subject to its own risks that may or may not be the same as the risks already disclosed herein.

Challenges relating to our ability to enter into new license agreements could cause our revenue and cash flow to decline.

We face challenges in entering into new patent license agreements. One of the most significant challenges we face is that most potential licensees do not voluntarily seek to enter into license agreements with us before they commence manufacturing and/or selling devices that use our patented inventions. As a result, we must approach companies that are reluctant to take licenses and attempt to establish license agreements with them. The process of identifying potential users of our inventions and negotiating license agreements with reluctant prospective licensees requires significant time, effort and expense. Once discussions with unlicensed companies have commenced, we face the additional challenges imposed by the significant negotiation issues that arise from time to time. Given these challenges relating to our ability to enter into new license agreements, we cannot ensure that all prospective licensees will be identified or, if they are identified, will be persuaded during negotiations to enter into a patent license agreement with us, either at all or on terms acceptable to us, and, as a result, our revenue and cash flow could materially decline. The length of time required to negotiate a license agreement also leads to delays in the receipt of the associated revenue stream, which could also cause our revenue and cash flow to decline.

In addition, as discussed more fully above in these Risk Factors, we are currently operating in a challenging regulatory and judicial environment, which may, under certain circumstances, lead to delays in the negotiation of and entry into new patent license agreements. Also, as discussed above in these Risk Factors and in Item 3, Legal Proceedings, in this Form 10-K, we are also currently, and may in the future be, involved in legal proceedings with potential licensees, with whom we do not yet have a patent license agreement. Any such delays in the negotiation or entry into new patent license agreements and receipt of the associated revenue stream could cause our revenue and cash flow to decline.

Our revenues are derived primarily from a limited number of licensees or customers.

We earn a significant amount of our revenues from a limited number of licensees or customers, and we expect that a significant portion of our revenues will continue to come from a limited number of licensees or customers for the foreseeable future. For example, in 2019, Apple, Samsung and LG Electronics accounted for approximately 35%, 25% and 10% of our total revenues, respectively. In the event that we are unable to renew one or more of such license agreements upon expiration, our future revenue and cash flow could be materially adversely affected. In the event that one or more of our significant licensees or customers fail to meet their payment or reporting obligations (for example, due to a credit issue or in connection with a legal dispute or similar proceeding) under their respective license agreements, our future revenue and cash flow could be materially adversely affected. In addition, in the event that there is a material decrease in shipments of licensed products by one of our per-unit licensees, our revenues from such licensee could significantly decline and our future revenue and cash flow could be adversely affected.

Our strategy to diversify our patent-based revenue by pursuing alternative patent licensing arrangements and patent sales may not be successful.

There is no guarantee that we will succeed in our pursuit of select patent licensing arrangements or patent sales, and, if we are successful, there is no guarantee that the revenue and cash flow generated through such alternative licensing arrangements (such as the Signal Trust and the Avanci licensing platform) or patent sales will be greater than the revenue and cash flow we would have generated if we had retained and/or licensed the patents ourselves. In addition, potential licensees may be reluctant to enter into new patent license agreements, and current licensees may be reluctant to renew their agreements, either at all or on terms acceptable to the Company, based on the fact that we have sold portions of our patent portfolio or the belief that we plan to sell or transfer some of the patents we are asking them to license.

A portion of our revenue and cash flow are dependent upon our licensees' sales and market conditions and other factors that are beyond our control or are difficult to forecast.

A portion of our licensing revenues is dependent on sales by our licensees that are outside our control and that could be negatively affected by a variety of factors, including global, regional and/or country-specific economic conditions and/or public health concerns (e.g., the current coronavirus outbreak), country-specific natural disasters impacting licensee manufacturing and sales, demand and buying patterns of end users, which are often driven by replacement and innovation cycles, the service life of products incorporating our technologies, competition for our licensees' products, supply chain disruptions, and any decline in the sale prices our licensees receive for their covered products. In addition, our operating results also could be affected by general economic and other conditions that cause a downturn in the market for the licensees of our products or technologies. Our revenue and cash flow also could be affected by (i) the unwillingness of any licensee to satisfy all of their royalty obligations on the terms or within the timeframe

we expect, (ii) a decline in the financial condition or market position of any licensee or (iii) the failure of sales to meet market forecasts due to global or regional economic conditions, political instability, natural disasters, competitive technologies, lower demand or otherwise. It is also difficult to predict the

timing, nature and amount of licensing revenue associated with past infringement (including as a result of the unwillingness of our licensees to compensate us for such past infringement) and new licenses, strategic relationships and the resolution of legal proceedings. The foregoing factors are difficult to forecast and could adversely affect both our quarterly and annual operating results and financial condition. In addition, some of our patent license agreements provide for upfront fixed payments or prepayments that cover our licensees' future sales for a specified period and reduce future cash receipts from those licensees. As a result, our cash flow has historically fluctuated from period to period. Depending upon the payment structure of any new patent license agreements into which we may enter, such cash flow fluctuations may continue in the future.

Our revenue may be affected by the deployment of future-generation wireless standards in place of 3G, 4G and 5G technologies or future-generation video standards, by the timing of such deployment, or by the need to extend or modify certain existing license agreements to cover subsequently issued patents.

Although we own an evolving portfolio of issued and pending patents related to 3G, 4G and 5G cellular technologies and non-cellular technologies including video coding technologies, our patent portfolio licensing program for future-generation wireless standards or video coding standards may not be as successful in generating licensing income as our current licensing programs. Although we continue to participate in worldwide standards bodies and contribute our intellectual property to future-generation wireless and video coding standards, including standards that will define 5G, our technologies might not be adopted by the relevant standards. In addition, we may not be as successful in the licensing of future-generation products as we have been in licensing products deploying existing wireless and video coding standards, or we may not achieve a level of royalty revenues on such products that is comparable to that which we have historically received on products deploying existing wireless and video coding standards. Furthermore, if there is a delay in the standardization and/or deployment of 5G or future video coding standards, our business and revenue could be negatively impacted.

The licenses that we grant under our patent license agreements typically only cover products designed to operate in accordance with specified technologies and that were manufactured or deployed or anticipated to be manufactured or deployed at the time of entry into the agreement. Also, we have patent license agreements with licensees that now offer for sale types of products that were not sold by such licensees at the time the patent license agreements were entered into and, thus, are not licensed by us. We do not derive patent licensing revenue from the sale of products by our licensees that are not covered by a patent license agreement. In order to grant a patent license for any such products, we will need to extend or modify our patent license agreements or enter into new license agreements with such licensees. We may not be able to extend or modify these license agreements, or enter into new license agreements, on financial terms acceptable to us, without affecting the other material terms and conditions of our license agreements with such licensees or at all. Further, such extensions, modifications or new license agreements may adversely affect our revenue on the sale of products covered by the license prior to any extension, modification or new license.

We face risks from doing business and maintaining offices in international markets.

A significant portion of our licensees, potential licensees and customers are international, and our licensees, potential licensees and customers sell their products to markets throughout the world. In addition, in recent years, we have expanded, and we may continue to expand, our international operations, opening offices in France, the United Kingdom, China, Belgium and Germany. Accordingly, we are subject to the risks and uncertainties of operating internationally and could be affected by a variety of uncontrollable and changing factors, including, but not limited to: difficulty in protecting our intellectual property in foreign jurisdictions; enforcing contractual commitments in foreign jurisdictions or against foreign corporations; government regulations, tariffs and other applicable trade barriers; biased enforcement of foreign laws and regulations to promote industrial or economic policies at our expense; retaliatory practices by foreign actors; currency control regulations and variability in the value of the U.S. dollar against foreign currency; export license requirements and restrictions on the use of technology; social, economic and political instability; natural disasters, acts of terrorism, widespread illness and war; potentially adverse tax consequences; general delays in remittance of and difficulties collecting non-U.S. payments; foreign labor regulations; anti-corruption laws; and difficulty in staffing and managing operations remotely. In addition, we also are subject to risks specific to the individual countries in which we and our licensees, potential licensees and customers do business.

We depend on key senior management, engineering, patent and licensing resources.

Our future success depends largely upon the effectiveness of our executive officers and other key personnel, including our ability to put in place adequate succession plans for such key personnel, and/or organizational strategies related to the departure of such key personnel. Our success also depends in part on our ability to continue to attract, retain and motivate qualified personnel with specialized patent, licensing, engineering and other skills. The market for such talent in our industry is extremely competitive as a result in part of the specialized nature of our industry. In particular, competition exists for qualified individuals with expertise in patents and in licensing and with significant engineering experience in cellular and air interface , video coding, and artificial intelligence technologies. Our ability to attract and retain qualified personnel could be affected by any adverse

decisions in any litigation, arbitration or regulatory proceeding, by our ability to offer competitive cash and equity compensation, the perception of our company both internally and externally, and work environment conditions and by the

geographic location of our various offices. The failure to attract and retain such persons with relevant and appropriate experience or to have in place adequate succession plans and/or organizational strategies related to the departure of certain key personnel could interfere with our ability to enter into new license agreements and undertake additional technology and product development efforts, as well as our ability to meet our strategic objectives.

We face competition from companies developing other or similar technologies.

We face competition from companies developing other and similar technologies that are competitive with our products and solutions that we may market or set forth, including into the standards-setting arena. Due to competing products and solutions, our products and solutions may not find a viable commercial marketplace or, where applicable, be adopted by the relevant standards. In addition, in licensing our patent portfolio, we may compete with other companies, many of whom also claim to hold SEPs, for a share of the royalties that certain licensees may argue to be the total royalty that is supported by a certain product or products. In any device or piece of equipment that contains intellectual property, the manufacturer may need to obtain a license from multiple holders of intellectual property. To the extent that multiple parties all seek royalties on the same product, the manufacturers could claim to have difficulty in meeting the financial requirements of each patent holder.

Our industry is subject to rapid technological change, uncertainty and shifting market opportunities.

Our success depends, in part, on our ability to define and keep pace with changes in industry standards, technological developments and varying customer requirements. Changes in industry standards and needs could adversely affect the development of, and demand for, our technology, rendering our technology currently under development obsolete and unmarketable. The patents and applications comprising our portfolio have fixed terms, and, if we fail to anticipate or respond adequately to these changes through the development or acquisition of new patentable inventions, patents or other technology, we could miss a critical market opportunity, reducing or eliminating our ability to capitalize on our patents, technology solutions or both.

Concentration and consolidation in the wireless communications industry could adversely affect our business.

There is some concentration among participants in the wireless communications industry, and the industry has experienced consolidation of participants and sales of participants or their businesses, and these trends may continue. For example, in 2019, Samsung, Apple and Huawei collectively accounted for approximately 40% of worldwide shipments of 3G and 4G handsets and 50% of worldwide smartphone shipments. Although the rollout of 5G handsets is still in its early stages, we anticipate a similar level of concentration in worldwide shipments of those units as well. Any further concentration or sale within the wireless industry among handset providers and/or original design manufacturers ("ODMs") may reduce the number of licensing opportunities or, in some instances, result in the reduction, loss or elimination of existing royalty obligations. We may also face a reduction in the number of licensing opportunities or existing royalty obligations as a result of government-imposed bans or other restrictions on the importation, manufacture and/or sale of cellular handsets by certain companies. In addition, acquisitions of, or consolidation among, ODMs could cause handset providers who outsource manufacturing to make supply chain changes, which in turn could result in the reduction, loss or elimination of existing royalty obligations (for example, if manufacturing is moved from an ODM with which we have a patent license agreement to an ODM with which we do not). Further, if wireless carriers consolidate with companies that utilize technologies that are competitive with our technologies or that are not covered by our patents, we could lose market opportunities, which could negatively impact our revenues and financial condition.

Our use of open source software could materially adversely affect our business, financial condition, operating results and cash flow.

Certain of our technology and our suppliers' technology may contain or may be derived from "open source" software, which, under certain open source licenses, may offer accessibility to a portion of a product's source code and may expose related intellectual property to adverse licensing conditions. Licensing of such technology may impose certain obligations on us if we were to distribute derivative works of the open source software. For example, these obligations may require us to make source code for derivative works available or license such derivative works under a particular type of license that is different from what we customarily use to license our technology. While we believe we have taken appropriate steps and employ adequate controls to protect our intellectual property rights, our use of open source software presents risks that, if we inappropriately use open source software, we may be required to re-engineer our technology, discontinue the sale of our technology, release the source code of our proprietary technology to the public at no cost or take other remedial actions, which could adversely affect our business, operating results and financial condition. There is a risk that open source licenses could be construed in a way that could impose unanticipated conditions or restrictions on our ability to commercialize our solutions, which could adversely affect our business, operating results and

financial condition. In addition, developing open source products, while adequately protecting the intellectual property rights upon which our licensing business depends, may prove burdensome and time-consuming under certain circumstances, thereby placing us at a competitive disadvantage.

Changes to our tax assets or liabilities could have an adverse effect on our consolidated financial condition or results of operations.

The calculation of tax assets and liabilities involves significant judgment in estimating the impact of uncertainties in the application of complex tax laws. We are subject to examinations by the Internal Revenue Service ("IRS") and other taxing jurisdictions on various tax matters, including challenges to various positions we assert in our filings and foreign tax liability and withholding. Pursuant to the guidance for accounting for uncertainty in income taxes, certain tax contingencies are recognized when they are determined to be more likely than not to occur. Although we believe we have adequately recorded tax assets and accrued for tax contingencies that meet this criterion, we may not fully recover our tax assets or may be required to pay taxes in excess of the amounts we have accrued. As of December 31, 2019, and 2018, there were certain tax contingencies that did not meet the applicable criteria to record an accrual. In the event that the IRS or another taxing jurisdiction levies an assessment in the future, it is possible the assessment could have an adverse effect on our consolidated financial condition or results of operations.

Changes in financial accounting standards or policies may affect our reported financial condition or results of operations and, in certain cases, could cause a decline and/or fluctuations in the price of our common stock.

From time to time the Financial Accounting Standards Board (the "FASB") and the Staff of the Securities and Exchange Commission (the "SEC") change their guidance governing the form and content of our external financial statements. In addition, accounting standard setters and those who interpret U.S. generally accepted accounting principles ("GAAP"), such as the FASB and the SEC, may change or even reverse their previous interpretations or positions with regard to how these standards should be applied. A change in accounting principles or their interpretation can have a significant effect on our reported results. In certain cases, we could be required to apply new or revised guidance retroactively or apply existing guidance differently. Potential changes in reporting standards could substantially change our reporting practices in a number of areas, including revenue recognition and recording of assets and liabilities, and affect our reported financial condition or results of operations.

The high amount of capital required to obtain radio frequency licenses, deploy and expand wireless networks and obtain new subscribers, as well as the cost of new handsets could slow the growth of the wireless communications industry and adversely affect our business.

Our growth is partially dependent upon the increased use of wireless communications services and cellular handsets that utilize our technology. In order to provide wireless communications services, wireless operators must obtain rights to use specific radio frequencies. The allocation of frequencies is regulated in the United States and other countries throughout the world, and limited spectrum space is allocated to wireless communications services. Industry growth may be affected by the amount of capital required to obtain licenses to use new frequencies, deploy wireless networks to offer voice and data services, expand wireless networks to grow voice and data services and obtain new subscribers. The significant cost of licenses, wireless networks and subscriber additions may slow the growth of the industry if wireless operators are unable to obtain or service the additional capital necessary to implement or expand advanced wireless networks. Growth in the number of cellular handsets may slow as the number of people worldwide without a cellular handset declines. In addition, if the cost of cellular handsets increases, customers may be less likely to replace their existing devices with new devices. The growth of our business could be adversely affected if either of these events occur.

Market projections and data are forward-looking in nature.

Our strategy is based on our own projections and on analyst, industry observer and expert projections, which are forward-looking in nature and are inherently subject to risks and uncertainties. The validity of their and our assumptions, the timing and scope of wireless markets, economic conditions, customer buying patterns, timeliness of equipment development, pricing of products, growth in wireless telecommunications services that would be delivered on wireless devices and availability of capital for infrastructure improvements could affect these predictions. In addition, market data upon which we rely is based on third party reports that may be inaccurate. The inaccuracy of any of these projections and/or market data could adversely affect our operating results and financial condition.

Our engineering services business could subject us to specific costs and risks that we might fail to manage adequately.

We derive a portion of our revenues from engineering services. Any mismanagement of, or negative development in, a number of areas, including, among others, the perceived value of our intellectual property portfolio, our ability to convince customers of the value of our engineering services and our reputation for performance under our service contracts, could cause our revenues from engineering services to decline, damage our reputation and harm our ability to attract future licensees, which would in turn harm

our operating results. If we fail to deliver as required under our service contracts, we could lose revenues and become subject to liability for breach of contract. We need to monitor these services adequately in order to ensure that we

do not incur significant expenses without generating corresponding revenues. Our failure to monitor these services adequately may harm our business, financial position, results of operations or cash flows.

It can be difficult for us to verify royalty amounts owed to us under our per-unit licensing agreements, and this may cause us to lose potential revenue.

The standard terms of our per-unit license agreements require our licensees to document the sale of licensed products and report this data to us on a quarterly basis. Although our standard license terms give us the right to audit books and records of our licensees to verify this information, audits can be expensive, time consuming, incomplete and subject to dispute. From time to time, we audit certain of our licensees to verify independently the accuracy of the information contained in their royalty reports in an effort to decrease the likelihood that we will not receive the royalty revenues to which we are entitled under the terms of our license agreements, but we cannot give assurances that these audits will be numerous enough and/or effective to that end.

Our plans to expand our revenue opportunities through commercializing our market-ready technologies and acquiring and/or developing new technology with commercial applicability may not be successful and could materially adversely affect our long-term business, financial condition and operating results.

As part of our business strategy, we are seeking to expand our revenue opportunities through the continued development, commercialization and licensing of technology projects, including in the IoT space. Our technology development and acquisition activities may experience delays, or the markets for our technology solutions may fail to materialize to the extent or at the rate we expect, if at all, each of which could reduce our opportunities for technology sales and licensing. In addition, there could be fewer applications for our technology and products than we expect. Technology markets also could be affected by general economic conditions, customer buying patterns, timeliness of equipment development, and the availability of capital for, and the high cost of, infrastructure improvements. Additionally, investing in technology development is costly and may require structural changes to the organization that could require additional costs, including without limitation legal and accounting fees. Furthermore, delays or failures to enter into additional partnering relationships to facilitate technology development efforts and secure support for our technologies or delays or failures to enter into technology licensing agreements to secure integration of additional functionality could impair our ability to introduce into the market portions of our technology and resulting products, cause us to miss critical market windows, or decrease our ability to remain competitive.

We have in the past and may in the future make investments that may fail to enhance shareholder value or produce the anticipated benefits.

We have in the past and may in the future make investments in other entities by purchasing minority equity interests or corporate bonds/notes in publicly traded or privately held companies. Most strategic investments entail a high degree of risk and may not become liquid for a period of time, if ever. In some cases, strategic investments may serve as consideration for a license in lieu of cash royalties. In addition, other investments may not generate financial returns or may result in losses due to market volatility, the general level of interest rates and inflation expectations. We have made in the past and may make in the future strategic investments in early-stage companies, which may require us to consolidate or record our share of the earnings or losses of those companies. Our share of any such losses may adversely affect our financial results until we exit from or reduce our exposure to these investments.

Our investments in new commercial initiatives may not be successful or generate meaningful revenues.

We have invested, and may continue to invest, in new businesses focused on commercializing technology that we have developed, incubated internally and/or acquired, such as video coding technology and other technologies for use on consumer electronics devices. Commercial success depends on many factors, including the demand for the technology, the highly competitive markets for our technology products, regulatory issues associated with such technology products, and effective marketing and licensing or product sales. In addition, our new technology offerings may require robust ecosystems of customers and service providers that may fail to materialize. Further, the establishment and operation of these commercial initiatives requires significant support, including technical, legal and financial resources. It is possible that these commercial initiatives will not be successful and/or will not achieve meaningful revenues for a number of years, if at all. Further, we may attempt to develop technologies or services that we believe we would be able to sell or license commercially using inside or outside technical, legal and financial resources. If our new commercial initiatives are not successful, or are not successful in the timeframe we anticipate, we may incur significant costs, our business may not grow as anticipated and/or our reputation may be harmed. In the event that any of these risks materialize, our long-term business, financial condition and operating results may be materially adversely affected.

Our technology development activities may experience delays.

We may experience technical, financial, resource or other difficulties or delays related to the further development of our technologies. Delays may have adverse financial effects and may allow competitors with comparable technology offerings to gain an advantage over us in the marketplace or in the standards setting arena. There can be no assurance that we will continue to have adequate staffing or that our development efforts will ultimately be successful. Moreover, certain of our technologies have not been fully tested in commercial use, and it is possible that they may not perform as expected. In such cases, our business, financial condition and operating results could be adversely affected, and our ability to secure new licensees and other business opportunities could be diminished.

We rely on relationships with third parties to develop and deploy technology solutions.

Successful exploitation of our technology solutions is partially dependent on the establishment and success of relationships with equipment producers and other industry participants. Delays or failure to enter into licensing or other relationships to facilitate technology development efforts or delays or failure to enter into technology licensing agreements to secure integration of additional functionality could impair our ability to introduce into the market portions of our technology and resulting products, cause us to miss critical market windows or impair our ability to remain competitive.

Currency fluctuations could negatively affect future product sales or royalty revenues or increase the U.S. dollar cost of our activities and international strategic investments.

We are exposed to risk from fluctuations in currencies, which may change over time as our business practices evolve, that could impact our operating results, liquidity and financial condition. We operate and invest globally. Adverse movements in currency exchange rates may negatively affect our business due to a number of situations, including the following:

- If the effective price of products sold by our licensees were to increase as a result of fluctuations in the exchange rate of the relevant currencies, demand for the products could fall, which in turn would reduce our royalty revenues.
- Assets or liabilities of our consolidated subsidiaries may be subject to the effects of currency fluctuations, which may affect our reported earnings. Our exposure to foreign currencies may increase as we expand into new markets.
- Certain of our operating and investing costs, such as foreign patent prosecution, are based in foreign currencies. If these costs are not subject to foreign exchange hedging transactions, strengthening currency values in selected regions could adversely affect our near-term operating expenses, investment costs and cash flows. In addition, continued strengthening of currency values in selected regions over an extended period of time could adversely affect our future operating expenses, investment costs and cash flows.
- If, as a result of tax treaty procedures, the U.S. government reaches an agreement with certain foreign governments to whom we have paid foreign taxes, resulting in a partial refund of foreign taxes paid with a related reduction in our foreign tax credits, such agreement could result in foreign currency gain or loss.

Our business and operations could suffer in the event of security breaches and our business is subject to a variety of domestic and international laws, rules and policies and other obligations regarding data protection.

Attempts by others to gain unauthorized access to information technology systems are becoming more sophisticated. These attempts, which in some cases could be related to industrial or other espionage, include covertly introducing malware to computers and networks and impersonating authorized users, among others. We seek to detect and investigate all security incidents and to prevent their recurrence, but, in some cases, we might be unaware of an incident or its magnitude and effects. While we have not identified any material incidents of unauthorized access to date, the theft, unauthorized use or publication of our intellectual property and/or confidential business or personal information (whether through a breach of our own systems or the breach of a system of a third party that provides services to us) could harm our competitive or negotiating positions, reduce the value of our investment in research and development and other strategic initiatives, compromise our patent enforcement strategies or outlook, damage our reputation or otherwise adversely affect our business. In addition, to the extent that any future security breach results in inappropriate disclosure of our employees’, licensees’, or customers’ confidential and /or personal information, we may incur liability or additional costs to remedy any damages caused by such breach.

We could also be affected by existing and proposed laws and regulations, as well as government policies and practices related to cybersecurity, privacy and data protection. For example, the European General Data Protection Regulation ("GDPR") adopted by the European Commission became effective in May 2018, the California Consumer Privacy

Act of 2018 (the "CCPA") adopted by the California State Legislature became effective in January 2020, and China adopted a new cybersecurity law as of June 2017. Complying with the GDPR, the CCPA and other existing and emerging and changing requirements could cause us to incur substantial costs or require us to change our business practices. Non-compliance could result in monetary penalties or significant legal liability.

If wireless handsets are perceived to pose health and safety risks, demand for products of our licensees could decrease.

Media reports and certain studies have suggested that radio frequency emissions from wireless handsets may be linked to health concerns, such as brain tumors, other malignancies and genetic damage to blood, and may interfere with electronic medical devices, such as pacemakers, telemetry and delicate medical equipment. Growing concerns over radio frequency emissions, even if unfounded, could discourage the use of wireless handsets and cause a decrease in demand for the products of our licensees. In addition, concerns over safety risks posed by the use of wireless handsets while driving and the effect of any resulting legislation could reduce demand for the products of our licensees.

The United Kingdom's ("U.K.") decision to exit the European Union ("E.U.") will continue to have uncertain effects and could adversely impact our business, results of operations and financial condition.

The U.K. held a referendum on June 23, 2016 in which a majority voted for the U.K.'s withdrawal from the E.U. (commonly referred to as "Brexit"), and on January 31, 2020, the U.K. formally withdrew from the E.U. The U.K. is now in a transition period until December 31, 2020 during which time the final terms of Brexit are to be negotiated with the E.U. The terms of Brexit and the U.K.'s relationship with the E.U. going forward continue to be uncertain for global companies like ours that conduct business both in the U.K. and the E.U. The actual and perceived impacts of Brexit may adversely affect business activity and economic and market conditions in the U.K., the Eurozone and globally, and could contribute to instability in global financial and foreign exchange markets. In addition, Brexit could lead to additional political, legal and economic instability in the U.K and the E.U. While we have not experienced any material financial impacts from Brexit on our business to date, we cannot predict its future implications. Any impact on our business from Brexit in the long term will depend, in part, on the outcome of tariff, tax treaties, trade, regulatory and other negotiations that the U.K and the E.U. conduct.

Risks Relating to Our Common Stock and our Convertible Notes

The price of our common stock is volatile and may decline regardless of our operating performance.

Historically, we have had large fluctuations in the price of our common stock, and such fluctuations could continue. From January 2, 2018 to February 18, 2020, the trading price of our common stock has ranged from a low of $47.02 per share to a high of $85.85 per share. The market price for our common stock is volatile and may fluctuate significantly in response to a number of factors, most of which we cannot control, including:

- the public's response to press releases or other public announcements by us or third parties, including our filings with the SEC and announcements relating to licensing, technology development, litigation, arbitration and other legal proceedings in which we are involved and intellectual property impacting us or our business;
- announcements concerning strategic transactions, such as commercial initiatives, joint ventures, strategic investments, acquisitions or divestitures;
- financial projections we may provide to the public, any changes in these projections or our failure to meet these projections;
- changes in GAAP, including new accounting standards that may materially affect our revenue recognition;
- changes in financial estimates or ratings by any securities analysts who follow our common stock, our failure to meet these estimates or failure of those analysts to initiate or maintain coverage of our common stock;
- investor perceptions as to the likelihood of achievement of near-term goals;
- changes in market share of significant licensees;
- changes in operating performance and stock market valuations of other wireless communications companies generally; and
- market conditions or trends in our industry or the economy as a whole.

In the past, shareholders have instituted securities class action litigation following periods of market volatility. If we were involved in securities litigation, we could incur substantial costs and our resources and the attention of management could be diverted from our business.

Our shareholders may not receive the level of dividends provided for in our dividend policy or any dividend at all, and any decrease in or suspension of the dividend could cause our stock price to decline.

Our current dividend policy contemplates the payment of a regular quarterly cash dividend of $0.35 per share on our outstanding common stock. We expect to continue to pay quarterly cash dividends on our common stock at the rate set forth in our current dividend policy. However, the dividend policy and the payment and timing of future cash dividends under the policy are subject to the final determination each quarter by our Board of Directors that (i) the dividend will be made in

compliance with laws applicable to the declaration and payment of cash dividends, including Section 1551(b) of the Pennsylvania Business Corporation Law, and (ii) the policy remains in our best interests, which determination will be based on a number of factors, including our earnings, financial condition, capital resources and capital requirements, alternative uses of capital, restrictions imposed by any existing debt, economic conditions and other factors considered relevant by the Board of Directors. Given these considerations, our Board of Directors may increase or decrease the amount of the dividend at any time and may also decide to vary the timing of or suspend or discontinue the payment of dividends in the future. Any decrease in the amount of the dividend, or suspension or discontinuance of payment of a dividend, could cause our stock price to decline.

Approved stock repurchase programs may not result in a positive return of capital to shareholders.

Our board-approved stock repurchase program may not return value to shareholders because the market price of the stock may decline significantly below the levels at which we repurchased shares of stock. Stock repurchase programs are intended to deliver shareholder value over the long term, but stock price fluctuations can reduce the effectiveness of such programs.

Our indebtedness could adversely affect our business, financial condition and results of operations and our ability to meet our payment obligations under such indebtedness.

Our total indebtedness as of December 31, 2019 was approximately $516.0 million, inclusive of debt resulting from the Technicolor Patent Acquisition (refer to Note 5, "*Business Combinations and Other Transactions*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information). This level of debt could have significant consequences on our future operations, including:

- making it more difficult for us to meet our payment and other obligations under our 1.50% Senior Convertible Notes due 2020 (the "2020 Notes") and our 2.00% Senior Convertible Notes due 2024 (the "2024 Notes" and, together with the 2020 Notes, the "Convertible Notes");
- reducing the availability of our cash flow to fund working capital, capital expenditures, acquisitions and other general corporate purposes, and limiting our ability to obtain additional financing for these purposes;
- limiting our flexibility in planning for, or reacting to, and increasing our vulnerability to, changes in our business, the industry in which we operate and the general economy; and
- placing us at a competitive disadvantage compared to our competitors that have less debt or are less leveraged.

Any of the above-listed factors could have an adverse effect on our business, financial condition and results of operations and our ability to meet our payment obligations under the Convertible Notes.

Our ability to meet our payment and other obligations under the Convertible Notes depends on our ability to generate significant cash flow in the future. This, to some extent, is subject to general economic, financial, competitive, legislative and regulatory factors as well as other factors that are beyond our control. We cannot be certain that our business will generate cash flow from operations, or that future borrowings will be available to us, in an amount sufficient to enable us to meet our payment obligations under the Convertible Notes and to fund other liquidity needs. If we are not able to generate sufficient cash flow to service our debt obligations, we may need to refinance or restructure our debt, including the 2020 Notes and/or the 20204 Notes, sell assets, reduce or delay capital investments, or seek to raise additional capital. If we are unable to implement one or more of these alternatives, we may not be able to meet our payment obligations under the 2020 Notes and/or the 2024 Notes, and this default could cause us to be in default on any other currently existing or future outstanding indebtedness.

The convertible note hedge transactions and warrant transactions that we entered into in connection with the offering of the 2020 Notes and the 2024 Notes may affect the value of the 2020 Notes and/or the 2024 Notes, respectively, and the market price of our common stock.

In connection with each offering of the Convertible Notes, we entered into convertible note hedge transactions with certain financial institutions (the “option counterparties”) and sold warrants to the respective option counterparties. These transactions will be accounted for as an adjustment to our shareholders’ equity. The convertible note hedge transactions are expected to reduce the potential equity dilution upon any conversion of the Convertible Notes. The warrants will have a dilutive effect on our earnings per share to the extent that the market price of our common stock exceeds the applicable strike price of the warrants on any expiration date of the warrants.

In addition, the respective option counterparties (and/or their affiliates) may modify their respective hedge positions from time to time (including during any observation period related to a conversion of the 2020 Notes and/or the 2024 Notes) by entering into or unwinding various derivative transactions with respect to our common stock and/or by purchasing or selling our common stock in open market transactions and/or privately negotiated transactions.

The potential effect, if any, of any of these transactions and activities on the market price of our common stock will depend in part on market conditions and cannot be ascertained at this time, but any of these activities could adversely affect the market price of our common stock.

Provisions of the Convertible Notes could discourage an acquisition of us by a third party.

Certain provisions of the Convertible Notes could make it more difficult or more expensive for a third party to acquire us. Upon the occurrence of certain transactions constituting a fundamental change under the respective Convertible Notes, holders of the 2020 Notes and/or the 2024 Notes will have the right, at their option, to require us to repurchase all of their applicable Convertible Notes or any portion of the principal amount of such Convertible Notes at a price of 100% of the principal amount of the Convertible Notes being repurchased, plus accrued and unpaid interest. We may also be required to issue additional shares upon conversion in the event of certain fundamental change transactions. These provisions could limit the price that some investors might be willing to pay in the future for shares of our common stock.

We are subject to counterparty risk with respect to the convertible note hedge transactions.

The respective option counterparties are financial institutions or affiliates of financial institutions, and we will be subject to the risk that such option counterparties may default under the respective convertible note hedge transactions. Our exposure to the credit risk of the option counterparties is not secured by any collateral. If an option counterparty becomes subject to insolvency proceedings, we will become an unsecured creditor in those proceedings with a claim equal to our exposure at that time under the applicable convertible note hedge transactions. Our exposure will depend on many factors but, generally, the increase in our exposure will be correlated to the increase in our common stock market price and in volatility of our common stock. In addition, upon a default by an option counterparty, we may suffer adverse tax consequences and dilution with respect to our common stock. We can provide no assurance as to the financial stability or viability of the option counterparties.

The accounting method for convertible debt securities, such as the Convertible Notes, could have a material adverse effect on our reported financial results.

In May 2008, the FASB, issued ASC 470-20. Under ASC 470-20, an entity must separately account for the liability and equity components of convertible debt instruments, such as the Convertible Notes, that may be settled partially in cash upon conversion in a manner that reflects the issuer's economic interest cost. ASC 470-20 requires the fair value of the conversion option of the 2020 Notes and the 2024 Notes be reported as a component of shareholders' equity and included in the additional paid-in-capital on our consolidated balance sheet. The value of the conversion option of the 2020 Notes and the 2024 Notes will be reported as discount to the 2020 Notes and the 2024 Notes, respectively. We will report lower net income in our financial results because ASC 470-20 will require interest to include both the current period's amortization of the debt discount (non-cash interest) and the instrument's cash interest, which could adversely affect our reported or future financial results, the trading price of our common stock and the trading price of the 2020 Notes and/or the 2024 Notes.

Future sales or other dilution of our equity could depress the market price of our common stock.

Sales of our common stock in the public market, or the perception that such sales could occur, could negatively impact the market price of our common stock. We also have several institutional shareholders that own significant blocks of our common stock. If one or more of these shareholders were to sell large portions of their holdings in a relatively short time, for liquidity or other reasons, the prevailing market price of our common stock could be negatively affected.

Under certain circumstances, shares of our common stock could be issued upon conversion of the 2020 Notes and/or the 2024 Notes, which would dilute the ownership interest of our existing shareholders. In addition, the issuance of additional common stock, or issuances of securities convertible into or exercisable for our common stock or other equity linked securities, including preferred stock or warrants, would dilute the ownership interest of our common shareholders and could depress the market price of our common stock and impair our ability to raise capital through the sale of additional equity securities.

If securities or industry analysts fail to continue publishing research about our business, our stock price and trading volume could decline.

The trading market for our common stock is influenced by the research and reports that industry or securities analysts publish about us or our business. If one or more of these analysts cease coverage of our company or fail to publish reports on us regularly, we could lose visibility in the financial markets, which in turn could cause our stock price or trading volume to decline.

Item 1B. *UNRESOLVED STAFF COMMENTS.*

None.

Item 2. *PROPERTIES.*

Our headquarters are located in Wilmington, Delaware, USA. Our research and development activities are conducted primarily in facilities located in Berlin, Germany; Conshohocken, Pennsylvania, USA; London, United Kingdom; Montreal, Canada; New York, New York, USA; Palo Alto, California, USA; Rennes, France; and San Diego, California, USA.

The following table sets forth information with respect to our principal properties:

Location	Approximate Square Feet	Principal Use	Lease Expiration Date
Melville, New York	44,800	Office and research space	February 2020
Wilmington, Delaware	36,200	Corporate headquarters	November 2022
Conshohocken, Pennsylvania	30,300	Office and research space	September 2026
Montreal, Quebec	17,300	Office and research space	June 2021
New York, New York	19,400	Office and research space	May 2030+
Palo Alto, California	4,900	Office and research space	June 2027†
San Diego, California	10,600	Office and research space	September 2025
Rennes, France	50,000	Office and research space	June 2019*
Princeton, New Jersey	16,900	Office space	February 2025

+ Based on an estimated commencement date in June 2020. This office space will replace our Melville, New York location.
† Based on an estimated commencement date in April 2020.
* We sublease our facility in Rennes from Thomson Licensing SAS.

We are also a party to leases for several smaller research and/or office spaces, including in Brussels, Belgium; Buffalo, New York, USA; Indianapolis, Indiana, USA; Paris, France; San Francisco, California, USA; and Shanghai, China. In addition, we own a building in Washington, District of Columbia, USA, that houses administrative office space.

We believe that the facilities described above are suitable and adequate for our present purposes and our needs in the near future.

Item 3. *LEGAL PROCEEDINGS.*

See Note 12, “Litigation and Legal Proceedings,” to the Notes to Consolidated Financial Statements included below in Part II, Item 8 of this Form 10-K for a description of our material legal proceedings, which is incorporated herein by reference.

Item 4. *MINE SAFETY DISCLOSURES.*

Not applicable.

PART II

Item 5. *MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES.*

Market Information

The NASDAQ Stock Market ("NASDAQ") is the principal market for our common stock, which is traded under the symbol "IDCC."

Holders

As of February 18, 2020, there were 510 holders of record of our common stock.

Dividends

Cash dividends on outstanding common stock declared in 2019 and 2018 were as follows (in thousands, except per share data):

2019	Per Share	Total	Cumulative by Fiscal Year
First quarter	$ 0.35	$ 11,180	$ 11,180
Second quarter	0.35	10,895	22,075
Third quarter	0.35	10,897	32,972
Fourth quarter	0.35	10,746	43,718
	$ 1.40	$ 43,718	
2018			
First quarter	$ 0.35	$ 12,124	$ 12,124
Second quarter	0.35	12,192	24,316
Third quarter	0.35	11,996	36,312
Fourth quarter	0.35	11,610	47,922
	$ 1.40	$ 47,922	

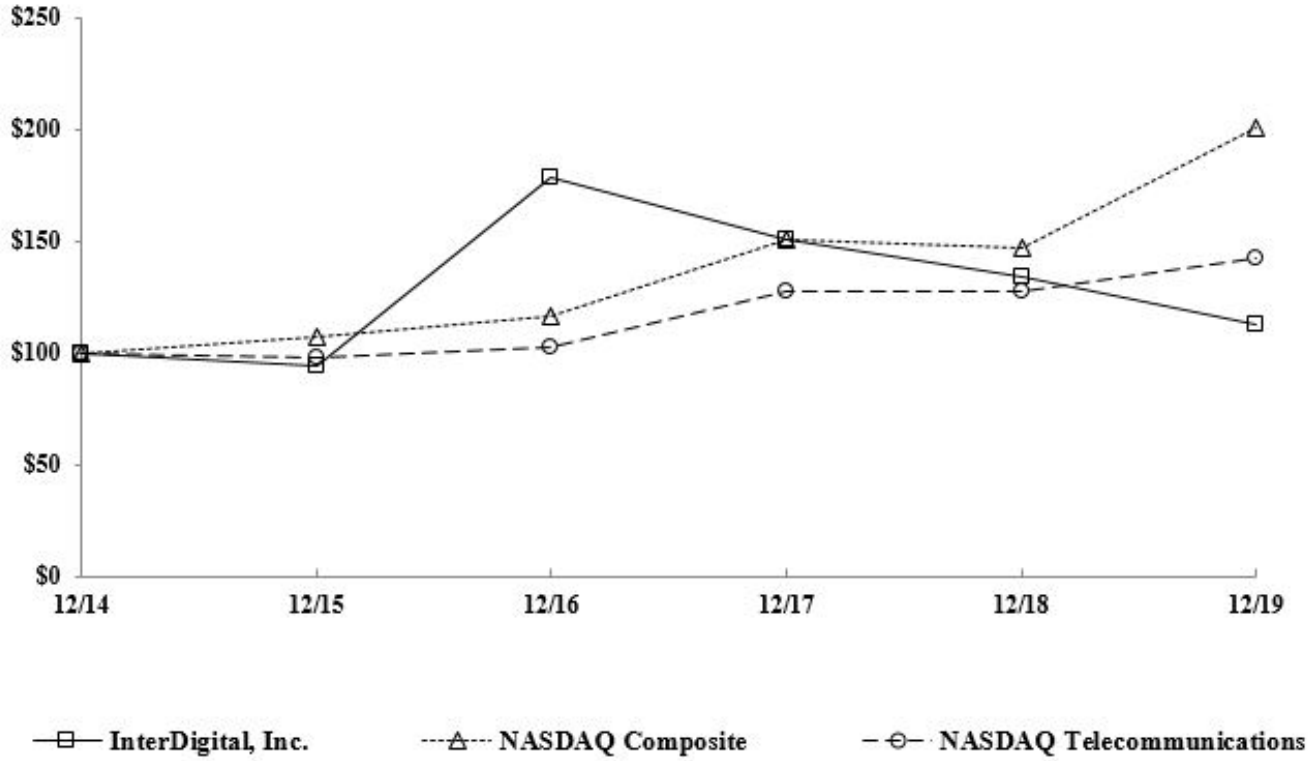
In September 2017, we announced that our Board of Directors had approved an increase in the Company's quarterly cash dividend to $0.35 per share. We currently expect to continue to pay dividends comparable to our quarterly $0.35 per share cash dividend in the future; however, continued payment of cash dividends and changes in the Company's dividend policy will depend on the Company's earnings, financial condition, capital resources and capital requirements, alternative uses of capital, restrictions imposed by any existing debt, economic conditions and other factors considered relevant by our Board of Directors.

Performance Graph

The following graph compares five-year cumulative total returns of the Company, the NASDAQ Composite Index and the NASDAQ Telecommunications Stock Index. The graph assumes $100 was invested in the common stock of InterDigital and each index as of December 31, 2014 and that all dividends were re-invested. Such returns are based on historical results and are not intended to suggest future performance.

COMPARISON OF 5 YEAR CUMULATIVE TOTAL RETURN*

Among InterDigital, Inc., the NASDAQ Composite Index
and the NASDAQ Telecommunications Index



*$100 invested on 12/31/14 in stock or index, including reinvestment of dividends. Fiscal year ending December 31.

	12/14	12/15	12/16	12/17	12/18	12/19
InterDigital, Inc.	**100.00**	**94.14**	**178.08**	**150.75**	**133.84**	**112.28**
NASDAQ Composite	**100.00**	**106.96**	**116.45**	**150.96**	**146.67**	**200.49**
NASDAQ Telecommunications	**100.00**	**97.52**	**102.36**	**127.62**	**127.16**	**142.60**

The above performance graph shall not be deemed "filed" for purposes of Section 18 of the Exchange Act, or incorporated by reference into any filing of InterDigital under the Securities Act of 1933, as amended, or the Exchange Act, except as shall be expressly set forth by specific reference in such filing.

Issuer Purchases of Equity Securities

Repurchase of Common Stock

The following table provides information regarding Company purchases of its common stock during fourth quarter 2019.

Period	Total Number of Shares (or Units) Purchased (1)	Average Price Paid Per Share (or Unit)	Total Number of Shares (or Units) Purchases as Part of Publicly Announced Plans or Programs (2)	Maximum Number (or Approximate Dollar Value) of Shares (or Units) That May Yet Be Purchased Under the Plans or Programs (3)
October 1, 2019 - October 31, 2019	—	$ —	—	$ 96,813,533
November 1, 2019 - November 30, 2019	432,849	$ 57.74	432,849	$ 71,813,695
December 1, 2019 - December 31, 2019	—	$ —	—	$ 71,813,695
Total	432,849	$ 57.74	432,849	$ 71,813,695

(1) Total number of shares purchased during each period reflects share purchase transactions that were completed (i.e., settled) during the period indicated.

(2) Shares were purchased pursuant to the Company's $700 million share repurchase program (the "2014 Repurchase Program"), $300 million of which was authorized by the Company's Board of Directors in June 2014, with an additional $100 million authorized by the Company's Board of Directors in each of June 2015, September 2017, December 2018, and May 2019, respectively. The 2014 Repurchase Program has no expiration date. The Company may repurchase shares under the 2014 Repurchase Program through open market purchases, pre-arranged trading plans, or privately negotiated purchases.

(3) Amounts shown in this column reflect the amounts remaining under the 2014 Repurchase Program.

Item 6. *SELECTED FINANCIAL DATA.*

The following data should be read in conjunction with the Consolidated Financial Statements, related Notes and other financial information contained in this Form 10-K. We adopted new revenue guidance, ASC 606, effective January 1, 2018 using the modified retrospective method. As such, revenue and other related accounts are presented in accordance with ASC 606 for the years ended December 31, 2019 and 2018, and in accordance with ASC 605 for all prior periods presented. Refer to Note 3, "*Revenue Recognition*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information regarding our adoption of ASC 606. Additionally, effective January 1, 2019, we adopted ASU 2016-02, "Leases (Topic 842)" or ("ASC 842"), which outlines a comprehensive change to the lease accounting model and supersedes prior lease guidance. Refer to Note 17, "*Leases*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information regarding our adoption of ASC 842.

	2019	2018	2017	2016	2015
			(in thousands except per share data)		
Consolidated statements of operations data:					
Revenues (a)	$ 318,924	$ 307,404	$ 532,938	$ 665,854	$ 441,435
Income from operations	$ 37,835	$ 62,595	$ 301,495	$ 437,306	$ 208,549
Income tax benefit (provision) (b)	$ (10,991)	$ 27,417	$ (121,676)	$ (116,791)	$ (64,621)
Net income applicable to InterDigital, Inc. common shareholders (c)	$ 20,928	$ 65,031	$ 176,220	$ 310,741	$ 120,803
Net income per common share — basic (c)	$ 0.66	$ 1.89	$ 5.09	$ 9.00	$ 3.35
Net income per common share — diluted (c)	$ 0.66	$ 1.84	$ 4.93	$ 8.83	$ 3.31
Weighted average number of common shares outstanding — basic	31,546	34,491	34,605	34,526	36,048
Weighted average number of common shares outstanding — diluted	31,785	35,307	35,779	35,189	36,463
Cash dividends declared per common share (d)	$ 1.40	$ 1.40	$ 1.30	$ 1.00	$ 0.80
Consolidated balance sheets data:					
Cash, cash equivalents and restricted cash (e)	$ 757,098	$ 488,733	$ 433,014	$ 404,074	$ 510,207
Short-term investments	179,204	470,724	724,981	548,687	423,501
Working capital	710,774	844,855	1,019,353	795,639	610,994
Total assets	1,612,082	1,626,558	1,854,420	1,727,853	1,474,485
Total debt	444,758	317,377	285,126	272,021	486,769
Total InterDigital, Inc. shareholders' equity (c)	761,557	936,729	863,808	746,323	515,393
Noncontrolling interest (c)	24,724	1,284	9,340	8,045	6,502
Total shareholders' equity	$ 786,281	$ 938,013	$ 873,148	$ 754,368	$ 521,895

(a) In 2019, 2018, 2017, 2016, and 2015, our revenues included $19.8 million, $26.3 million, $162.9 million, $309.7 million, and $65.8 million of non-current patent royalties, respectively.

(b) In 2018, our income tax benefit includes an $18.0 million tax benefit due to our income qualifying as foreign derived intangible income ("FDII"), as well as a $14.7 million benefit as a result of anticipated filings of amended tax returns in connection with the Korea Competent Authority Proceeding defined and discussed below. In 2017, our income tax provision was impacted by the U.S. Tax Cuts and Jobs Act (the "TCJA") as discussed in our results of operations. For more information, refer to Note 14, "*Taxes*" in the Notes to the Consolidated Financial Statements included in Part II, Item 8, of this Form 10-K. In 2016, our income tax provision included the impact of a $23.6 million net tax benefit primarily related to domestic activity production deductions for prior years.

(c) As discussed in Note 1 within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K, "*Background and Basis of Presentation*," we revised our prior period presentation of noncontrolling interest.

(d) In September 2017, we announced that our Board of Directors had approved an increase in the Company's quarterly cash dividend to $0.35 per share. In September 2016, we announced that our Board of Directors had approved an increase in the Company's quarterly cash dividend to $0.30 per share.

(e) Includes restricted cash which is included within "*Prepaid and other current assets*" or "*Other non-current assets*" in the consolidated balance sheets.

Item 7. *MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS.*

OVERVIEW

The following discussion should be read in conjunction with the Selected Financial Data, the Consolidated Financial Statements and the Notes thereto contained in this Form 10-K.

Effective January 1, 2018, we adopted FASB Accounting Standards Codification 606, Revenue from Contracts with Customers ("ASC 606"), which affected our recognition of revenue from both our fixed-fee and per-unit license agreements beginning in first quarter 2018. All periods prior to January 1, 2018 are presented in accordance with ASC Topic 605, Revenue

Recognition (“ASC 605”). Refer to Note 3, "*Revenue Recognition*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information.

Throughout the following discussion and elsewhere in this Form 10-K, we refer to “recurring revenues” and “non-current patent royalties.” For all periods presented, recurring revenues are comprised of “current patent royalties” and “current technology solutions revenue.” For 2019 and 2018, non-current patent royalties are comprised of “past patent royalties” and “static fixed-fee" agreement royalties. For periods prior to 2018, non-current patent royalties are comprised of just past patent royalties, whereas static fixed-fee agreement royalties are included as part of recurring revenues.

Business

InterDigital, Inc. ("InterDigital") is a research and development company that licenses its innovations to the global wireless and consumer electronics industries. We design and develop advanced technologies that enable connected, immersive experiences in a broad range of communications and entertainment products and services. Since our founding in 1972, our engineers have designed and developed a wide range of innovations that are used in wireless products and networks, from the earliest digital cellular systems to 5G and, today, solutions that we believe will shape the world beyond 5G. With the acquisition of the patent licensing business of visual technology industry leader Technicolor SA ("Technicolor") in 2018 (the "Technicolor Patent Acquisition"), followed by the acquisition of their Research & Innovation unit in 2019 (the "R&I Acquisition" and, together with the Technicolor Patent Acquisition, the "Technicolor Acquisitions"), we are now a leader in video processing, encoding/decoding, and display technology, with a significant Artificial Intelligence ("AI") research effort that intersects with both wireless and visual technologies.

InterDigital is one of the largest pure research & development and licensing companies in the world, with one of the most significant patent portfolios in the technology industry. As of December 31, 2019, InterDigital's wholly owned subsidiaries held a portfolio of approximately 32,000 patents and patent applications related to wireless communications, video coding, display technology, and other areas relevant to the wireless and consumer electronics industries. Our portfolio includes numerous patents and patent applications that we believe are or may be essential or may become essential to standards established by many Standards Development Organizations ("SDOs"), including cellular and other wireless communications and video technology standards. Those wireless standards include 3G, 4G and the IEEE 802 suite of standards, as well as patents and patent applications that we believe are or may become essential to 5G standards that currently exist and as they continue to develop. Our video technology portfolio includes patents and applications relating to standards established by ISO/IEC Moving Picture Expert Group (MPEG), the ITU-T Video Coding Expert Group (VCEG), the Joint Collaborative Team on Video Coding (JCT-VC) and the Joint Video Expert Team (JVET), among others.

Our wireless portfolio has largely been built through internal development, supplemented by joint development projects with other companies, as well as select acquisitions of patents and companies. Products incorporating our patented inventions in wireless include: mobile devices, such as cellular phones, tablets, notebook computers and wireless personal digital assistants; wireless infrastructure equipment, such as base stations; components, dongles and modules for wireless devices; and Internet of Things ("IoT") devices and software platforms. Our video technology portfolio largely represents patents and applications that came to InterDigital as a result of the Technicolor Patent Acquisition, supplemented by internal development. Our patented inventions in video are incorporated in a range of products and services, including cellular phones, notebook computers, televisions, gaming consoles, set-top boxes, streaming devices and other consumer electronics.

Acquisition of Technicolor's Research & Innovation Unit

On May 31, 2019, we completed the acquisition of the R&I unit of Technicolor SA, which we refer to as the R&I Acquisition. R&I is a premier research lab that conducts fundamental research into video coding, IoT and smart home, imaging sciences, augmented reality and virtual reality, and artificial intelligence and machine learning technologies.

As consideration for the R&I Acquisition, the parties agreed to terminate the jointly funded R&D collaboration agreement that was entered into as part of the Technicolor Patent Acquisition. In addition, InterDigital assumed certain employment-related obligations and Technicolor agreed to reduce its rights to a revenue-sharing arrangement announced as part of the Technicolor Patent Acquisition. There was no cash consideration. The R&I Acquisition resulted in a net gain of approximately $14.2 million, inclusive of a $20.5 million gain from the derecognition of a contingent consideration liability, all of which is included within “*Other Income (Expense), Net*” within the consolidated statement of income for the year ended December 31, 2019.

Refer to Note 5, "*Business Combinations and Other Transactions*" within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further discussion of both the Technicolor Patent Acquisition and the R&I Acquisition.

Revenue

As previously discussed, we adopted new revenue guidance, ASC 606, effective January 1, 2018 using the modified retrospective method. Consistent with the modified retrospective adoption method, our results of operations for periods prior to our adoption of ASC 606 remain unchanged. As such, revenue is presented in accordance with ASC 606 for the years ended December 31, 2019 and 2018 and in accordance with ASC 605 for all prior periods presented. Refer to Note 3, "*Revenue Recognition*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information regarding our adoption of ASC 606.

In 2019, 2018, and 2017, our total revenues were $318.9 million, $307.4 million and $532.9 million, respectively. Our recurring revenues in 2019, 2018 and 2017 were $298.2 million, $280.3 million and $370.0 million, respectively. In each of the years presented, we recognized between $19.8 million and $162.9 million of non-current patent royalties as more fully discussed below. In 2019, fixed-fee royalties accounted for approximately 86% of our recurring revenues. These fixed-fee revenues are not affected by the related licensees' success in the market or the general economic climate. The majority of the remaining portion of our recurring revenue was variable in nature due to the per-unit structure of the related license agreements.

New Agreements

Direct Licenses

During fourth quarter 2019, we entered into a multi-year, worldwide, non-exclusive, royalty-bearing patent license and settlement agreement with ZTE. The agreement covers the sale of ZTE's 3G, 4G and 5G handset and tablet products, as well as 802.11 and HEVC technologies incorporated into such products.

Also during fourth quarter 2019, we entered into a multi-year, worldwide, non-exclusive, royalty-bearing patent license agreement with u-blox AG ("u-blox"). The agreement covers the sale of u-blox's 3G and 4G machine to machine modules and certain consumer modules.

During second quarter 2019, we entered into a multi-year, worldwide, non-exclusive, royalty-bearing patent license agreement with Teltronic S.A.U. ("Teltronic"). The agreement covers the sale of Teltronic's 4G terminal units as well as 3G and 4G infrastructure equipment.

Also during second quarter 2019, we entered into a Settlement Agreement and First Amendment to the Patent License Agreement with Asustek Computer Incorporated ("Asus"). The agreement provides for, among other things, a multi-year amendment to our 2008 patent license agreement with Asus (the "2008 Asus PLA") that adds coverage for 4G technologies and amends certain other terms of the 2008 Asus PLA.

Licenses Through Platforms

During fourth quarter 2019, Google was granted a multi-year, worldwide, non-exclusive, royalty-bearing patent license covering the sale of certain of its 3G and 4G mobile communication devices. We entered into this agreement through a licensing platform.

Also during fourth quarter 2019, as part of the Madison Arrangement, we entered into a multi-year, non-exclusive, royalty-bearing patent license agreement with Funai Electric Co., Ltd. ("Funai"). The agreement covers the U.S. sales of Funai's DTVs.

During 2019, Avanci announced that it entered into several patent license agreements with new licensees, including Audi and Porsche, the Volkswagen Group Companies and Volvo Cars.

Refer to the "*Critical Accounting Policies and Estimates — Revenue Recognition*" section below for details of our revenue recognition accounting policies and additional information on agreements with multiple performance obligations, as well as the estimates and methods used to determine the fair value of patents acquired, when applicable.

Expiration of License Agreements

Our patent license agreements with two licensees expired during 2019. No revenue was recognized under either agreement in 2019.

Our patent license agreement with LG is scheduled to expire at the end of 2020. LG contributed $31.8 million, or approximately 11%, of our recurring revenue in 2019.

Our patent license agreements with seven other licensees are scheduled to expire during 2020. Collectively with LG, all eight agreements expiring in 2020 accounted for $35.1 million, or approximately 12%, of our recurring revenue in 2019.

Intellectual Property Rights Enforcement

If we believe a party is required to license our patents in order to manufacture, use and/or sell certain products and such party refuses to do so, we typically offer such party to have royalty rates, or other terms, set by third party adjudicators (such as arbitrators). If the party refuses that offer and we believe they are unwilling to agree to a patent license on a fair, reasonable and non-discriminatory basis, we may have no other viable recourse but to institute legal action against them to enforce our patent rights. This legal action has typically taken the form of a patent infringement lawsuit or an administrative proceeding. In addition, we and our licensees, in the normal course of business, might seek to resolve disagreements as to the rights and obligations of the parties under the applicable license agreement through arbitration or litigation.

During 2019, we filed patent infringement actions in the United Kingdom against Lenovo and Huawei as more fully discussed in Note 12, "*Litigation and Legal Proceedings*," to the Notes to Consolidated Financial Statements included below in Part II, Item 8 of this Form 10-K. We filed each of these actions after lengthy periods of negotiation and after the refusal by both Lenovo and Huawei to accept our various proposals to each of them, including our proposal to have a third party adjudicator set a royalty rate and resolve certain other terms that we could not mutually agree upon with Lenovo and Huawei, respectively.

In 2019, our intellectual property enforcement costs increased to $25.4 million from $17.6 million and $15.2 million in 2018 and 2017, respectively. These costs represented 16% of our total patent administration and licensing costs of $154.9 million in 2019. Intellectual property enforcement costs will vary depending upon activity levels, and it is likely they will continue to be a significant expense for us in the future.

Hillcrest Sale

On July 19, 2019, we completed the sale of our Hillcrest product business to a subsidiary of CEVA, Inc. In connection with the sale, we received initial proceeds of $10.0 million, with a customary portion of the purchase price placed in escrow to secure potential indemnification claims. As part of the transaction, we retained substantially all of the Hillcrest patent assets that we acquired in 2016. As a result of this transaction, we recorded an $8.5 million gain on sale which is included within "*Other Income (Expense), Net*" in the consolidated statement of income for the year ended December 31, 2019.

2024 Senior Convertible Notes

On June 3, 2019, we issued the $400.0 million aggregate principal 2024 Notes. The net proceeds from the offering of the 2024 Notes were approximately $391.6 million after deducting the initial purchasers' fees and estimated offering expenses. Additionally, on May 29 and May 31, 2019, in connection with the offering of the 2024 Notes, we entered into the 2024 Call Spread Transactions.

The net proceeds from the issuance of the 2024 Notes, after deducting fees and offering expenses, were used for the following: (i) $232.7 million was used to repurchase $221.1 million in aggregate principal amount of the 2020 Notes in privately negotiated transactions concurrently with the offering of the 2024 Notes (ii) $19.6 million was used to repurchase shares of common stock at $62.53 per share, the closing price of the stock on May 29, 2019; and (iii) $24.4 million, in addition to the proceeds from the 2024 Warrant Transactions, was used to fund the cost of the 2024 Call Spread Transactions.

The 2024 Notes will be convertible into cash, shares of our common stock or a combination thereof, at our election, at an initial conversion rate of 12.3018 shares of common stock per $1,000 principal amount of 2024 Notes (which is equivalent to an initial conversion price of approximately $81.29 per share), as adjusted pursuant to the terms of the Indenture.

Refer to Note 10, "*Obligations*" within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for defined terms and further discussion of the 2024 Notes and related 2024 Call Spread Transactions.

Cash and Short-Term Investments

As of December 31, 2019, we had $0.9 billion of cash, restricted cash and short-term investments and up to an additional $404.7 million of payments due under signed agreements, including $28.3 million recorded in accounts receivable which includes estimates related to our fourth quarter 2019 variable patent royalty revenue. A portion of our cash and short-term investments include fixed royalty payments we have received related to revenue we will record in the future. As a result, our future cash receipts from existing licenses subject to fixed patent royalties will be lower than if the royalty payments were structured to coincide with the underlying sales. During 2019, we recorded $295.2 million of cash receipts related to patent

licensing and technology solutions agreements as follows (in thousands):

	Cash In
Patent royalties	$ 288,123
Technology solutions	7,053
	$ 295,176

As of December 31, 2019, approximately $267.6 million of our $270.3 million deferred revenue balance as of December 31, 2019 related to dynamic fixed-fee royalty payments that were scheduled to amortize as follows (in thousands):

2020	$ 143,652
2021	76,534
2022	47,430
2023	—
2024	—
Thereafter	—
	$ 267,616

Refer to "*New Accounting Guidance*" below for a discussion regarding our adoption of ASC 606 effective January 1, 2018.

Repurchase of Common Stock

In June 2014, our Board of Directors authorized a $300 million share repurchase program (the “2014 Repurchase Program”). In June 2015, September 2017, December 2018, and May 2019, our Board of Directors authorized four $100 million increases to the program, respectively, bringing the total amount of the 2014 Repurchase Program to $700 million. The Company may repurchase shares under the 2014 Repurchase Program through open market purchases, pre-arranged trading plans or privately negotiated purchases.

The table below sets forth the total number of shares repurchased and the dollar value of shares repurchased under the 2014 Repurchase Program (in thousands). As of December 31, 2019, there was approximately $71.8 million remaining under the stock repurchase authorization.

	2014 Repurchase Program	
	# of Shares	Value
2019	2,962	$ 196,269
2018	1,478	110,505
2017	107	7,693
2016	1,304	64,685
2015	1,836	96,410

2014	3,554	152,625
Total	11,241	$ 628,187

Comparability of Financial Results

When comparing our 2019 financial results against the financial results of other periods, the following items should be taken into consideration:

- the Technicolor Patent Acquisition and the R&I Acquisition, which closed on July 30, 2018 and May 31, 2019, respectively, contributed $32.0 million to our 2019 revenue and $63.0 million to our 2019 operating expenses. The $63.0 million of operating expenses is comprised of $48.3 million of recurring costs, of which $16.6 million relates to patent amortization, $8.4 million relates to one-time transaction-related and integration costs, and $6.3 million relates to revenue sharing from the Madison Arrangement;

- the R&I Acquisition resulted in a net gain of approximately $14.2 million, inclusive of a $20.5 million gain from the derecognition of a contingent consideration liability, all of which is included within "*Other Income (Expense), Net*" within our consolidated statement of income;
- in connection with the offering of the 2024 Notes, we repurchased approximately $221.1 million in aggregate principal amount of our 2020 Notes, which resulted in the recognition of a $5.5 million loss on extinguishment of debt that is included in "*Other Income (Expense), Net*" within our consolidated statement of income;
- 2019 "*Other Income (Expense), Net*" also includes an $8.5 million gain on sale of our Hillcrest product business, as well as a net loss of $2.6 million resulting from the partial impairment of one of our strategic investments partially offset by a gain on sale of a separate strategic investment;
- our 2019 results include a $5.5 million net charge as contra non-recurring revenue related to a recently restructured licensing arrangement with a long-term customer; and
- our 2019 income tax provision includes a $2.2 million tax benefit as a result of filing amended tax returns in connection with the Korea Competent Authority Proceeding, as defined and discussed below.

Critical Accounting Policies and Estimates

Our consolidated financial statements are based on the selection and application of GAAP, which require us to make estimates and assumptions that affect the amounts reported in both our consolidated financial statements and the accompanying notes. Future events and their effects cannot be determined with absolute certainty. Therefore, the determination of estimates requires the exercise of judgment. Actual results could differ from these estimates and any such differences may be material to the financial statements. Our significant accounting policies are described in Note 2 within our Consolidated Financial Statements and are included in Item 8 of Part II of this Form 10-K. We believe the accounting policies that are of particular importance to the portrayal of our financial condition and results and that may involve a higher degree of complexity and judgment in their application compared to others are those relating to revenue recognition, compensation, business combinations and goodwill, and income taxes. If different assumptions were made or different conditions existed, our financial results could have been materially different.

Revenue Recognition

On January 1, 2018, we adopted ASU No. 2014-09, "Revenue from Contracts with Customers (Topic 606)" (ASC 606) using the modified retrospective method. Refer to Note 3, "*Revenue Recognition*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information regarding our adoption of this guidance. The discussion that follows below is a description of our revenue recognition practices in effect beginning January 1, 2018 under ASC 606.

We derive the vast majority of our revenue from patent licensing. The timing and amount of revenue recognized from each licensee depends upon a variety of factors, including the specific terms of each agreement and the nature of the deliverables and obligations. Such agreements are often complex and include multiple performance obligations. These agreements can include, without limitation, performance obligations related to the settlement of past patent infringement liabilities, patent and/or know-how licensing royalties on covered products sold by licensees, access to a portfolio of technology as it exists at a point in time, and access to a portfolio of technology at a point in time along with a promises to provide any technology updates to the portfolio during the term.

In accordance with US GAAP, we use a five-step model to achieve the core underlying principle that an entity should recognize revenue to depict the transfer of goods or services to customers at an amount that the entity expects to be entitled to in exchange for those goods or services. These steps include (1) identifying the contract with the customer, (2) identifying the performance obligations, (3) determining the transaction price, (4) allocating the transaction price to the performance obligations, and (5) recognizing revenue as the entity satisfies the performance obligation(s). Additionally, we have elected to utilize certain practical expedients in the application of ASC 606. In evaluating the presence of a significant financing component in our agreements, we utilize the practical expedient to exclude any contracts wherein the gap between payment by our customers and the delivery of our performance obligation is less than one year. We have also elected to utilize the practical expedient related to costs of obtaining a contract where an entity may recognize the incremental costs of obtaining a contract as an expense when incurred if the amortization period of the asset that the entity otherwise would have recognized is one year or less. Timing of revenue recognition may differ significantly from the timing of invoicing to customers. Contract assets are included in accounts receivable and represent unbilled amounts expected to be received from customers in future periods, where the revenue recognized to date exceeds the amount billed, and right to payment is subject to the underlying contractual terms. Contract assets are classified as long-term assets if the payments are expected to be received more than one year from the reporting date. Contract assets due within less than twelve months of the balance sheet date are included within accounts receivable in our consolidated balance sheets. Contract assets due more than twelve months after the balance sheet date are included within other non-current assets.

Patent License Agreements

Upon signing a patent license agreement, we provide the licensee permission to use our patented inventions in specific applications. We account for patent license agreements in accordance with the guidance indicated above. Certain patent license agreements contain revenue from non-financial sources in the form of patents received from the customer. Under our patent license agreements, we typically receive one or a combination of the following forms of payment as consideration for permitting our licensees to use our patented inventions in their applications and products:

Consideration for Past Patent Royalties

Consideration related to a licensee's product sales from prior periods may result from a negotiated agreement with a licensee that utilized our patented inventions prior to signing a patent license agreement with us or from the resolution of a disagreement or arbitration with a licensee over the specific terms of an existing license agreement. We may also receive consideration for past patent royalties in connection with the settlement of patent litigation where there was no prior patent license agreement. In each of these cases, we record the consideration as revenue as prescribed by the five-step model.

Fixed-Fee Agreements

Fixed-fee license agreements include fixed, non-refundable royalty payments that fulfill the licensee's obligations to us under a patent license agreement for a specified time period or for the term of the agreement for specified products, under certain patents or patent claims, for sales in certain countries, or a combination thereof - in each case for a specified time period (including for the life of the patents licensed under the agreement).

Dynamic fixed-fee license agreements contain a single performance obligation that represents ongoing access to a portfolio of technology over the license term, since our promise to transfer to the licensee access to the portfolio as it exists at inception of the license, along with promises to provide any technology updates to the portfolio during the term, are not separately identifiable. Upon entering a new agreement, we allocate the transaction price to the performance obligations delivered at signing (e.g. our existing patent portfolio) and future performance obligations (e.g. the technology updates). We use a time-based input method of progress to determine the timing of revenue recognition, and as such we recognize the future deliverables on a straight-line basis over the term of the agreement. We utilize the straight-line method as we believe that it best depicts efforts expended to develop and transfer updates to the customer evenly throughout the term of the agreement.

Static fixed-fee license agreements are fixed-price contracts that generally do not include updates to technology we create after the inception of the license agreement or in which the customer does not stand to substantively benefit from those updates during the term. Although we have few static fixed-fee license agreements, we generally satisfy our performance obligations under such agreements at contract signing, and as such revenue is recognized at that time.

Variable Agreements

Upon entering a new variable patent license agreement, the licensee typically agrees to pay royalties or license fees on licensed products sold during the term of the agreement. We utilize the sales- or usage- based royalty exception for these agreements and recognize revenues during the contract term when the underlying sale or usage occurs.

Our licensees under variable agreements provide us with quarterly royalty reports that summarize their sales of covered products and their related royalty obligations to us. We typically receive these royalty reports subsequent to the period in which our licensees' underlying

sales occurred. As a result, we are required to estimate revenues, subject to the constraint on our ability to estimate such amounts.

Technology Solutions

Technology solutions revenue consists of revenue from royalty payments, software licenses, engineering services and product sales. The nature of these contracts and timing of payments vary. We recognize revenue from royalty payments and license agreements using the same methods described above under our policy for recognizing revenue from patent license agreements. We recognize revenue from engineering services using percentage of completion method.

Patent Sales

Our business strategy of monetizing our intellectual property includes the sale of select patent assets. As patent sales executed under this strategy represent a component of our ongoing major or central operations and activities, we will record the related proceeds as revenue. We will recognize the revenue in accordance with the five-step model, generally upon closing of the patent sale transaction.

Agreements with Multiple Performance Obligations

During 2019, we signed three new agreements that had multiple performance obligations. Consistent with the revenue recognition policies disclosed above under ASC 606, we (1) identified the contract with the customer, (2) identified the performance obligations, (3) determined the transaction price, (4) allocated the transaction price to the performance obligations, and (5) recognized revenue as we satisfy the performance obligations. We allocated the transaction price to each performance obligation for accounting purposes using our best estimate of the term and value. The process for determining the value of the standalone selling prices of identified performance obligations in dynamic fixed-fee license agreements requires the exercise of significant judgment when evaluating the valuation methods and assumptions, including the assumed royalty rates, projected sales volumes, discount rate, and comparable market transactions which are not directly observable and other relevant factors. Changes in any of a number of these assumptions could have had a substantial impact on the relative fair value assigned to each performance obligation for accounting purposes. These inputs and assumptions represent management's best estimates at the time of the transaction.

The impact that a five percent change in the aggregate amount allocated to past patent royalties under these agreements would have had on 2019 revenue is summarized in the following table (in thousands):

	Change in amount allocated	
Allocation to past patent royalties	**+5%**	**-%5**
Change in Revenue	$ 1,618	$ (1,618)

Revenue from Non-financial Sources

During 2019, 2018 and 2017, our patent licensing royalties were derived from patent license agreements ("PLAs") with 69, 66 and 27 independent licensees, respectively. The number of independent licensees largely increased from 2017 to 2018 due to the Technicolor Patent Acquisition. We recognized revenue from four, three and five PLAs in 2019, 2018 and 2017, respectively, for which patents generally comprised less than forty-percent of the total consideration paid or due to us under those agreements. In addition, during 2019, 2018 and 2017, we recognized revenue from one PLA that was executed in 2014 in connection with a patent purchase agreement ("PPA") with the licensee. Total cash paid to our licensee under this PPA is approximately 56% of the total cash due to us under this licensee's PLA. During 2019, 2018 and 2017, approximately 6%, 3% and 4%, respectively, of our total revenue was based on the estimated fair value of the patents in the above transactions.

The process for determining the value of revenue from non-financial sources requires estimating the fair value of patents received. We estimated the fair value of the patents in the above transactions using one of, or a combination of, an analysis of comparable market transactions (the market approach), a discounted cash flow analysis (the

income approach) and/or by quantifying the amount of money required to replace the future service capability of the assets (the cost approach). For the market approach, judgment was applied as to which market transactions were most comparable to the transaction. For the income approach, the inputs and assumptions used to develop these estimates were based on a market participant perspective and included estimates of projected royalties, discount rates, economic lives and income tax rates, among others. For the cost approach, we utilized the historical cost of assets of similar technologies to determine the estimated replacement cost, including research, development, testing and patent application fees. The development of a number of these inputs and assumptions requires a significant amount of management judgment and is based upon a number of factors, including comparable market transactions, assumed royalty rates, projected sales volumes, economic lives of the patents and other relevant factors. Changes

in any of a number of these assumptions could have had a substantial impact on the fair value assigned to the patents for accounting purposes. These inputs and assumptions represent management's best estimates at the time of the transaction.

The impact that a five-percent change in the estimated aggregate value of the patents acquired would have had on 2019 revenue, patent amortization and pre-tax income is summarized in the following table (in thousands).

	Change in estimate	
Estimated value of patents acquired in connection with PLAs	**+5%**	**-%5**
Revenue	$ 1,070	$ (1,070)
Less: Patent amortization	672	(672)
Pre-tax income	$ 398	$ (398)

Compensation Programs

We use a variety of compensation programs to attract, retain and motivate our employees, and to more closely align employee compensation with company performance. These programs include, but are not limited to, short-term incentives tied to performance goals, cash awards to inventors for filed patent applications and patent issuances, and long-term incentives in the form of stock option awards, time-based restricted stock unit (“RSU”) awards, performance-based awards and cash awards, noting equity awards are granted pursuant to the terms and conditions of our Equity Plans (as defined within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K). Our long-term incentives, including equity awards, typically include annual equity and cash award grants with three- to five-year vesting periods; as a result, in any one year, we are typically accounting for at least three active cycles.

The aggregate amount of performance compensation expense we record in a period, under both short-term and long-term incentive compensation programs, requires the input of subjective assumptions and is a function of our estimated progress toward performance compensation goals at both the beginning and the end of the period. Our estimated progress toward goals under performance equity grants is based on meeting a minimum confidence level in accordance with accounting rules for share-based compensation. Achievement rates can vary by performance cycle and from period to period, resulting in variability in our compensation expense.

If we had accrued all performance compensation cost throughout 2019 on the assumption that all plans and active cycles thereunder would be paid out at 100%, we would have recorded approximately $12.5 million more in compensation expense in 2019 than we actually recorded.

We account for compensation costs associated with share-based compensation based on the fair value of the instruments issued. The estimated value of stock options includes assumptions around expected life, stock volatility and dividends. The expected life of our stock option awards is based on the simplified method as prescribed by Staff Accounting Bulletin Topic 14. In all periods, our policy has been to set the value of RSUs and restricted stock awards equal to the value of our underlying common stock on the date of measurement. For grants with graded vesting, we amortize the associated unrecognized compensation cost using an accelerated method. For grants that cliff vest, we amortize the associated unrecognized compensation cost on a straight-line basis over their vesting term.

In the event of canceled awards, we adjust compensation expense recognized to date as they occur. Tax windfalls and shortfalls related to the tax effects of employee share-based compensation are included in our tax provision. On the consolidated statements of cash flows, tax windfalls and shortfalls related to employee share-based compensation awards are included within operating activities and cash paid to tax authorities for shares withheld are included within financing activities. The inclusion of windfalls and shortfalls

in the tax provision could increase our earnings volatility between periods. Tax windfalls related to share-based compensation for the years ended 2019, 2018 and 2017 were $0.2 million, $1.8 million and $12.1 million, respectively.

The below table summarizes our supplemental compensation expense for 2019, 2018 and 2017, in thousands:

	2019	2018	2017
Short-term incentive compensation	$ 14,129	$ 13,045	$ 13,994
Time-based awards (a)	6,327	5,985	6,958
Performance-based awards (a) (b)	299	1,415	6,883
Other share-based compensation	1,307	1,768	4,999
Total supplemental compensation expense	$ 22,062	$ 22,213	$ 32,834

(a) For 2019, 2018 and 2017, approximately 5%, 28%, and 6%, respectively, of the aggregate expense associated with time-based and performance-based awards related to cash awards. The increase in cash awards in 2018 is primarily related to certain cash-based executive retirement awards.

(b) Includes a charge of $0.4 million in 2017 to increase the accrual rates under our long-term incentive programs driven by the Company's success toward achieving goals for the related cycles. There were no changes to the accrual rates under our long-term incentive programs during 2019 or 2018.

Business Combinations and Goodwill

Acquisitions that qualify as a business combination are accounted for using the acquisition method of accounting. The fair value of consideration transferred for an acquisition is allocated to the assets acquired and liabilities assumed based on their fair value as of the acquisition date. Goodwill is recorded as the difference, if any, between the aggregate consideration paid for an acquisition and the fair value of the net tangible and identified intangible assets acquired under a business combination.

Under the acquisition method of accounting, the Company completes valuation procedures for an acquisition to determine the fair value of the assets acquired and liabilities assumed. These valuation procedures require management to make assumptions and apply significant judgment to estimate the fair value of the assets acquired and liabilities assumed. If the estimates or assumptions used should significantly change, the resulting differences could materially affect the fair value of net assets. We estimate the fair value of the intangible assets acquired generally through a combination of a discounted cash flow analysis (the income approach) and an analysis of comparable market transactions (the market approach). For the income approach, we base the inputs and assumptions used to develop these estimates on a market participant perspective which includes estimates of projected revenues, discount rates, economic lives and income tax rates, among others, all of which require significant management judgment. For the market approach, we apply judgment to identify the most comparable market transactions to the transaction. Definite-lived intangible assets, which are primarily comprised of patents, are amortized over their estimated useful lives using the straight-line method and are assessed for impairment whenever events or changes in circumstances indicate the carrying value of the asset may not be recoverable.

Goodwill is not amortized but is reviewed for impairment annually on the first day of the fourth quarter, or when events or changes in the business environment indicate that the carrying value of a reporting unit may exceed its fair value. We first assess qualitative factors to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying amount as a basis for determining whether a quantitative goodwill impairment test is necessary. If we conclude it is more likely than not that the fair value of a reporting unit exceeds its carrying amount, we need not perform the quantitative assessment. If based on the qualitative assessment we believe it is more likely than not that the fair value of a reporting unit is less than its carrying value, a quantitative assessment test is required to be performed. This assessment requires us to compare the fair value of each reporting unit to its carrying value including allocated goodwill. We determine the fair value of our reporting units generally using a combination of the income and market approaches. If the carrying value of a reporting unit exceeds the reporting unit’s fair value, a goodwill impairment charge will be recorded for the difference up to the carrying value of goodwill.

Income Taxes

Income taxes are accounted for under the asset and liability method. Under this method, deferred tax assets and liabilities are recognized for the estimated future tax consequences attributable to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases, and operating loss and tax credit carry forwards. Deferred tax assets and liabilities are measured using enacted tax rates in effect for the year in which those temporary differences are expected to be recovered or settled. The effect on deferred tax assets and liabilities of a change in tax rates is recognized in the Consolidated Statement of Income in the period in which the change was enacted. A valuation allowance is recorded to reduce the carrying amounts of deferred tax assets if management has determined that it is more likely than not that such assets will not be realized.

In addition, the calculation of tax liabilities involves significant judgment in estimating the impact of uncertainties in the application of complex tax laws. We are subject to examinations by the U.S. IRS and other taxing jurisdictions on various tax matters, including challenges to various positions we assert in our filings. In the event that the IRS or another taxing jurisdiction levies an assessment in the future, it is possible the assessment could have a material adverse effect on our consolidated financial condition or results of operations.

The financial statement recognition of the benefit for a tax position is dependent upon the benefit being more likely than not to be sustainable upon audit by the applicable tax authority. If this threshold is met, the tax benefit is then measured and recognized at the largest amount that is greater than 50 percent likely of being realized upon ultimate settlement. In the event that the IRS or another taxing jurisdiction levies an assessment in the future, it is possible the assessment could have a material adverse effect on our consolidated financial condition or results of operations.

Between 2006 and 2019, we paid approximately $177.4 million in foreign taxes to foreign governments that have tax treaties with the U.S., for which we have claimed foreign tax credits against our U.S. tax obligations, and for which the tax treaty procedures are still open. It is possible that as a result of tax treaty procedures, the U.S. government may reach an agreement with the related foreign governments that will result in a partial refund of foreign taxes paid with a related reduction in our foreign tax credits. Due to foreign currency fluctuations, any such agreement could result in foreign currency gain or loss.

On November 8, 2019, the Company received notification that its request for competent authority pertaining to Article 25 (Mutual Agreement Procedure) of the United States-Republic of Finland Income Tax Convention had been reviewed by the IRS and an agreement has been reached (the “Finland Competent Authority Proceeding”). As a result of this agreement, the Company does not anticipate any tax consequences.

On July 24, 2018, the Company received notification that its request for competent authority pertaining to Article 27 (Mutual Agreement 14 Table of Contents Procedure) of the United States-Republic of Korea Income Tax Convention had been reviewed by the IRS and an agreement had been reached (the "Korea Competent Authority Proceeding"). As a result of this agreement, the Company received refunds of $97.4 million, inclusive of interest. In addition, we have recorded a net tax benefit of $14.7 million in our full year 2018. In September 2019 the amended tax returns for tax years covered by this agreement were filed and an additional benefit of $2.2 million was recorded related to the final refund the Company expects to receive.

New Accounting Guidance

Refer to Note 2, "*Summary of Significant Accounting Policies and New Accounting Guidance*" within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for a discussion of recently issued accounting guidance.

Legal Proceedings

We are routinely involved in disputes associated with enforcement and licensing activities regarding our intellectual property, including litigations, arbitrations and other proceedings. These litigations, arbitrations and other proceedings are important means to enforce our intellectual property rights. We are a party to other disputes and legal actions not related to our intellectual property, but also arising in the ordinary course of our business. Refer to Note 12, “*Litigation and Legal Proceedings*,” to the Notes to Consolidated Financial Statements included below in Part II, Item 8 of this Form 10-K for a description of our material legal proceedings.

FINANCIAL POSITION, LIQUIDITY AND CAPITAL RESOURCES

Our primary sources of liquidity are cash, cash equivalents and short-term investments, as well as cash generated from operations. We believe we have the ability to obtain additional liquidity through debt and equity financings. Based on our past performance and current expectations, we believe our available sources of funds, including cash, cash

equivalents and short-term investments and cash generated from our operations, will be sufficient to finance our operations, capital requirements, debt obligations (including the repayment of the remaining $94.9 million of our 2020 Notes), existing stock repurchase program and dividend program for the next twelve months.

Cash, cash equivalents, restricted cash and short-term investments

As of December 31, 2019 and December 31, 2018, we had the following amounts of cash, cash equivalents, restricted cash and short-term investments (in thousands):

	December 31, 2019	December 31, 2018	Increase / (Decrease)
Cash and cash equivalents	$ 745,491	$ 475,056	$ 270,435
Restricted cash included within prepaid and other current assets	10,526	13,677	(3,151)
Restricted cash included within other non-current assets	1,081	—	1,081
Short-term investments	179,204	470,724	(291,520)
Total cash, cash equivalents, restricted cash and short-term investments	$ 936,302	$ 959,457	$ (23,155)

The net decrease in cash, cash equivalents, restricted cash and short-term investments was attributable to cash used in financing activities of $89.3 million and cash used in investing activities, excluding sales and purchases of short-term investments, of $28.3 million. These uses were partially offset by cash provided by operating activities of $89.4 million. Cash used in financing activities primarily related to share repurchases, dividend payments and cash payments for payroll taxes upon vesting of restricted stock units, partially offset by net proceeds from the debt refinancing and related expenses and proceeds received from non-controlling interests. Cash used in investing activities, excluding sales and purchases of short-term investments, primarily related to capital investments for patents and fixed assets, partially offset by proceeds received from the sale of our Hillcrest product business. Refer to the sections below for further discussion of these items.

Cash flows from operations

We generated the following cash flows from our operating activities in 2019 and 2018 (in thousands):

	For the Year Ended December 31,		
	2019	2018	Increase / (Decrease)
Cash flows provided by operating activities	$ 89,433	$ 146,792	$ (57,359)

Our cash flows provided by operating activities are principally derived from cash receipts from patent license and technology solutions agreements, offset by cash operating expenses and income tax payments. The decrease in cash flows provided by operating activities of $57.4 million was due to both a decrease in cash receipts primarily attributable to the timing of cash receipts from our dynamic fixed-fee royalty agreements for existing licensees and higher cash operating expenses driven by the Technicolor Acquisitions. The table below provides the significant items comprising our cash flows provided by operating activities during the years ended December 31, 2019 and 2018 (in thousands).

	For the Year Ended December 31,		
	2019	2018	Increase / (Decrease)
Cash Receipts:			
Patent royalties	$ 288,123	$ 322,835	$ (34,712)
Technology solutions	7,053	2,537	4,516
Total cash receipts	$ 295,176	$ 325,372	$ (30,196)
Cash Outflows:			
Cash operating expenses (a)	(195,682)	(167,728)	(27,954)
Income taxes paid, net of refunds (b)	(24,229)	(16,426)	(7,803)

Total cash outflows	(219,911)	(184,154)	(35,757)
Other working capital adjustments	14,168	5,574	8,594
Cash flows provided by operating activities	$ 89,433	$ 146,792	$ (57,359)

(a) Cash operating expenses include operating expenses less depreciation of fixed assets, amortization of patents, non-cash compensation and non-cash changes in fair value.
(b) Income taxes paid include foreign withholding taxes. For the year ended December 31, 2018, this amount includes a net cash benefit of $17.5 million related to the Korea Competent Authority Proceeding discussed further above and within Note 14, "*Income Taxes*," in the consolidated financial statements.

Cash provided by or used in investing and financing activities

Net cash provided by investing activities in 2019 was $268.3 million, a $198.3 million change from $70.0 million net cash provided by investing activities in 2018. During 2019, we sold $296.6 million of short-term marketable securities, net of purchases. We also received initial proceeds of $10.0 million related to the sale of our Hillcrest product business, with a customary portion of the purchase price placed in escrow to secure potential indemnification claims. During 2018, we sold $256.6 million of short-term marketable securities, net of purchases, and applied a substantial portion of the proceeds from our sale of short-term marketable securities toward the $143.0 million, net of cash acquired, paid for the Technicolor Patent Acquisition. Long-term investments decreased by $6.3 million due to a decrease in strategic investment activity.

Net cash used in financing activities for 2019 was $89.3 million, a $71.7 million decrease from net cash used in financing activities of $161.1 million in 2018. This change was attributable to several offsetting factors. The second quarter 2019 debt refinancing, including the repayment of approximately 70% of our 2020 Notes, and related expenses resulted in net proceeds of $140.2 million during 2019. Additionally, proceeds from noncontrolling interests were $15.7 million and there was a $4.4 million decrease in payroll taxes paid upon the vesting of restricted stock units during 2019 as compared to 2018. Lastly, there was a $3.9 million decrease in dividends paid attributable to repurchases of common stock. These increases in cash were offset by a $85.8 million increase in repurchases of common stock and a $6.7 million decrease in proceeds received from the exercise of stock options.

Other

Our combined short-term and long-term deferred revenue balance at December 31, 2019 was approximately $270.3 million, an increase of $1.0 million from December 31, 2018. Based on current license agreements, we expect the amortization of dynamic fixed-fee royalty payments to reduce the December 31, 2019 deferred revenue balance of $270.3 million by $143.7 million over the next twelve months.

Convertible Notes

Our Convertible Notes are included in the dilutive earnings per share calculation using the treasury stock method. Under the treasury stock method, we must calculate the number of shares of common stock issuable under the terms of the Convertible Notes based on the average market price of our common stock during the applicable reporting period and include that number in the total diluted shares figure for the period. At the time we issued the Convertible Notes, we entered into the 2024 Call Spread Transactions and the 2020 Call Spread Transactions, respectively, (each as defined in the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K). The 2024 Call Spread Transactions and the 2020 Call Spread Transactions were designed to have the economic effect of reducing the net number of shares that will be issued in excess of the principal amount of converted Notes in the event of conversion of the Convertible Notes if the market price per share of our common stock is greater than the strike price of the 20204 Note Hedge Transactions or 2020 Note Hedge Transactions, as applicable, by, in effect, increasing the conversion price of the Convertible Notes from our economic standpoint. However, under GAAP, since the impact of the 2024 Note Hedge Transactions and 2020 Note Hedge Transactions (together, the "Note Hedge Transactions") is anti-dilutive, we exclude from the calculation of fully diluted shares the number of shares of our common stock that we would receive from the counterparties to these agreements upon settlement.

During periods in which the average market price of our common stock is above the applicable conversion price of the Convertible Notes ($81.29 per share for the 2024 Notes and $70.64 per share for the 2020 Notes as of December 31, 2019) or above the strike price of the warrants ($109.43 per share for the 2024 Warrant Transactions and $86.34 per share for the 2020 Warrant Transactions as of December 31, 2019), the impact of conversion or exercise, as applicable, would be dilutive and such dilutive effect is reflected in diluted earnings per share. As a result, in periods where the average market price of our common stock is above the conversion price or strike price, as applicable, under the treasury stock method, we calculate the number of shares issuable under the terms of the Convertible Notes and the warrants based on the average market price of the stock during the period, and includes that number in the total diluted shares outstanding for the period.

Under the treasury stock method, changes in the price per share of our common stock can have a significant impact on the number of shares that we must include in the fully diluted earnings per share calculation. As described in Note 10, "*Obligations*," it is our current intent and policy to settle all conversions of the Convertible Notes through a combination settlement of cash and shares of common stock, with a specified dollar amount of $1,000 per $1,000 principal amount of the Convertible Notes and any remaining amounts in shares ("net share settlement"). Assuming net share settlement upon conversion, the following table illustrates how, based on the $400.0 million aggregate principal amount of the 2024 Notes and the $94.9 million remaining aggregate principal amount of the 2020 Notes as of December 31, 2019, and the approximately 4.9 million warrants

related to the 2024 Notes and the 1.3 million remaining warrants related to the 2020 Notes, outstanding as of the same date, changes in our stock price would affect (i) the number of shares issuable upon conversion of the Convertible Notes, (ii) the number of shares issuable upon exercise of the warrants subject to the 2024 Warrant Transactions and 2020 Warrant Transactions (together, the "Warrant Transactions"), (iii) the number of additional shares deemed outstanding with respect to the Convertible Notes, after applying the treasury stock method, for purposes of calculating diluted earnings per

share ("Total Treasury Stock Method Incremental Shares"), (iv) the number of shares of common stock deliverable to us upon settlement of the Note Hedge Transactions, and (v) the number of shares issuable upon concurrent conversion of the Convertible Notes, exercise of the warrants subject to the Warrant Transactions, and settlement of the Note Hedge Transactions:

2024 Notes

Market Price Per Share	Shares Issuable Upon Conversion of the 2024 Notes	Shares Issuable Upon Exercise of the 2024 Warrant Transactions	Total Treasury Stock Method Incremental Shares	Shares Deliverable to InterDigital upon Settlement of the 2024 Note Hedge Transactions	Incremental Shares Issuable [(a)]
			(Shares in thousands)		
$85	215	—	215	(215)	—
$90	476	—	476	(476)	—
$95	710	—	710	(710)	—
$100	921	—	921	(921)	—
$105	1,111	—	1,111	(1,111)	—
$110	1,284	25	1,309	(1,284)	25
$115	1,442	238	1,680	(1,442)	238
$120	1,587	433	2,020	(1,587)	433
$125	1,721	613	2,334	(1,721)	613
$130	1,844	779	2,623	(1,844)	779

2020 Notes

Market Price Per Share	Shares Issuable Upon Conversion of 2020 Notes	Shares Issuable Upon Exercise of Warrants	Total Treasury Stock Method Incremental Shares	Shares Deliverable to InterDigital upon Settlement of the Hedge Agreements	Incremental Shares Issuable [(a)]
			(Shares in thousands)		
$75	78	—	78	(78)	—
$80	157	—	157	(157)	—
$85	227	—	227	(227)	—
$90	289	55	344	(289)	55
$95	344	122	466	(344)	122
$100	394	184	578	(394)	184
$105	440	239	679	(440)	239
$110	481	289	770	(481)	289
$115	518	335	853	(518)	335
$120	553	377	930	(553)	377

(a) Represents incremental shares issuable upon concurrent conversion of convertible notes, exercise of warrants and settlement of the hedge agreements.

Contractual Obligations

As discussed above, on June 3, 2019 we issued $400.0 million in aggregate principal amount of 2.00% Senior Convertible Notes due 2024, or the 2024 Notes. The 2024 Notes bear interest at a rate of 2.00% per year, payable in cash on June 1 and December 1 of each year, commencing on December 1, 2019, and mature on June 1, 2024, unless earlier converted or repurchased.

On March 11, 2015, we issued $316.0 million in aggregate principal amount of 1.50% Senior Convertible Notes due 2020, or the 2020 Notes. The 2020 Notes bear interest at a rate of 1.50% per year, payable in cash on March 1 and September 1 of each year, commencing September 1, 2015, and mature on March 1, 2020, unless earlier converted or repurchased. We used a portion of the proceeds from the issuance of the 2024 Notes to repurchase $221.1 million in aggregate principal amount of the 2020 Notes. As a result, $94.9 million in aggregate principal amount of the 2020 Notes remains outstanding as of December 31, 2019.

For more information on the 2024 Notes and 2020 Notes, see Note 10, "*Obligations*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K.

The following table summarizes our contractual obligations as of December 31, 2019 (in thousands):

	Payments Due by Period				
	Total	Less Than 1 year	1-3 Years	3-5 Years	Thereafter
2020 Notes	$ 94,909	$ 94,909	$ —	$ —	$ —
Contractual interest payments on the 2020 Notes	712	712	—	—	—
2024 Notes	400,000	—	—	400,000	—
Contractual interest payments on the 2024 Notes	35,356	8,000	16,000	11,356	—
Operating lease obligations	36,556	5,535	10,652	9,014	11,355
Defined benefit plan obligations (a)	3,261	94	416	816	1,935
Purchase obligations (b)	16,074	16,074	—	—	—
Total contractual obligations	$ 586,868	$ 125,324	$ 27,068	$ 421,186	$ 13,290

(a) Refer to Note 5, "*Business Combinations and Other Transactions*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for details of our defined benefit plan obligations. Estimated future benefit payments included above are through 2029.

(b) Purchase obligations consist of agreements to purchase goods and services that are legally binding on us, as well as accounts payable. Our consolidated balance sheet as of December 31, 2019 includes a $4.5 million noncurrent liability for uncertain tax positions. The future payments related to uncertain tax positions have not been presented in the table above due to the uncertainty of the amounts and timing of cash settlement with the taxing authorities.

As of December 31, 2019, we have recorded long-term debt of $21.1 million related to the Technicolor Patent Acquisition. Additionally, as part of the Technicolor Patent Acquisition, we committed to contributing cash, subject to certain requirements, of up to a maximum of $25.0 million to fund a collaborative arrangement related to the transaction. Lastly, we are subject to a revenue-sharing arrangement with Technicolor resulting from the Technicolor Acquisitions.

Refer to Note 5, "*Business Combinations and Other Transactions*," within the Notes to the Consolidated Financial Statements included in Part II, Item 8 of this Form 10-K for further information. Due to the uncertainty regarding the timing and amount of future payments related to these items, the amounts are excluded from the contractual obligations table above.

Off-Balance Sheet Arrangements

We do not have any off-balance sheet arrangements as defined by Item 303(a)(4) of Regulation S-K.

RESULTS OF OPERATIONS

2019 Compared with 2018

Revenues

The following table compares 2019 revenues to 2018 revenues (in thousands). Amounts below for the years ended December 31, 2019 and 2018 are presented in accordance with ASC 606.

	For the Year Ended December 31,			
	2019	2018	Total Increase/(Decrease)	
Variable patent royalty revenue	$ 30,428	$ 36,384	$ (5,956)	(16)%
Fixed-fee royalty revenue	257,221	239,347	17,874	7 %
Current patent royalties [a]	287,649	275,731	11,918	4 %
Non-current patent royalties [b]	19,782	26,329	(6,547)	(25)%
Total patent royalties	307,431	302,060	5,371	2 %
Current technology solutions revenue [a]	10,518	4,594	5,924	129 %
Patent sales	975	750	225	30 %
Total revenue	$ 318,924	$ 307,404	$ 11,520	4 %

(a) Recurring revenues are comprised of current patent royalties, inclusive of Dynamic Fixed-Fee Agreement royalties, and current technology solutions revenue.
(b) Non-recurring revenues are comprised of non-current patent royalties, which primarily include past patent royalties and royalties from static agreements, as well as patent sales.

The $11.5 million increase in total revenue was driven by an increase in recurring revenue of $17.8 million, primarily attributable to fixed-fee royalties and current technology solutions revenue. Fixed-fee royalty revenue increased by $17.9 million, primarily resulting from a dynamic fixed-fee agreement signed in each of the fourth quarters of 2018 and 2019. The increase in current technology solutions revenue related to the inclusion of engineering services revenue attributable to our on-going relationship with Technicolor. These increases were partially offset by a decrease variable patent royalties, which was primarily due to a restructured licensing arrangement with a long-term customer in first quarter 2019 whose revenues are now classified as fixed-fee royalty revenue and have declined as compared to prior year. This decrease in variable patent royalties was partially negated by the inclusion of variable patent royalties assumed as part of the Technicolor Patent Acquisition. Additionally, non-current patent royalties decreased by $6.5 million primarily due to a $5.5 million net charge recorded as contra non-recurring revenue during first quarter 2019 related to a restructured licensing arrangement with a long-term customer.

In 2019 and 2018, 70% and 71% of our total revenues, respectively, were attributable to companies that individually accounted for 10% or more of our total revenues. In 2019 and 2018, the following licensees or customers accounted for 10% or more of our total revenues:

	For the Year Ended December 31,	
	2019	2018
Apple	35%	36%
Samsung	25%	25%
LG	10%	10%

Operating Expenses

The following table summarizes the change in operating expenses by category (in thousands):

For the Year Ended December 31,		
2019	2018	Increase/(Decrease)

Patent administration and licensing	$ 154,940	$ 124,081	$ 30,859	25%
Development	74,860	69,698	5,162	7%
Selling, general and administrative	51,289	51,030	259	1%
Total operating expenses	$ 281,089	$ 244,809	$ 36,280	15%

Operating expenses increased 15% to $281.1 million in 2019 from $244.8 million in 2018. The $36.3 million increase in total operating expenses was primarily due to increases/(decreases) in the following items (in thousands):

	Increase/(Decrease)
Recurring operations of the Technicolor Acquisitions	$ 32,137
One-time costs related to the Technicolor Acquisitions	(9,325)
Revenue sharing for Madison Arrangement	6,260
Intellectual property enforcement and non-patent litigation	7,089
Personnel-related costs	1,203
Other	(1,084)
Total increase in operating expenses	$ 36,280

The $36.3 million increase in operating expenses was primarily driven by the Technicolor Acquisitions, which contributed $63.0 million to 2019 operating expenses following our May 2019 R&I Acquisition. This compares to $34.0 million of operating expenses in 2018 following our July 2018 Technicolor Patent Acquisition. The $63.0 million of operating expenses in 2019 resulting from the Technicolor Acquisitions is comprised of $48.3 million of recurring costs, of which $16.6 million relates to patent amortization, $8.4 million relates to transaction and integration costs during 2019, and the remaining $6.3 million relates to revenue sharing for the Madison Arrangement. The $34.0 million of operating expenses in 2018 resulting from the Technicolor Patent Acquisition was comprised of $16.2 million for five months of recurring costs, of which $6.8 million related to patent amortization, and the remaining $17.8 million related to transaction and integration costs. The $7.1 million increase in intellectual property enforcement and non-patent litigation was primarily due to the enforcement proceedings we initiated against Lenovo and Huawei in second half 2019. The increase in personnel-related costs was primarily related to severance and related expenses associated with ongoing efforts to optimize our cost structure, as well as one-time costs associated with the sale of our Hillcrest product business.

Patent administration and licensing expense: The $30.9 million increase in patent administration and licensing expense primarily resulted from the above-noted increases related to the Technicolor Acquisitions and intellectual property enforcement costs.

Development expense: The $5.2 million increase in development expense primarily resulted from the above-noted increases related to the Technicolor Acquisitions, as discussed above, partially offset by reduced spending on development of commercial solutions driven by the sale our Hillcrest product business.

Selling, general and administrative expense: The $0.3 million increase in selling, general and administrative expense primarily resulted from the above-noted increases related to the Technicolor Acquisitions and increased personnel-related costs, discussed above.

Non-Operating Income (Expense)

The following table compares 2019 non-operating income (expense) to 2018 non-operating income (expense) (in thousands):

	For the Year Ended December 31,			
	2019	2018	Increase / (Decrease)	
Interest expense	$ (40,955)	$ (35,956)	$ (4,999)	(14)%
Interest and investment income	14,991	14,590	401	3 %
Gain on asset acquisition and sale of business	22,690	—	22,690	— %

Loss on extinguishment of long-term debt	(5,488)	—	(5,488)	— %
Other income (expense), net	(3,131)	(9,171)	6,040	(66)%
Total non-operating income (expense)	$ (11,893)	$ (30,537)	$ 18,644	61 %

The change in non-operating income (expense) between periods was primarily driven by the recognition of an aggregate $22.7 million gain on asset acquisition and sale of business during the year ended December 31, 2019, of which $14.2 million relates to the R&I Acquisition in second quarter 2019 and $8.5 million relates to the gain on sale of our

Hillcrest product business in third quarter 2019. These gains were partially offset by the recognition of a $5.5 million loss on extinguishment of debt recognized in connection with the settlement of a portion of our 2020 Notes in second quarter 2019.

Additionally, during the year ended December 31, 2019, we recognized a net loss of $2.6 million resulting from the partial impairment of one of our strategic investments partially offset by a gain on sale of a separate strategic investment. During the year ended December 31, 2018, we recognized an aggregate $8.4 million loss resulting from the sale of our entire ownership interest in one of our strategic investments and the impairment of a separate strategic investment. These items are included in the "*Other income (expense), net*" caption in the table above. Higher interest expense relates to interest on the 2024 Notes and interest incurred on long-term debt resulting from the Technicolor Patent Acquisition.

Income Taxes

In 2019, based on the statutory federal tax rate net of discrete federal and state taxes, our effective tax rate was a provision of 42.4%. The effective tax rate for 2019 was unfavorably impacted by an $8.0 million provision associated with valuation allowances on the Company's losses in jurisdictions for which the Company receives no benefit. As a result of the difference in timing between US GAAP revenue and tax revenue, the Company's estimate of current taxable income is zero. The Company was unable to benefit from favorable rates associated with Foreign Derived Intangible Income ("FDII") as a result of having zero taxable income.

This is compared to an effective tax rate benefit of 85.5% in 2018, based on the statutory federal tax rate net of discrete federal and state taxes. The effective tax rate for 2018 was impacted by an $18.0 million benefit associated with the FDII deduction provisions contained within the Tax Cuts and Jobs Act, or TCJA, and a $14.7 million benefit from expected amended returns related to the Korea Competent Authority Proceeding settlement discussed above.

On March 6, 2019, the IRS issued proposed regulations for FDII. We are currently evaluating the impact of the proposed regulations and will record the impact, if any, as applicable when the regulations become finalized.

2018 Compared with 2017

Revenues

The following table compares 2018 revenues to 2017 revenues (in thousands). Amounts below for the year ended December 31, 2018 are presented in accordance with ASC 606 and amounts below for the year ended December 31, 2017 are presented in accordance with ASC 605.

	For the Year Ended December 31,				Components of Increase/(Decrease)		
	2018	2017	Total Increase/(Decrease)		Due to ASC 606	Operational	Total
Variable patent royalty revenue	$ 36,384	$ 47,840	$ (11,456)	(24)%	$ (461)	$ (10,995)	$ (11,456)
Fixed-fee royalty revenue	239,347	301,628	(62,281)	(21)%	(79,341)	17,060	(62,281)
Current patent royalties [a]	275,731	349,468	(73,737)	(21)%	(79,802)	6,065	(73,737)
Non-current patent royalties [b]	26,329	162,890	(136,561)	(84)%	10,000	(146,561)	(136,561)
Total patent royalties	302,060	512,358	(210,298)	(41)%	(69,802)	(140,496)	(210,298)
Current technology solutions revenue [a]	4,594	20,580	(15,986)	(78)%	(4,907)	(11,079)	(15,986)
Patent sales	750	—	750	— %	—	750	750
Total revenue	$ 307,404	$ 532,938	$ (225,534)	(42)%	$ (74,709)	$ (150,825)	$ (225,534)

(a) Recurring revenues are comprised of current patent royalties, inclusive of Dynamic Fixed-Fee Agreement royalties, and current technology solutions revenue.

(b) For the year ended December 31, 2018, non-recurring revenues are comprised of non-current patent royalties, which primarily include past patent royalties and royalties from static agreements, as well as patent sales. For the year ended December 31, 2017, non-current royalties consist of past patent royalties.

As discussed above, we adopted new revenue guidance, ASC 606, effective January 1, 2018. Consistent with the modified retrospective adoption method, our results of operations for periods prior to our adoption of ASC 606 remain unchanged. As a result, the difference in accounting principles attributable to the adoption of ASC 606 accounted for $74.7 million of the decrease in net revenue. This decrease was primarily related to pre-existing static fixed-fee license agreements.

The $150.8 million "Operational" decrease in total revenue was primarily driven by a decrease in non-current patent royalties. In 2017, non-current patent royalties were primarily attributable to the LG agreement, the recognition of a

prepayment balance remaining under a patent license agreement that expired in fourth quarter 2017 and our second quarter 2017 settlement agreement with Microsoft Corporation. The decreases in current technology solutions revenue and variable patent royalties primarily related to the expiration at the end of 2017 of certain royalty obligations under a technology solutions agreement and decreased shipments by certain of our variable licensees, respectively. These decreases were partially offset by the LG dynamic fixed-fee agreement signed in fourth quarter 2017 and new dynamic fixed-fee agreements signed during 2018.

In 2018 and 2017, 71% and 61% of our total revenues, respectively, were attributable to companies that individually accounted for 10% or more of our total revenues. In 2018 and 2017, the following licensees or customers accounted for 10% or more of our total revenues:

	For the Year Ended December 31,	
	2018	**2017**
Apple	36%	21%
Samsung	25%	13%
LG	10%	< 10%
Huawei [a]	—%	14%
BlackBerry [b]	—%	13%

(a) 2017 revenues included $8.4 million of non-current patent royalties.

(b) 2017 revenues included $70.7 million of non-current patent royalties.

Operating Expenses

The following table summarizes the change in operating expenses by category (in thousands):

	For the Year Ended December 31,			
	2018	**2017**	**Increase/(Decrease)**	
Patent administration and licensing	$ 124,081	$ 102,651	$ 21,430	21 %
Development	69,698	75,724	(6,026)	(8)%
Selling, general and administrative	51,030	53,068	(2,038)	(4)%
Total operating expenses	$ 244,809	$ 231,443	$ 13,366	6 %

Operating expenses increased 6% to $244.8 million in 2018 from $231.4 million in 2017. The $13.4 million increase in total operating expenses was primarily due to increases/(decreases) in the following items (in thousands):

	Increase/(Decrease)
Recurring operations of the Technicolor Patent Acquisition	$ 16,242
One-time costs related to the Technicolor Patent Acquisition	15,804
Intellectual property enforcement and non-patent litigation	2,605

Depreciation and amortization	2,072
Performance-based incentive compensation	(7,921)
Consulting services	(7,127)
Commercial initiatives	(3,738)
Personnel-related costs	(2,912)
Patent maintenance and evaluation	(2,067)
Other	408
Total increase in operating expenses	$ 13,366

The $13.4 million increase in operating expenses was primarily driven by the Technicolor Patent Acquisition, which increased 2018 operating expenses by $32.0 million. One-time transaction-related costs associated with the Technicolor Patent Acquisition increased $15.8 million. Additionally, the Technicolor Patent Acquisition contributed an additional $16.2 million for five months of operating expenses for the acquired Technicolor business, of which $6.8 million relates to patent

amortization. The $2.6 million increase in intellectual property enforcement and non-patent litigation was primarily due to increased activity related to existing licensee disputes. The $2.1 million increase of depreciation and amortization, which does not include the previously mentioned amortization from the Technicolor Patent Acquisition, was primarily related to the growth in our patent portfolio driven by both internal patent generation and patent acquisitions. The $7.9 million decrease in performance-based incentive compensation was primarily driven by higher accrual rates in the prior year. Consulting services decreased by $7.1 million, primarily related to spending on corporate initiatives, including the implementation of a new enterprise resource planning system in 2017. The $2.9 million decrease in personnel-related costs and the $3.7 million decrease in commercial initiatives were due to a reduction in headcount and reduced spending on the development of commercial solutions in an ongoing effort to optimize our cost structure. The $2.1 million decrease in patent maintenance and evaluation costs was a result of our initiatives to more efficiently prosecute and maintain our patent portfolio.

Patent administration and licensing expense: The $21.4 million increase in patent administration and licensing expense primarily resulted from the above-noted increases related to the Technicolor Patent Acquisition, intellectual property enforcement costs and patent amortization expense. These increases were partially offset by a decrease in performance-based compensation and patent maintenance costs.

Development expense: The $6.0 million decrease in development expense primarily resulted from the above-noted decreases in performance-based incentive compensation, personnel-related costs, commercial initiatives, as well as consulting services related to development projects.

Selling, general and administrative expense: The $2.0 million decrease in selling, general and administrative expense primarily resulted from the above-noted decreases in performance-based incentive compensation, consulting services, and personnel-related costs. These decreases were partially offset by the above-noted increases related to the Technicolor Patent Acquisition.

Non-Operating Income (Expense)

The following table compares 2018 non-operating income (expense) to 2017 non-operating income (expense) (in thousands):

	For the Year Ended December 31,			
	2018	2017	Increase / (Decrease)	
Interest expense	$ (35,956)	$ (17,845)	$ (18,111)	(101)%
Other income (expense), net	(9,171)	252	(9,423)	(3,739)%
Interest and investment income	14,590	8,488	6,102	72 %
Total non-operating income (expense)	$ (30,537)	$ (9,105)	$ (21,432)	(235)%

In 2018, non-operating expense was $30.5 million as compared to $9.1 million in 2017. The year ended December 31, 2018 includes $16.7 million of interest expense related to significant financing components of patent license agreements resulting from the adoption of ASC 606. Interest expense also increased by $0.7 million due to interest incurred on long-term debt resulting from the Technicolor Patent Acquisition. Non-operating expense for 2018 also includes an aggregate $8.4 million loss related to the sale of one of our strategic long-term investments and the impairment of a separate strategic long-term investment during the year, which is included in the "*Other income (expense), net*" caption in the table above. The remaining change between periods was primarily due to an increase in interest and investment income of $6.1 million primarily due to higher average investment balances and higher returns during 2018 as compared to 2017.

Income Taxes

In 2018, based on the statutory federal tax rate net of discrete federal and state taxes, our effective tax rate was a benefit of 85.5%. The effective tax rate for 2018 was favorably impacted by an $18.0 million benefit associated with the FDII deduction provisions contained within the Tax Cuts and Jobs Act ("TCJA") and a $14.7 million benefit from expected amended returns related to the Korea Competent Authority Proceeding settlement discussed above.

This is compared to an effective tax rate provision of 41.6% in 2017, based on the statutory federal tax rate net of discrete federal and state taxes. The effective tax rate for 2017 was impacted by a $42.6 million tax charge for the revaluation of our net deferred tax assets at the new statutory tax rate of 21% due to the TCJA signed into law in December 2017. The revaluation of our net deferred tax assets contributed approximately 14.6% to the rate increase, which was partially offset by a

contribution of approximately 4.0% due to our adoption of ASU 2016-09, "Improvements to Employee Share-Based Payment Accounting", as well as by a contribution of 2.7% as a result of the release of unrecognized tax benefits related to the conclusion of the IRS audits for tax years 2011 through 2015.

STATEMENT PURSUANT TO THE PRIVATE SECURITIES LITIGATION REFORM ACT OF 1995 - FORWARD-LOOKING STATEMENTS

This Annual Report on Form 10-K contains forward-looking statements within the meaning of Section 21E of the Securities Exchange Act of 1934, as amended. Such statements include certain information in “Part I, Item 1. Business” and “Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations” and other information regarding our current beliefs, plans and expectations, including, without limitation, the matters set forth below. Words such as “anticipate,” “estimate,” “expect,” “project,” “intend,” “plan,” “forecast,” “believe,” “could,” “would,” “should,” “if,” “may,” “might,” “future,” “target,” “goal,” “trend,” “seek to,” “will continue,” “predict,” “likely,” “in the event,” variations of any such words or similar expressions contained herein are intended to identify such forward-looking statements. Forward-looking statements in this Annual Report on Form 10-K include, without limitation, statements regarding:

(i) our objective to continue to be a leading designer and developer of technology solutions and innovation for the mobile, video and consumer electronics industries and to monetize those solutions and innovations through a combination of licensing, sales and other revenue opportunities;

(ii) our plans for executing on our business strategy, including our plans to develop and source innovative technologies related to wireless and video, establish and grow our patent-based revenue, pursue commercial opportunities for our advanced platforms and solutions, and maintain a collaborative relationship with key industry players and worldwide standards bodies;

(iii) our belief that our portfolio includes a number of patents and patent applications that are or may be essential or may become essential to cellular, other wireless and video standards, including 3G, 4G, 5G and the IEEE 802 suite of standards, as well as patents and patent applications that we believe may become essential to standards that are under development;

(iv) our belief that a number of our CDMA and OFDM/OFDMA inventions are, may be or may become essential to the implementation of CDMA and OFDM/OFDMA-based systems in use today;

(v) our belief that companies making, importing, using or selling products compliant with the standards covered by our patent portfolio require a license under our patents and will require licenses under patents that may issue from our pending patent applications;

(vi) our belief that our ongoing research efforts and associated patenting activities enable us to sell patent assets that are not vital to our core licensing programs, as well as to execute patent swaps that can strengthen our overall portfolio;

(vii) our belief that our commercial initiatives are potential revenue opportunities;

(viii) the estimated growth of the IoT market, including the size of the connected device installed base and number of connected device shipments, over the next several years;

(ix) the types of licensing arrangements and various royalty structure models that we anticipate using under our future license agreements;

(x) the possible outcome of audits of our license agreements when underreporting or underpayment is revealed;

(xi) our belief that our facilities are suitable and adequate for our present purposes and our needs in the near future;

(xii) our expectations and estimations regarding the income tax effects, and the impact on the Company, of the Tax Cuts and Jobs Act, or TCJA, and our belief that we currently expect a significant portion of our income to qualify as FDII and thus be subject to the 13.1% tax rate;

(xiii) our expectation that we will continue to pay a quarterly cash dividend on our common stock comparable to our quarterly $0.35 per share cash dividend in the future;

(xiv) our belief that intellectual property enforcement costs, including litigation costs, will likely continue to be a significant expense for us in the future;

(xv) our belief that we have the ability to obtain additional liquidity through debt and equity financings;

(xvi) our belief that our available sources of funds will be sufficient to finance our operations, capital requirements, debt obligations, existing stock repurchase program and dividend program for the next twelve months;

(xvii) our expectations regarding the potential effects of new accounting standards on our financial statements or results of operations;

(xviii) our expectation that the amortization of fixed-fee royalty payments will reduce our deferred revenue balance over the next twelve months;

(xix) our belief in our ability to continue to expand into the consumer electronics market, and the opportunities that market presents;

(xx) our projections of amounts to be owed to Technicolor under our revenue sharing arrangement;

(xxi) the expected timing, outcome and impact of our various litigation, arbitration and administrative matters; and

(xxii) our belief that there will be a level of concentration in worldwide shipments of 5G handsets similar to the current level of concentration in worldwide shipments of 3G and 4G handsets.

Although the forward-looking statements in this Form 10-K reflect the good faith judgment of our management, such statements can only be based on facts and factors currently known by us. Consequently, forward-looking statements concerning our business, results of operations and financial condition are inherently subject to risks and uncertainties. We caution readers that actual results and outcomes could differ materially from those expressed in or anticipated by such forward-looking statements due to a variety of factors, including, without limitation, the following:

(i) uncertainty and decline in U.S.-China relations and/or increased economic uncertainty in China;

(ii) unanticipated difficulties or delays related to the further development of our technologies;

(iii) the failure of the markets for our technologies to materialize to the extent or at the rate that we expect;

(iv) changes in our plans, strategy or initiatives;

(v) the challenges related to entering into new and renewed patent license agreements and unanticipated delays, difficulties or acceleration in the negotiation and execution of patent license agreements;

(vi) our ability to leverage our strategic relationships and secure new patent license and technology solutions agreements on acceptable terms;

(vii) the impact of current trends in the industry that could result in reductions in and/or caps on royalty rates under new patent license agreements;

(viii) changes in the market share and sales performance of our primary licensees, delays in product shipments of our licensees, delays in the timely receipt and final reviews of quarterly royalty reports from our licensees, delays in payments from our licensees and related matters;

(ix) the timing and/or outcome of our various litigation, arbitration, regulatory or administrative proceedings, including any awards or judgments relating to such proceedings, additional legal proceedings, changes in the schedules or costs associated with legal proceedings or adverse rulings in such legal proceedings;

(x) the determination of royalty rates, or other terms, under our patent license agreements through arbitration or other third-party adjudications, or the establishment by arbitrators or other third-party adjudicators of patent royalty rates at levels lower than our agreed or historical rates;

(xi) the impact of potential patent legislation, USPTO rule changes and international patent rule changes on our patent prosecution and licensing strategies;

(xii) the impact of rulings in legal proceedings, potential legislation affecting the jurisdiction and authority of the USITC and potential changes to the IPR policies of worldwide standards bodies on our investments in research and development and our strategies for patent prosecution, licensing and enforcement;

(xiii) changes in our interpretations of, and assumptions and calculations with respect to the impact on the Company of, the Tax Cuts and Jobs Act, or TCJA, as well as further guidance that may be issued regarding the TCJA;

(xiv) the timing and/or outcome of any state or federal tax examinations or audits, changes in tax laws and the resulting impact on our tax assets and liabilities;

(xv) the effects of any dispositions, acquisitions or other strategic transactions by the Company;

(xvi) decreased liquidity in the capital markets; and

(xvii) unanticipated increases in our cash needs or decreases in available cash.

You should carefully consider these factors as well as the risks and uncertainties outlined in greater detail in Part I, Item 1A, in this Form 10-K before making any investment decision with respect to our common stock. These factors, individually or in the aggregate, may cause our actual results to differ materially from our expected and historical results. You should understand that it is not possible to predict or identify all such factors. In addition, you should not place undue reliance on the forward-looking statements contained herein, which are made only as of the date of this Form 10-K. We undertake no obligation to revise or update publicly any forward-looking statement for any reason, except as otherwise required by law.

Item 7A. *QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK.*

Cash, cash equivalents, restricted cash and short-term investments

The primary objectives of our investment activities are to preserve principal and maintain liquidity while at the same time capturing a market rate of return. To achieve these objectives, we maintain our portfolio of cash, cash equivalents, restricted cash and short-term and long-term investments in a variety of securities, including government obligations, corporate bonds and commercial paper.

Interest Rate Risk — We invest our cash in a number of diversified high quality investment-grade fixed and floating rate securities with a fair value of $0.9 billion as of December 31, 2019. Our exposure to interest rate risks is not significant due to the short average maturity, quality and diversification of our holdings. We do not hold any derivative, derivative commodity instruments or other similar financial instruments in our investment portfolio. The risk associated with fluctuating interest rates is generally limited to our investment portfolio. We believe that a hypothetical 10% change in period-end interest rates would not have a significant impact on our results of operations or cash flows.

The following table provides information about our interest-bearing securities that are sensitive to changes in interest rates as of December 31, 2019. The table presents principal cash flows, weighted-average yield at cost and contractual maturity dates. Additionally, we have assumed that these securities are similar enough within the specified categories to aggregate these securities for presentation purposes.

Interest Rate Sensitivity
Principal Amount by Expected Maturity
Average Interest Rates
(in thousands)

	2020	2021	2022	2023	2024	Thereafter	Total
Money market and demand accounts	$757,098	—	—	—	—	—	$757,098
Short-term investments	$163,108	$16,096	—	—	—	—	$179,204
Average Interest rate	1.9%	2.2%	—%	—%	—%	—%	1.9%

Cash and cash equivalents and available-for-sale securities are recorded at fair value.

Bank Liquidity Risk — As of December 31, 2019, we had approximately $757.1 million in operating accounts that are held with domestic and international financial institutions. The majority of these balances are held with domestic financial institutions. While we monitor daily cash balances in our operating accounts and adjust the cash

balances as appropriate, these cash balances could be lost or become inaccessible if the underlying financial institutions fail or if they are unable to meet the liquidity requirements of their depositors. Notwithstanding, we have not incurred any losses and have had full access to our operating accounts to date.

Foreign Currency Exchange Rate Risk — We are exposed to limited risk from fluctuations in currencies, which might change over time as our business practices evolve, that could impact our operating results, liquidity and financial condition. We operate and invest globally. Adverse movements in currency exchange rates might negatively affect our business due to a number of situations. Currently, our international licensing agreements are typically made in U.S. dollars and are generally not subject to foreign currency exchange rate risk. We do not engage in foreign exchange hedging transactions at this time.

Between 2006 and 2019, we paid approximately $177.4 million in foreign taxes to foreign governments that have tax treaties with the U.S., for which we have claimed foreign tax credits against our U.S. tax obligations, and for which the tax treaty procedures are still open. It is possible that as a result of tax treaty procedures, the U.S. government may reach an

agreement with the related foreign governments that will result in a partial refund of foreign taxes paid with a related reduction in our foreign tax credits. Due to foreign currency fluctuations, any such agreement could result in foreign currency gain or loss.

Investment Risk — We are exposed to market risk as it relates to changes in the market value of our short-term and long-term investments in addition to the liquidity and creditworthiness of the underlying issuers of our investments. We hold a diversified investment portfolio, which includes, fixed and floating-rate, investment-grade marketable securities, mortgage and asset-backed securities and U.S. government and other securities. The instruments included in our portfolio meet high credit quality standards, as specified in our investment policy guidelines. This policy also limits our amount of credit exposure to any one issue, issuer and type of instrument. Given that the guidelines of our investment policy prohibit us from investing in anything but highly rated instruments, our investments are not subject to significant fluctuations in fair value due to the volatility of the credit markets and prevailing interest rates for such securities. Our marketable securities, consisting of government obligations, corporate bonds and commercial paper, are primarily classified as available-for-sale with a fair value of $179.2 million as of December 31, 2019.

Equity Risk — We are exposed to changes in the market-traded price of our common stock as it influences the calculation of earnings per share. In connection with the offering of the 2024 Notes and the 2020 Notes, we entered into convertible note hedge transactions with option counterparties. We also sold warrants to the option counterparties. These transactions have been accounted for as an adjustment to our shareholders' equity. The convertible note hedge transactions are expected to reduce the potential equity dilution upon conversion of the 2024 Notes and the 2020 Notes. The warrants along with any shares issuable upon conversion of the 2024 Notes and the 2020 Notes will have a dilutive effect on our earnings per share to the extent that the average market price of our common stock for a given reporting period exceeds the applicable strike price or conversion price of the warrants or convertible 2024 Notes and 2020 Notes, respectively.

Item 8. ***FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA.***

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All other schedules are omitted because they are either not required or applicable or equivalent information has been included in the financial statements and notes thereto.

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Shareholders of InterDigital, Inc.

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of InterDigital, Inc. and its subsidiaries (the "Company") as of December 31, 2019 and 2018, and the related consolidated statements of income, comprehensive income, shareholders' equity, and cash flows for each of the three years in the period ended December 31, 2019, including the related notes and financial statements schedule listed in the accompanying index (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2019, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2019 and 2018, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2019 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2019, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the COSO.

Changes in Accounting Principles

As discussed in Note 2 to the consolidated financial statements, the Company changed the manner in which it accounts for leases in 2019 and the manner in which it accounts for revenue in 2018.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in Management's Annual Report on Internal Control Over Financial Reporting appearing under Item 9A. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and

expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Revenue Recognition - Determination of the Value of Revenue from Non-Financial Sources and of Standalone Selling Prices of Identified Performance Obligations in Dynamic Fixed-Fee License Agreements

As described in Notes 2 and 3 to the consolidated financial statements, dynamic fixed-fee license agreements include fixed, non-refundable royalty payments that fulfill the licensee's obligations to the Company under a patent license agreement for a specified time period or for the term of the agreement. Additionally, certain patent license agreements contain revenue from non-financial sources in the form of patents received from the customer. Total fixed-fee royalty revenue and non-current patent royalties were $257.2 million and $19.8 million, respectively, for the year ended December 31, 2019, of which a significant portion relates to dynamic fixed-fee agreements. As disclosed by management, management's process for determining the value of the standalone selling prices of identified performance obligations in dynamic fixed-fee license agreements requires the exercise of significant judgment when evaluating the valuation methods and assumptions, including the assumed royalty rates, projected sales volumes, discount rate and comparable market transactions which are not directly observable and other relevant factors. Management's process for determining the value of revenue from non-financial sources requires estimating the fair value of patents received using one, or a combination of, an analysis of comparable market transactions (the market approach), a discounted cash flow analysis (the income approach), and/or by quantifying the amount of money required to replace the future service capability of the assets (the cost approach). The development of a number of these inputs and assumptions requires a significant amount of management judgment and is based upon a number of factors, including comparable market transactions, assumed royalty rates, projected sales volumes, economic lives of the patents and other relevant factors.

The principal considerations for our determination that performing procedures relating to the determination of the value of revenue from non-financial sources and of standalone selling prices of identified performance obligations in dynamic fixed-fee license agreements is a critical audit matter are there was significant judgment by management in determining the value of the revenue from non-financial sources and standalone selling prices. This in turn led to a high degree of auditor judgment, subjectivity and effort in performing procedures and in evaluating evidence related to the significant assumptions made by management to establish the value of revenue from non-financial sources and standalone selling prices, including the assumed royalty rates, projected sales volumes, discount rate and comparable market transactions. In addition, the audit effort involved the use of professionals with specialized skill and knowledge to assist in performing these procedures and evaluating the audit evidence obtained.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the revenue recognition process, including controls over the determination of the value of revenue from non-financial sources and standalone selling prices of identified performance obligations in dynamic fixed-fee license agreements. These procedures also included, among others (i) obtaining and reading a selection of new dynamic fixed-fee license agreements entered into during the year and testing management's process for determining the value of revenue from non-financial sources and standalone selling prices of identified performance obligations in dynamic fixed-fee license agreements and (ii) evaluating the appropriateness of the valuation methods and reasonableness of significant assumptions used in determining the value of revenue from non-financial sources and developing the standalone selling prices, including assumed royalty rates, projected sales volumes, discount rate and comparable market transactions. Evaluating the reasonableness of management's significant

assumptions related to assumed royalty rates, discount rate and comparable market transactions involved considering prospective third-party market data and previous license agreements entered into by the Company and the consistency of the projected sales volume with historical sales data. Professionals with specialized skill and knowledge were used to assist in the

evaluation of the valuation methods and certain significant assumptions, including comparable market transactions used to estimate the value of revenue from non-financial sources.

/s/ PricewaterhouseCoopers LLP

Philadelphia, Pennsylvania
February 20, 2020

We have served as the Company's auditor since 2002.

INTERDIGITAL, INC. AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS
(in thousands, except per share data)

	DECEMBER 31, 2019	DECEMBER 31, 2018
ASSETS		
CURRENT ASSETS:		
Cash and cash equivalents	$ 745,491	$ 475,056
Short-term investments	179,204	470,724
Accounts receivable, less allowances of $537 and $693	28,272	35,032
Prepaid and other current assets	63,365	43,438
Total current assets	1,016,332	1,024,250
PROPERTY AND EQUIPMENT, NET	10,217	10,051
PATENTS, NET	436,339	454,567
DEFERRED TAX ASSETS	73,168	77,225
OTHER NON-CURRENT ASSETS	76,026	60,465
	595,750	602,308
TOTAL ASSETS	$ 1,612,082	$ 1,626,558
LIABILITIES AND SHAREHOLDERS' EQUITY		
CURRENT LIABILITIES:		
Current portion of long-term debt	$ 94,170	$ —
Accounts payable	13,393	19,367
Accrued compensation and related expenses	29,162	26,838
Deferred revenue	146,654	111,672
Taxes payable	51	1,508
Dividend payable	10,746	11,627
Other accrued expenses	11,382	8,383
Total current liabilities	305,558	179,395
LONG-TERM DEBT	350,588	317,377
LONG-TERM DEFERRED REVENUE	123,653	157,634
OTHER LONG-TERM LIABILITIES	46,002	34,139
TOTAL LIABILITIES	825,801	688,545
COMMITMENTS AND CONTINGENCIES		

SHAREHOLDERS' EQUITY:		
Preferred Stock, $0.10 par value, 14,399 shares authorized, 0 shares issued and outstanding	—	—
Common Stock, $0.01 par value, 100,000 shares authorized, 71,268 and 71,134 shares issued and 30,701 and 33,529 shares outstanding	712	711
Additional paid-in capital	727,402	685,512
Retained earnings	1,412,779	1,435,970
Accumulated other comprehensive loss	(74)	(2,471)
	2,140,819	2,119,722
Treasury stock, 40,567 and 37,605 shares of common held at cost	1,379,262	1,182,993
Total InterDigital, Inc. shareholders' equity	761,557	936,729
Noncontrolling interest	24,724	1,284
Total equity	786,281	938,013
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY	$ 1,612,082	$ 1,626,558

The accompanying notes are an integral part of these statements.

INTERDIGITAL, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF INCOME
(in thousands, except per share data)

	FOR THE YEAR ENDED DECEMBER 31,		
	2019	**2018**	**2017**
REVENUES:			
Patent licensing royalties	$ 307,431	$ 302,060	$ 512,358
Patent sales	975	750	—
Technology solutions	10,518	4,594	20,580
Total Revenue	318,924	307,404	532,938
OPERATING EXPENSES:			
Patent administration and licensing	154,940	124,081	102,651
Development	74,860	69,698	75,724
Selling, general and administrative	51,289	51,030	53,068
Total Operating Expenses	281,089	244,809	231,443
Income from operations	37,835	62,595	301,495
Interest expense	(40,955)	(35,956)	(17,845)
OTHER INCOME (EXPENSE), NET	29,062	5,419	8,740
Income before income taxes	25,942	32,058	292,390
INCOME TAX BENEFIT (PROVISION)	(10,991)	27,417	(121,676)
NET INCOME	$ 14,951	$ 59,475	$ 170,714
Net loss attributable to noncontrolling interest	(5,977)	(5,556)	(5,506)
NET INCOME ATTRIBUTABLE TO INTERDIGITAL, INC.	$ 20,928	$ 65,031	$ 176,220
NET INCOME PER COMMON SHARE — BASIC	$ 0.66	$ 1.89	$ 5.09
WEIGHTED AVERAGE NUMBER OF COMMON SHARES OUTSTANDING — BASIC	31,546	34,491	34,605
NET INCOME PER COMMON SHARE — DILUTED	$ 0.66	$ 1.84	$ 4.93
WEIGHTED AVERAGE NUMBER OF COMMON SHARES OUTSTANDING — DILUTED	31,785	35,307	35,779
CASH DIVIDENDS DECLARED PER COMMON SHARE	$ 1.40	$ 1.40	$ 1.30

The accompanying notes are an integral part of these statements.

INTERDIGITAL, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME
(in thousands)

	For the Year Ended December 31,		
	2019	**2018**	**2017**
Net income	$ 14,951	$ 59,475	$ 170,714
Unrealized gain (loss) on investments, net of tax	2,397	61	(1,569)
Comprehensive income	$ 17,348	$ 59,536	$ 169,145
Comprehensive loss attributable to noncontrolling interest	(5,977)	(5,556)	(5,506)
Total comprehensive income attributable to InterDigital, Inc.	$ 23,325	$ 65,092	$ 174,651

The accompanying notes are an integral part of these statements.

INTERDIGITAL, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF SHAREHOLDERS' EQUITY
(in thousands, except per share data)

	Common Stock		Additional Paid-In Capital	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Treasury Stock		Non-Controlling Interest	Total Shareholders' Equity
	Shares	Amount				Shares	Amount		
BALANCE, DECEMBER 31, 2016	**70,318**	**$ 703**	**$ 683,549**	**$ 1,127,380**	**$ (514)**	**36,020**	**$ (1,064,795)**	**$ 8,045**	**$ 754,368**
Net income attributable to InterDigital, Inc.	—	—	—	176,220	—	—	—	—	176,220
Proceeds from noncontrolling interests	—	—	—	—	—	—	—	6,801	6,801
Net (loss) income attributable to noncontrolling interest	—	—	—	—	—	—	—	(5,506)	(5,506)
Net change in unrealized gain (loss) on short-term investments	—	—	—	—	(1,569)	—	—	—	(1,569)
Dividends Declared ($1.30 per share)	—	—	846	(45,968)	—	—	—	—	(45,122)
Exercise of Common Stock options	9	1	381	—	—	—	—	—	382
Issuance of Common Stock, net	422	3	(22,798)	—	—	—	—	—	(22,795)
Amortization of unearned compensation	—	—	18,062	—	—	—	—	—	18,062
Repurchase of Common Stock	—	—	—	—	—	107	(7,693)	—	(7,693)
BALANCE, DECEMBER 31, 2017	**70,749**	**$ 707**	**$ 680,040**	**$ 1,257,632**	**$ (2,083)**	**36,127**	**$ (1,072,488)**	**$ 9,340**	**$ 873,148**
Cumulative effect of change in accounting principle	—	—	—	161,701	(449)	—	—	—	161,252
Net income attributable to InterDigital, Inc.	—	—	—	65,031	—	—	—	—	65,031
Distribution preference	—	—	—	—	—	—	—	(2,500)	(2,500)
Net (loss) income attributable to noncontrolling interest	—	—	—	—	—	—	—	(5,556)	(5,556)
Net change in unrealized gain (loss) on short-term investments	—	—	—	—	61	—	—	—	61
Dividends Declared ($1.40 per share)	—	—	472	(48,394)	—	—	—	—	(47,922)
Exercise of Common Stock options and warrants	153	2	6,721	—	—	—	—	—	6,723
Issuance of Common Stock, net	232	2	(8,810)	—	—	—	—	—	(8,808)
Amortization of unearned compensation	—	—	7,089	—	—	—	—	—	7,089
Repurchase of Common Stock	—	—	—	—	—	1,478	(110,505)	—	(110,505)
BALANCE, DECEMBER 31, 2018	**71,134**	**$ 711**	**$ 685,512**	**$ 1,435,970**	**$ (2,471)**	**37,605**	**$ (1,182,993)**	**$ 1,284**	**$ 938,013**
Net income attributable to InterDigital, Inc.	—	—	—	20,928	—	—	—	—	20,928
Proceeds from and increases in noncontrolling interests	—	—	—	—	—	—	—	29,417	29,417
Net (loss) income attributable to noncontrolling interest	—	—	—	—	—	—	—	(5,977)	(5,977)
Net change in unrealized gain (loss) on short-term investments	—	—	—	—	2,397	—	—	—	2,397
Dividends Declared ($1.40 per share)	—	—	401	(44,119)	—	—	—	—	(43,718)
Exercise of common stock options	—	—	2	—	—	—	—	—	2
Issuance of Common Stock, net	134	1	(4,368)	—	—	—	—	—	(4,367)

Amortization of unearned compensation	—	—	7,603	—	—	—	—	—	7,603
Repurchase of Common Stock	—	—	—	—	—	2,962	(196,269)	—	(196,269)
Equity component of debt, net of tax	—	—	56,917	—	—	—	—	—	56,917
Net convertible note hedge transactions, net of tax	—	—	(49,740)	—	—	—	—	—	(49,740)
Net warrant transactions	—	—	43,416	—	—	—	—	—	43,416
Deferred financing costs allocated to equity, net of tax	—	—	(1,692)	—	—	—	—	—	(1,692)
Reacquisition of equity component of debt due to prepayment, net of tax	—	—	(10,649)	—	—	—	—	—	(10,649)
BALANCE, DECEMBER 31, 2019	**71,268**	**$ 712**	**$ 727,402**	**$ 1,412,779**	**$ (74)**	**40,567**	**$ (1,379,262)**	**$ 24,724**	**$ 786,281**

The accompanying notes are an integral part of these statements

INTERDIGITAL, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF CASH FLOWS
(in thousands)

	FOR THE YEAR ENDED DECEMBER 31,		
	2019	**2018**	**2017**
CASH FLOWS FROM OPERATING ACTIVITIES:			
Net income	$ 14,951	$ 59,475	$ 170,714
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation and amortization	77,094	66,108	57,053
Non-cash interest expense, net	18,709	13,509	13,105
Non-cash change in fair value	710	3,884	—
Gain on asset acquisition and sale of business	(22,690)	—	—
Change in deferred revenue	(7,749)	6,966	(36,892)
Deferred income taxes	4,123	(45,426)	64,950
Share-based compensation	7,603	7,089	18,062
Impairment of long-term investment	3,312	200	—
Loss on extinguishment of debt	5,488	—	—
Loss (gain) on disposal of assets	119	8,323	—
Other	623	(625)	(2)
(Increase) decrease in assets:			
Receivables	6,742	31,615	12,171
Deferred charges and other assets	(27,206)	(6,065)	19,426
Increase (decrease) in liabilities:			
Accounts payable	(638)	6,203	(3,789)
Accrued compensation and other expenses	9,699	254	(3,218)
Accrued taxes payable and other tax contingencies	(1,457)	(4,718)	4,220
Net cash provided by operating activities	89,433	146,792	315,800
CASH FLOWS FROM INVESTING ACTIVITIES:			
Purchases of short-term investments	(92,436)	(142,555)	(930,016)
Sales of short-term investments	389,032	399,105	751,308
Purchases of property and equipment	(4,509)	(2,576)	(2,071)
Capitalized patent costs	(33,481)	(32,069)	(34,933)
Acquisition of patents	—	(2,250)	—
Acquisition of business, net of cash acquired	—	(142,985)	—
Proceeds from sale of business	10,000	—	—
Long-term investments	(350)	(6,686)	(4,585)

Net cash provided by (used in) investing activities	268,256	69,984	(220,297)
CASH FLOWS FROM FINANCING ACTIVITIES:			
Net proceeds from exercise of stock options	2	6,723	382
Proceeds from issuance of senior convertible notes	400,000	—	—
Payments on long-term debt	(221,091)	—	—
Purchase of convertible bond hedge	(72,000)	—	—
Payment for warrant unwind	(4,184)	—	—
Prepayment penalty on long term debt	(10,763)	—	—
Proceeds from hedge unwind	9,038	—	—
Proceeds from issuance of warrants	47,600	—	—
Payments of debt issuance costs	(8,375)	—	—
Proceeds from non-controlling interests	15,666	—	6,801
Dividends paid	(44,580)	(48,468)	(43,255)
Shares withheld for taxes	(4,368)	(8,807)	(22,798)
Repurchase of common stock	(196,269)	(110,505)	(7,693)
Net cash used in financing activities	(89,324)	(161,057)	(66,563)
NET INCREASE (DECREASE) IN CASH, CASH EQUIVALENTS AND RESTRICTED CASH	268,365	55,719	28,940
CASH, CASH EQUIVALENTS AND RESTRICTED CASH, BEGINNING OF PERIOD	488,733	433,014	404,074
CASH, CASH EQUIVALENTS AND RESTRICTED CASH, END OF PERIOD	$ 757,098	$ 488,733	$ 433,014

Refer to Note 1, "Background and Basis of Presentation," for additional supplemental cash flow information. Additionally, refer to Note 6, "Cash, Cash Equivalents, Restricted Cash and Marketable Securities" for a reconciliation to the consolidated balance sheets and Note 17, "Leases" for information regarding the impact of our adoption of the new leases accounting standard, ASC 842, on January 1, 2019.

The accompanying notes are an integral part of these statements.

INTERDIGITAL, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
December 31, 2019

1. BACKGROUND AND BASIS OF PRESENTATION

InterDigital designs and develops advanced technologies that enable and enhance wireless communications and capabilities. Since our founding in 1972, our engineers have designed and developed a wide range of innovations that are used in digital cellular and wireless products and networks, including 2G, 3G, 4G and IEEE 802-related products and networks, as well as video processing, coding and display technology. We are a leading contributor of innovation to the wireless communications industry, as well as a leading holder of patents in the video industry.

Principles of Consolidation

The accompanying consolidated financial statements include all of our accounts and all entities in which we have a controlling interest and/or are required to be consolidated in accordance with the Generally Accepted Accounting Principles in the United States ("GAAP"). All significant intercompany accounts and transactions have been eliminated in consolidation.

In determining whether we are the primary beneficiary of a variable interest entity and therefore required to consolidate, we apply a qualitative approach that determines whether we have both the power to direct the economically significant activities of the entity and the obligation to absorb losses of, or the right to receive benefits from, the entity that could potentially be significant to that entity. These considerations impact the way we account for our existing collaborative relationships and other arrangements. We continuously assess whether we are the primary beneficiary of a variable interest entity as changes to existing relationships or future transactions may result in us consolidating or deconsolidating our partner(s) to collaborations and other arrangements.

Use of Estimates

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities, the disclosure of contingent assets and liabilities as of the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. Actual results could differ from these estimates. If different assumptions were made or different conditions had existed, our financial results could have been materially different.

Reclassifications

Certain reclassifications have been made to prior year amounts to conform to the current year presentation.

Prior Periods Financial Statement Revision

In connection with the preparation of the condensed consolidated financial statements for first quarter 2019, it was identified that we incorrectly attributed tax benefit to the net loss attributable to noncontrolling interest in our presentation of noncontrolling interest.

We assessed the materiality of this misstatement on prior periods' financial statements in accordance with ASC Topic 250, Accounting Changes and Error Corrections, ("ASC 250") and concluded it was not material to any prior annual or interim periods. In accordance with ASC 250, we have corrected our presentation of noncontrolling interest

for all prior periods presented in this Form 10-K by revising the consolidated financial statements and other consolidated financial information included herein. We will continue to present the prior periods on this revised basis to the extent we present such prior periods in future filings. Refer to Note 21, "*Revision to Noncontrolling Interest*" for additional information on the revision.

Supplemental Cash Flow Information

The following table presents additional supplemental cash flow information for the year ended December 31, 2019, 2018 and 2017 (in thousands):

SUPPLEMENTAL CASH FLOW INFORMATION:	FOR THE YEAR ENDED DECEMBER 31, 2019	2018	2017
Interest paid	$ 7,886	$ 4,740	$ 4,740
Income taxes paid, including foreign withholding taxes	24,229	33,904	66,793
Non-cash investing and financing activities:			
Dividend payable	10,746	11,627	12,156
Increases in noncontrolling interests	13,750	—	—
Non-cash acquisition of patents	22,500	—	32,500
Accrued capitalized patent costs and property and equipment	1,619	(2,789)	1

2. SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES AND NEW ACCOUNTING GUIDANCE

Foreign Currency Translation

The functional currency of substantially all of the Company's wholly-owned subsidiaries is the U.S. dollar. Certain subsidiaries have monetary assets and liabilities that are denominated in a currency that is different than the functional currency. The gains and losses resulting from this remeasurement and translation of monetary assets denominated in a currency that is different than the functional currency are reflected in the determination of net income (loss).

Cash, Cash Equivalents, Restricted Cash and Marketable Securities

We classify all highly liquid investment securities with original maturities of three months or less at date of purchase as cash equivalents. Cash that is held for a specific purpose and therefore not available to the Company for immediate or general business use is classified as restricted cash. Our investments are comprised of mutual and exchange traded funds, commercial paper, United States and municipal government obligations and corporate securities. Management determines the appropriate classification of our investments at the time of acquisition and re-evaluates such determination at each balance sheet date.

As of December 31, 2019 and 2018, the majority of our marketable securities have been classified as available-for-sale and are carried at fair value, with unrealized gains and losses reported net-of-tax as a separate component of shareholders' equity. Substantially all of our investments are investment grade government and corporate debt securities that have maturities of less than 2 years, and we have both the ability and intent to hold the investments until maturity.

Other-than-Temporary Impairments

We review our investment portfolio during each reporting period to determine whether there are identified events or circumstances that would indicate there is a decline in the fair value that is considered to be other-than-temporary. For non-public investments, if there are no identified events or circumstances that would have a significant adverse effect on the fair value of the investment, then the fair value is not estimated. If an investment is deemed to have experienced an other-than-temporary decline below its cost basis, we reduce the carrying amount of the investment to its quoted or estimated fair value, as applicable, and establish a new cost basis for the investment. We charge the impairment to the "*Other Income (Expense), Net"* line of our consolidated statements of income.

Intangible Assets

Patents

We capitalize external costs, such as filing fees and associated attorney fees, incurred to obtain issued patents and patent license rights. We expense costs associated with maintaining and defending patents subsequent to their issuance in the period incurred. We amortize capitalized patent costs for internally generated patents on a straight-line basis over 10 years, which represents the estimated useful lives of the patents. The ten-year estimated useful life for internally generated patents is based on our assessment of such factors as: the integrated nature of the portfolios being licensed, the overall makeup of the portfolio over time, and the length of license agreements for such patents. The estimated useful lives of acquired patents and patent rights, however, have been and will continue to be based on a separate analysis related to each acquisition and may differ from the estimated useful lives of internally generated patents. The average estimated useful life of acquired patents is 9.6 years. We assess the potential impairment to all capitalized net patent costs when events or changes in circumstances indicate that the carrying amount of our patent portfolio may not be recoverable.

Goodwill

Goodwill is recorded as the difference, if any, between the aggregate consideration paid for an acquisition and the fair value of the net tangible and identified intangible assets acquired under a business combination. We review impairment of goodwill annually on the first day of the fourth quarter. We first assess qualitative factors to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying amount as a basis for determining whether a quantitative goodwill impairment test is necessary. If we conclude it is more likely than not that the fair value of a reporting unit exceeds its carrying amount, we need not perform the quantitative assessment.

If based on the qualitative assessment we believe it is more likely than not that the fair value of a reporting unit is less than its carrying value, a quantitative assessment test is required to be performed. This assessment requires us to compare the fair value of each reporting unit to its carrying value including allocated goodwill. We determine the fair value of our reporting units generally using a combination of the income and market approaches. The income approach is estimated through the discounted cash flow method based on assumptions about future conditions such as future revenue growth rates, new product and technology introductions, gross margins, operating expenses, discount rates, future economic and market conditions, and other assumptions. The market approach estimates the fair value of our equity by utilizing the market comparable method which is based on revenue multiples from comparable companies in similar lines of business. If the carrying value of a reporting unit exceeds the reporting unit's fair value, a goodwill impairment charge will be recorded for the difference up to the carrying value of goodwill.

The Company acquired goodwill from our acquisition of the patent licensing business of Technicolor (the "Technicolor Patent Acquisition") in 2018 and from our acquisition of Hillcrest Laboratories, Inc. (the "Hillcrest product business") in 2016. Refer to Note 5, "*Business Combinations and Other Transactions*," for more information regarding these transactions.

The carrying value of goodwill as of December 31, 2019 and 2018 was $22.4 million, respectively, which was included within "*Other Non-Current Assets*" in the consolidated balance sheets. No impairments were recorded during 2019, 2018 or 2017 as a result of our annual goodwill impairment assessment.

Other Intangible Assets

We capitalize the cost of technology solutions and platforms we acquire or license from third parties when they have a future benefit and the development of these solutions and platforms is substantially complete at the time they are acquired or licensed.

Intangible assets consist of acquired patents, existing technology, and trade names. Refer to the above *Patents* section for more information on acquired patents and existing technology. Our intangible assets are amortized on a straight-line basis over their estimated useful lives, ranging from 9 to 10 years. We make judgments about the recoverability of purchased finite-lived intangible assets whenever facts and circumstances indicate that the useful life is shorter than originally estimated or that the carrying amount of assets may not be recoverable. If such facts and circumstances exist, we assess recoverability by comparing the projected undiscounted net cash flows associated with the related asset or group of assets over their remaining lives against their respective carrying amounts. Impairments, if any, are based on the excess of the carrying amount over the fair value of those assets. If the useful life is shorter than originally estimated, we would accelerate the rate of amortization and amortize the remaining carrying value over the new shorter useful life.

Property and Equipment

Property and equipment are stated at cost. Depreciation and amortization of property and equipment are provided using the straight-line method. The estimated useful lives for computer equipment, computer software, engineering and test equipment and furniture and fixtures are generally three to five years. Leasehold improvements are amortized over the lesser of their estimated useful lives or their respective lease terms, which are generally five to ten years. Buildings are being depreciated over twenty-five years. Expenditures for major improvements and betterments are capitalized, while minor repairs and maintenance are charged to expense as incurred. Upon the retirement or disposition of property, plant and equipment, the related cost and accumulated depreciation or amortization are removed, and a gain or loss is recorded.

Leases

In February 2016, the Financial Accounting Standards Board ("FASB") issued ASU 2016-02, "Leases (Topic 842)" or ("ASC 842"), which outlines a comprehensive change to the lease accounting model and supersedes prior lease guidance. The new guidance requires lessees to recognize lease liabilities and corresponding right-of-use assets for all leases with lease terms of greater than 12 months, and also changes the definition of a lease and expands the disclosure requirements of lease arrangements.

The Company adopted this guidance on January 1, 2019 using the modified retrospective transition effective date method. As part of that adoption, we have elected the package of three practical expedients, which includes the following: an entity may elect not to reassess whether expired or existing contracts contain a lease under the revised definition of a lease; an entity may elect not to reassess the lease classification for expired or existing leases; and an entity may elect not to reassess whether previously capitalized initial direct costs would qualify for capitalization. The Company has elected not to utilize the hindsight expedient in determining the lease term, and to not record leases with an initial term of 12 months or less on our balance sheet. Additionally, the Company has elected to account for lease components and non-lease components as a single lease component for all asset classes. Lease expense is recognized over the expected term on a straight-line basis.

Internal-Use Software Costs

We capitalize costs associated with software developed for internal use that are incurred during the software development stage. Such costs are limited to expenses incurred after management authorizes and commits to a computer software project, believes that it is more likely than not that the project will be completed, the software will be used to perform the intended function with an estimated service life of two years or more, and the completion of conceptual formulation, design and testing of possible software project alternatives (the preliminary design stage). Costs incurred after final acceptance testing has been successfully completed are expensed. Capitalized computer software costs are amortized over their estimated useful life of three years.

All computer software costs capitalized to date relate to the purchase, development and implementation of engineering, accounting and other enterprise software.

Impairment of Long-Lived Assets

We evaluate long-lived assets for impairment when factors indicate that the carrying value of an asset may not be recoverable. When factors indicate that such assets should be evaluated for possible impairment, we review whether we will be able to realize our long-lived assets by analyzing the projected undiscounted cash flows in measuring whether the asset is recoverable. We did not have any long-lived asset impairments in 2019, 2018 or 2017.

Investments in Other Entities

We may make strategic investments in companies that have developed or are developing technologies that are complementary to our business. We made an accounting policy election for a measurement alternative for our equity investments that do not have readily determinable fair values, specifically related to our strategic investments in other entities. Under the alternative, our strategic investments in other entities without readily determinable fair values are measured at cost, less any impairment, plus or minus changes resulting from observable price changes in orderly transactions for an identical or similar investment of the same issuer, if any. On a quarterly basis, we monitor items such as our

investment's financial position and liquidity, performance targets, business plans, and cost trends to assess whether there are any triggering events or indicators present that would be indicative of an impairment, or any other observable price changes as indicated above. We do not adjust our investment balance when the investee reports profit or loss.

Additionally, other investments may be accounted for under the equity method of accounting. Under this method, we initially record our investment in the stock of an investee at cost, and adjust the carrying amount of the investment to recognize our share of the earnings or losses of the investee after the date of acquisition. The amount of the adjustment is included in the

determination of net income, and such amount reflects adjustments similar to those made in preparing consolidated statements including adjustments to eliminate intercompany gains and losses, and to amortize, if appropriate, any difference between our cost and underlying equity in net assets of the investee at the date of investment. The investment is also adjusted to reflect our share of changes in the investee's capital. Dividends received from an investee reduce the carrying amount of the investment. When there are a series of operating losses by the investee or when other factors indicate that a decrease in value of the investment has occurred which is other than temporary, we recognize an impairment equal to the difference between the fair value and the carrying amount of our investment.

The carrying value of our investments in other entities are included within "*Other Non-Current Assets*" on our consolidated balance sheets. During 2019, 2018 and 2017, we made investments in other entities of 0.4 million, 6.7 million and 4.6 million, respectively. The carrying value of our investments in other entities as of December 31, 2019 and 2018 was $14.2 million and $17.4 million, respectively, the majority of which are accounted for under the measurement alternative for equity investments described above.

Revenue Recognition

Refer to Note 3, "*Revenue Recognition*," for further information regarding our adoption of ASU No. 2014-09, "*Revenue from Contracts with Customers (Topic 606)*", which we refer to as ASC 606, effective January 1, 2018. The discussion that follows below is a description of our revenue recognition practices which were in effect beginning January 1, 2018 under ASC 606.

We derive the vast majority of our revenue from patent licensing. The timing and amount of revenue recognized from each licensee depends upon a variety of factors, including the specific terms of each agreement and the nature of the deliverables and obligations. Such agreements are often complex and include multiple performance obligations. These agreements can include, without limitation, performance obligations related to the settlement of past patent infringement liabilities, patent and/or know-how licensing royalties on covered products sold by licensees, access to a portfolio of technology as it exists at a point in time, and access to a portfolio of technology at a point in time along with a promises to provide any technology updates to the portfolio during the term.

In accordance with US GAAP, we use a five-step model to achieve the core underlying principle that an entity should recognize revenue to depict the transfer of goods or services to customers at an amount that the entity expects to be entitled to in exchange for those goods or services. These steps include (1) identifying the contract with the customer, (2) identifying the performance obligations, (3) determining the transaction price, (4) allocating the transaction price to the performance obligations, and (5) recognizing revenue as the entity satisfies the performance obligation(s). Additionally, we have elected to utilize certain practical expedients in the application of ASC 606. In evaluating the presence of a significant financing component in our agreements, we utilize the practical expedient to exclude any contracts wherein the gap between payment by our customers and the delivery of our performance obligation is less than one year. We have also elected to utilize the practical expedient related to costs of obtaining a contract where an entity may recognize the incremental costs of obtaining a contract as an expense when incurred if the amortization period of the asset that the entity otherwise would have recognized is one year or less. Timing of revenue recognition may differ significantly from the timing of invoicing to customers. Contract assets are included in accounts receivable and represent unbilled amounts expected to be received from customers in future periods, where the revenue recognized to date exceeds the amount billed, and right to payment is subject to the underlying contractual terms. Contract assets are classified as long-term assets if the payments are expected to be received more than one year from the reporting date. Contract assets due within less than twelve months of the balance sheet date are included within accounts receivable in our consolidated balance sheets. Contract assets due more than twelve months after the balance sheet date are included within other non-current assets.

Patent License Agreements

Upon signing a patent license agreement, we provide the licensee permission to use our patented inventions in specific applications. We account for patent license agreements in accordance with the guidance indicated above. Certain patent license agreements contain revenue from non-financial sources in the form of patents received from the customer. Under our patent license agreements, we typically receive one or a combination of the following forms of payment as consideration for permitting our licensees to use our patented inventions in their applications and products:

Consideration for Past Patent Royalties

Consideration related to a licensee's product sales from prior periods may result from a negotiated agreement with a licensee that utilized our patented inventions prior to signing a patent license agreement with us or from the resolution of a disagreement or arbitration with a licensee over the specific terms of an existing license agreement. We may also receive consideration for past patent royalties in connection with the settlement of patent litigation where there was no prior patent license agreement. In each of these cases, we record the consideration as revenue as prescribed by the five-step model.

Fixed-Fee Agreements

Fixed-fee license agreements include fixed, non-refundable royalty payments that fulfill the licensee's obligations to us under a patent license agreement for a specified time period or for the term of the agreement for specified products, under certain patents or patent claims, for sales in certain countries, or a combination thereof - in each case for a specified time period (including for the life of the patents licensed under the agreement).

Dynamic fixed-fee license agreements contain a single performance obligation that represents ongoing access to a portfolio of technology over the license term, since our promise to transfer to the licensee access to the portfolio as it exists at inception of the license, along with promises to provide any technology updates to the portfolio during the term, are not separately identifiable. Upon entering a new agreement, we allocate the transaction price to the performance obligations delivered at signing (e.g. our existing patent portfolio) and future performance obligations (e.g. the technology updates). We use a time-based input method of progress to determine the timing of revenue recognition, and as such we recognize the future deliverables on a straight-line basis over the term of the agreement. We utilize the straight-line method as we believe that it best depicts efforts expended to develop and transfer updates to the customer evenly throughout the term of the agreement.

Static fixed-fee license agreements are fixed-price contracts that generally do not include updates to technology we create after the inception of the license agreement or in which the customer does not stand to substantively benefit from those updates during the term. Although we have few static fixed-fee license agreements, we generally satisfy our performance obligations under such agreements at contract signing, and as such revenue is recognized at that time.

Variable Agreements

Upon entering a new variable patent license agreement, the licensee typically agrees to pay royalties or license fees on licensed products sold during the term of the agreement. We utilize the sales- or usage- based royalty exception for these agreements and recognize revenues during the contract term when the underlying sale or usage occurs. Our licensees under variable agreements provide us with quarterly royalty reports that summarize their sales of covered products and their related royalty obligations to us. We typically receive these royalty reports subsequent to the period in which our licensees' underlying sales occurred. As a result, we are required to estimate revenues, subject to the constraint on our ability to estimate such amounts.

Technology Solutions

Technology solutions revenue consists of revenue from royalty payments, software licenses, engineering services and product sales. The nature of these contracts and timing of payments vary. We recognize revenue from royalty payments and license agreements using the same methods described above under our policy for recognizing revenue from patent license agreements. We recognize revenue from engineering services using percentage of completion method.

Patent Sales

Our business strategy of monetizing our intellectual property includes the sale of select patent assets. As patent sales executed under this strategy represent a component of our ongoing major or central operations and activities, we will record the related proceeds as revenue. We will recognize the revenue in accordance with the five-step model, generally upon closing of the patent sale transaction.

Collaborative Arrangements

We record the elements of our collaboration agreements that represent joint operating activities in accordance with ASC 808, *Collaborative Arrangements* ("ASC 808"). Accordingly, the elements of our collaboration agreements that represent activities in which both parties are active participants, and to which both parties are exposed to the significant risks and rewards that are dependent on the commercial success of the activities, are recorded as collaborative arrangements. Generally, the classification of a transaction under a collaborative arrangement is determined based on the nature and contractual terms of the arrangement along with the nature of the operations of the participants. For transactions that are deemed to be a collaborative arrangement under ASC 808, costs incurred and revenues generated on sales to third parties will be reported in our consolidated statement of operations on a gross basis if the Company is deemed to be the principal in the transaction, or on a net basis if the Company is instead deemed to be the agent in the transaction, consistent with the guidance in ASC 606-10-55-36, *Revenue From Contracts with Customers - Principal Agent Considerations*.

Deferred Charges

Direct costs of obtaining a contract or fulfilling a contract in a transaction that results in the deferral of revenue may be either expensed as incurred or capitalized, depending on certain criteria. In conjunction with our adoption of ASC 606 effective January 1, 2018, we made a policy election to utilize the practical expedient related to costs of obtaining a contract where an entity may recognize the incremental costs of obtaining a contract as an expense when incurred if the amortization period of the asset that the entity otherwise would have recognized is one year or less. If the amortization period is greater than one year, we capitalize direct costs incurred for the acquisition or fulfillment of a contract through the date of signing if they are directly related to a particular revenue arrangement and are expected to be recovered. The costs are amortized on a straight-line basis over the life of the patent license agreement.

For example, from time to time, we use sales agents to assist us in our licensing and/or patent sale activities. In such cases, we may pay a commission. The commission rate varies from agreement to agreement. Commissions are normally paid shortly after our receipt of cash payments associated with the patent license or patent sale agreements. We defer recognition of commission expense and amortize these expenses in proportion to our recognition of the related revenue. Commission expense is included within the "*Patent administration and licensing"* line of our consolidated statements of income and was immaterial for the years presented. There were no new direct contract costs incurred during 2019, 2018 or 2017.

Incremental direct costs incurred related to a debt financing transaction may be capitalized. In connection with our offering of the 2024 Notes and 2020 Notes, defined and discussed in detail within Note 10, "*Obligations"*, we incurred directly related costs. The initial purchasers' transaction fees and related offering expenses were allocated to the liability and equity components of the debt in proportion to the allocation of proceeds and accounted for as debt issuance costs. The debt issuance costs allocated to the liability component of the debt were capitalized as deferred financing costs and recorded as a direct reduction of the debt. These costs are being amortized over the term of the debt using the effective interest method and are included within the "*Interest expense"* line of our consolidated statements of income. The costs allocated to the equity component of the debt were recorded as a reduction of the equity component of the debt. The balance of unamortized deferred financing costs as of December 31, 2019 and 2018 was $5.9 million and $1.6 million, respectively. The Company incurred $6.4 million of new debt issuance costs during 2019 in conjunction with the issuance of the 2024 Notes, noting no new debt issuance costs were incurred in 2018 or 2017. Deferred financing expense was $1.5 million, $1.4 million and $1.4 million in 2019, 2018 and 2017, respectively.

Research and Development

Research and development expenditures are expensed in the period incurred, except certain software development costs that are capitalized between the point in time that technological feasibility of the software is established and when the product is available for general release to customers. We did not have any capitalized software costs related to research and development in any period presented. Research, development and other related costs were approximately $74.9 million, $69.7 million and $75.7 million in 2019, 2018 and 2017, respectively.

Compensation Programs

We use a variety of compensation programs to attract, retain and motivate our employees, and to more closely align employee compensation with company performance. These programs include, but are not limited to, short-term incentives tied to performance goals, cash awards to inventors for filed patent applications and patent issuances, and long-term incentives in the form of stock option awards, time-based restricted stock unit (“RSU”) awards, performance-based awards and cash awards, noting equity awards are granted pursuant to the terms and conditions of our Equity Plans (as defined in Note 13, "*Compensation Plans and Programs*"). Our long-term incentives, including equity awards, typically include annual equity and cash award grants with three- to five-year vesting periods; as a result, in any one year, we are typically accounting for at least three active cycles.

We account for compensation costs associated with share-based compensation based on the fair value of the instruments issued. The estimated value of stock options includes assumptions around expected life, stock volatility and dividends. The expected life of our stock option awards is based on the simplified method as prescribed by Staff Accounting Bulletin Topic 14. In all periods, our policy has been to set the value of RSUs and restricted stock awards equal to the value of our underlying common stock on the

date of measurement. For grants with graded vesting, we amortize the associated unrecognized compensation cost using an accelerated method. For grants that cliff vest, we amortize the associated unrecognized compensation cost on a straight-line basis over their vesting term.

In the event of canceled awards, we adjust compensation expense recognized to date as they occur. Tax windfalls and shortfalls related to the tax effects of employee share-based compensation are included in our tax provision. On the consolidated statements of cash flows, tax windfalls and shortfalls related to employee share-based compensation awards are

included within operating activities and cash paid to tax authorities for shares withheld are included within financing activities. The inclusion of windfalls and shortfalls in the tax provision could increase our earnings volatility between periods. Tax windfalls related to share-based compensation for the years ended 2019, 2018 and 2017 were $0.2 million, $1.8 million and $12.1 million, respectively.

Income Taxes

Income taxes are accounted for under the asset and liability method. Under this method, deferred tax assets and liabilities are recognized for the estimated future tax consequences attributable to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases, and operating loss and tax credit carry forwards. Deferred tax assets and liabilities are measured using enacted tax rates in effect for the year in which those temporary differences are expected to be recovered or settled. The effect on deferred tax assets and liabilities of a change in tax rates is recognized in the Consolidated Statement of Income in the period in which the change was enacted. A valuation allowance is recorded to reduce the carrying amounts of deferred tax assets if management has determined that it is more likely than not that such assets will not be realized.

In addition, the calculation of tax liabilities involves significant judgment in estimating the impact of uncertainties in the application of complex tax laws. We are subject to examinations by the U.S. IRS and other taxing jurisdictions on various tax matters, including challenges to various positions we assert in our filings. In the event that the IRS or another taxing jurisdiction levies an assessment in the future, it is possible the assessment could have a material adverse effect on our consolidated financial condition or results of operations.

The financial statement recognition of the benefit for a tax position is dependent upon the benefit being more likely than not to be sustainable upon audit by the applicable tax authority. If this threshold is met, the tax benefit is then measured and recognized at the largest amount that is greater than 50 percent likely of being realized upon ultimate settlement. In the event that the IRS or another taxing jurisdiction levies an assessment in the future, it is possible the assessment could have a material adverse effect on our consolidated financial condition or results of operations.

New Accounting Guidance

Accounting Standards Update: Leases

In February 2016, the FASB issued ASU 2016-02, "Leases (Topic 842)", or ASC 842, which outlines a comprehensive change to the lease accounting model and supersedes prior lease guidance. Refer to Note 17, "*Leases*," for information regarding our adoption of this guidance effective January 1, 2019 and a discussion of the impact to information presented herein, as well as additional required disclosures under the new guidance.

Accounting Standards Update: Improvements to Nonemployee Share-Based Payment Accounting

In June 2018, the FASB issued ASU No. 2018-07, "Stock Compensation (Topic 718): Improvements to Nonemployee Share-Based Payment Accounting," which is intended to reduce cost and complexity and to improve financial reporting for share-based payments issued to nonemployees. The guidance is effective for fiscal years beginning after December 15, 2018 and early adoption is permitted. We adopted this guidance in first quarter 2019 and it did not have a material impact on our consolidated financial statements.

Accounting Standards Update: Financial Instruments - Credit Losses

In June 2016, the FASB issued ASU No. 2016-13, "Financial Instruments - Credit Losses". This ASU introduces a new accounting model for recognizing credit losses on certain financial instruments and financial assets, including trade receivables, based upon an estimate of current expected credit losses, otherwise known as CECL. The new guidance requires the recognition of an allowance that reflects the current estimate of credit losses expected to be incurred over the life of the financial asset, based not only on historical experience and current conditions, but also on reasonable forecasts. Additionally, ASU No. 2016-13 made several changes to the available-for-sale impairment model. The

guidance is effective for fiscal years beginning after December 15, 2019, including interim periods within those fiscal years, and early adoption is permitted. We have concluded that this guidance will not have a material impact on our consolidated financial statements.

Accounting Standards Update: Cloud Computing Arrangements

In August 2018, the FASB issued ASU No. 2018-15 “Intangibles – Goodwill and Other – Internal-Use Software (Subtopic 350-40): Customer’s Accounting for Implementation Costs Incurred in a Cloud Computing Arrangement that is a Service Contract”. The amendments in this ASU align the requirements for capitalizing implementation costs incurred in a hosting arrangement that is a service contract with the requirements for capitalizing implementation costs incurred to develop or obtain internal-use software. The guidance is effective for fiscal years beginning after December 15, 2019, including interim

periods within those fiscal years, and early adoption is permitted. While we are still completing our accounting assessment, we do not expect this guidance to have a material impact on our consolidated financial statements.

Accounting Standards Update: Collaborative Arrangements

In November 2018, the FASB issued ASU No. 2018-18, "Collaborative Arrangements (Topic 808): Clarifying the Interaction Between Topic 808 and Topic 606". The amendments in this ASU provide guidance on how to assess whether certain transactions between collaborative arrangement participants should be accounted for within the revenue recognition standard. The amendments in this update are effective for fiscal years beginning after December 15, 2019, including interim periods within those fiscal years, and early adoption is permitted for entities who have previously adopted the new revenue recognition guidance. We have concluded that this guidance will not have a material impact on our consolidated financial statements.

Accounting Standards Update: Simplifying the Accounting for Income Taxes

In December 2019, the FASB issued ASU No. 2019-12, "Income Taxes (Topic 740): Simplifying the Accounting for Income Taxes" ("ASU 2019-12"). The amendments in this ASU are intended to simplify various aspects related to accounting for income taxes. ASU 2019-12 removes certain exceptions to the general principles in Topic 740 and also clarifies and amends existing guidance to improve consistent application. ASU 2019-12 is effective for fiscal years beginning after December 15, 2020 with early adoption allowed. The Company is currently evaluating the impact of the adoption of ASU 2019-12 on its consolidated financial statements.

3. REVENUE RECOGNITION

In May 2014, the Financial Accounting Standards Board ("FASB") issued ASU No. 2014-09, "R*evenue from Contracts with Customers (Topic 606)*" ("ASC 606") which superseded most prior revenue recognition guidance, including industry-specific guidance. The underlying principle is that an entity will recognize revenue to depict the transfer of goods or services to customers at an amount that the entity expects to be entitled to in exchange for those goods or services. The guidance also requires enhanced disclosures regarding the nature, amount, timing and uncertainty of revenue and cash flows arising from an entity's contracts with customers. We adopted the requirements of the new standard as of January 1, 2018 using the modified retrospective transition method applied to those contracts that were not completed as of January 1, 2018. Accordingly, all periods prior to January 1, 2018 are presented in accordance with ASC Topic 605, "Revenue Recognition" ("ASC 605"). Periods beginning January 1, 2018 are presented in accordance with ASC 606. See Note 2 "*Summary of Significant Accounting Policies and New Accounting Guidance*" for our revised revenue recognition accounting policy upon adoption of the new guidance.

For accounting purposes under ASC 606, we separate our fixed-fee license agreements into two categories: (i) those agreements that provide rights, over the term of the license, to future technologies that are highly interdependent or highly interrelated to the technologies provided at the inception of the agreement ("Dynamic Fixed-Fee Agreements") and (ii) those agreements that do not provide for rights to such future technologies ("Static Fixed-Fee Agreements"). After the fair value allocation between the past and future components of the agreement, we recognize the future components of revenue from Dynamic Fixed-Fee Agreements on a straight-line basis over the term of the related license agreement, while we recognize most or all of the revenue from Static Fixed-Fee Agreements in the quarter the license agreement is signed. We did not recognize any ongoing revenue from Static Fixed-Fee Agreements already in existence at the time the guidance was adopted. Additionally, in the event a significant financing component is determined to exist in any of our agreements, we recognize more or less revenue and corresponding interest expense or income, as appropriate.

In addition, we record per-unit royalty revenue in the same period in which the licensee's underlying sales occur. Because we generally do not receive the per-unit licensee royalty reports for sales during a given quarter within the time frame necessary to adequately review the reports and include the actual amounts in our quarterly results for such quarter, we accrue the related revenue based on estimates of our licensees' underlying sales, subject to certain constraints on our ability to estimate such amounts. As a result of accruing revenue for the quarter based on such estimates, adjustments are required in the following quarter to true-up revenue to the actual amounts reported by our licensees. In addition, to the extent we receive non-refundable prepayments related to per-unit license agreements that do not provide rights over the term of the license to future technologies

that are highly interdependent or highly interrelated to the technologies provided at the inception of the agreement, we recognize such prepayments as revenue in the period in which all remaining revenue recognition criteria have been met.

Finally, under ASC 606, we recognize a receivable, and any related deferred tax asset for foreign withholding taxes, for payments as they become due.

Timing of revenue recognition may differ significantly from the timing of invoicing to customers. Contract assets are included in accounts receivable and represent unbilled amounts expected to be received from customers in future periods, where the revenue recognized to date exceeds the amount billed, and right to payment is subject to the underlying contractual terms. Contract assets are classified as long-term assets if the payments are expected to be received more than one year from the reporting date.

Disaggregated Revenue

The following table presents the disaggregation of our revenue for the year ended December 31, 2019 and 2018 under ASC 606. Revenue for the year ended December 31, 2017 is presented in accordance with ASC 605. Amounts are in thousands.

	For the Year Ended December 31,		
	2019	**2018**	**2017**
Variable patent royalty revenue	$ 30,428	$ 36,384	$ 47,840
Fixed-fee royalty revenue	257,221	239,347	301,628
Current patent royalties [a]	287,649	275,731	349,468
Non-current patent royalties [b]	19,782	26,329	162,890
Total patent royalties	307,431	302,060	512,358
Current technology solutions revenue [a]	10,518	4,594	20,580
Patent sales	975	750	—
Total revenue	$ 318,924	$ 307,404	$ 532,938

a. Recurring revenues consist of current patent royalties, inclusive of Dynamic Fixed-Fee Agreement royalties, and current technology solutions revenue.

b. Non-current patent royalties for the year ended December 31, 2019 and 2018 consist of past patent royalties and royalties from static agreements. For the year ended December 31, 2017, non-current patent royalties consist of past patent royalties.

During the year ended December 31, 2019, we recognized $214.0 million of revenue that had been included in deferred revenue as of the beginning of the period. As of December 31, 2019, we had contract assets of $16.2 million and $10.2 million included within "*Accounts receivable*" and "*Other non-current assets*" in the consolidated balance sheet, respectively. As of December 31, 2018, we had contract assets of $19.7 million and $5.5 million included within "*Accounts receivable*" and "*Other non-current assets*" in the consolidated balance sheet, respectively.

Impact of Adoption of ASC 606

In accordance with the new revenue standard requirements, the disclosure of the impact of adoption on our current period consolidated income statement and balance sheet is presented below. We believe this additional information is vital to allow readers of our financial statements to compare financial results from the preceding financial years given the absence of restatement of the prior period. The adoption of ASC 606 did not affect our reported total amounts of cash flows from operating, investing and financing activities. Amounts contained in the tables below are in thousands, except per share data.

	For the Year Ended December 31,				
	2019	2018			2017
	As Reported ASC 606	As Reported ASC 606	Adjustment	ASC 605	As Reported ASC 605
REVENUES:					
Variable patent royalty revenue	$ 30,428	$ 36,384	$ 461	$ 36,845	$ 47,840
Fixed-fee royalty revenue	257,221	239,347	79,341	318,688	301,628
Current patent royalties	287,649	275,731	79,802	355,533	349,468
Non-current patent royalties	19,782	26,329	(10,000)	16,329	162,890
Total patent royalties	307,431	302,060	69,802	371,862	512,358
Patent sales	975	750	—	750	—
Current technology solutions revenue	10,518	4,594	4,907	9,501	20,580
	$ 318,924	$ 307,404	$ 74,709	$ 382,113	$ 532,938
OPERATING EXPENSES:	281,089	244,809	—	244,809	231,443
Income from operations	37,835	62,595	74,709	137,304	301,495
Interest expense	(40,955)	(35,956)	16,655	(19,301)	(17,845)
OTHER INCOME (EXPENSE), NET	29,062	5,419	—	5,419	8,740
Income before income taxes	25,942	32,058	91,364	123,422	292,390
INCOME TAX BENEFIT (PROVISION)	(10,991)	27,417	(6,686)	20,731	(121,676)
NET INCOME	$ 14,951	$ 59,475	$ 84,678	$ 144,153	$ 170,714
Net loss attributable to noncontrolling interest	(5,977)	(5,556)	—	(5,556)	(5,506)
NET INCOME ATTRIBUTABLE TO INTERDIGITAL, INC.	$ 20,928	$ 65,031	$ 84,678	$ 149,709	$ 176,220
NET INCOME PER COMMON SHARE — BASIC	$ 0.66	$ 1.89	$ 2.45	$ 4.34	$ 5.09
NET INCOME PER COMMON SHARE — DILUTED	$ 0.66	$ 1.84	$ 2.40	$ 4.24	$ 4.93

Contracted Revenue

Based on contracts signed and committed Dynamic Fixed-Fee Agreement payments as of December 31, 2019, we expect to recognize the following amounts of revenue over the term of such contracts (in thousands):

	Revenue
2020	$ 260,813
2021	191,146
2022	86,728

2023	—
2024	—
Thereafter	—
	$ 538,687

4. GEOGRAPHIC / CUSTOMER CONCENTRATION

We have one reportable segment. During 2019, 2018 and 2017, the majority of our revenue was derived from a limited number of licensees based outside of the United States, primarily in Asia. Substantially all of these revenues were paid in U.S. dollars and were not subject to any substantial foreign exchange transaction risk. The table below lists the countries of the headquarters of our licensees and customers and the total revenue derived from each country or region for the periods indicated (in thousands):

	For the Year Ended December 31,		
	2019	2018	2017
United States	$ 139,162	$ 119,159	$ 194,184
South Korea	113,189	112,291	113,059
Japan	35,614	29,525	25,210
China	11,103	309	77,087
Sweden	6,934	6,933	6,935
France	5,895	277	—
Other Europe	5,810	5,116	6,305
Taiwan	938	23,326	36,051
Other Asia	279	468	—
Finland	—	10,000	—
Canada	—	—	74,107
Total	$ 318,924	$ 307,404	$ 532,938

During 2019, 2018 and 2017, the following licensees or customers accounted for 10% or more of total revenues:

	2019	2018	2017
Apple	35%	36%	21%
Samsung	25%	25%	13%
LG	10%	10%	< 10%
Blackberry [(a)]	—%	—%	13%
Huawei [(b)]	—%	—%	14%

(a) 2017 revenue include $70.7 million of non-current patent royalties.
(b) 2017 revenue include $8.4 million of non-current patent royalties.

As of December 31, 2019, 2018 and 2017, we held $446.6 million, $464.6 million and $336.1 million, respectively, of our property, equipment and patents, net of accumulated depreciation and amortization, of which greater than 97% of the total was within the United States in each of the years presented. As of December 31, 2019, we held $1.7 million of property and equipment, net of accumulated depreciation, collectively, in Canada, Europe and Asia.

5. BUSINESS COMBINATIONS AND OTHER TRANSACTIONS

Acquisition of Technicolor's Patent Licensing Business

On July 30, 2018, we completed our acquisition of the patent licensing business of Technicolor SA ("Technicolor"), a worldwide technology leader in the media and entertainment sector, which we refer to as the Technicolor Patent Acquisition. The Technicolor Patent Acquisition included the acquisition by InterDigital of approximately 18,000

patents and applications, across a broad range of technologies, including approximately 3,000 worldwide video coding patents and applications. The acquisition of Technicolor's portfolio greatly expanded InterDigital's technology footprint in the mobile industry, and opens new markets in consumer home electronics, display technology and video. Under the terms of the original agreement, the portfolio was to be supplemented by a jointly funded R&D collaboration. This jointly funded R&D collaboration was terminated in conjunction with the acquisition of Technicolor's Research & Innovation unit (the "R&I Acquisition"), which is discussed below. Members of Technicolor's licensing, legal and other support teams in offices in Rennes and Paris, France; Princeton, New Jersey, USA; and other locations joined InterDigital's team of more than 300 R&D and other staff in locations around the world. In addition, we have assumed Technicolor's rights and obligations under a joint licensing program with Sony Corporation ("Sony") relating to digital televisions and standalone computer display monitors (the "Madison Arrangement"), including Technicolor's role as sole licensing agent for the Madison Arrangement. We account for our assumption of Technicolor's rights and obligations under the Madison Arrangement as a collaborative arrangement. As part of this transaction, we also granted back to Technicolor a perpetual license for patents acquired in the transaction.

The Technicolor Patent Acquisition met the definition of a business combination and, as such, was accounted for using the acquisition method of accounting. Under the terms of the agreement, in third quarter 2018, we paid Technicolor $158.9 million in cash, inclusive of $15.9 million of cash acquired, yielding net cash consideration of $143.0 million. We funded this

payment with cash on hand. Under the terms of the original agreement, Technicolor was to receive 42.5% of all of InterDigital's future cash receipts (net of estimated operating expenses) from InterDigital's new licensing efforts in the consumer electronics field; there was no revenue sharing associated with InterDigital's mobile industry licensing efforts. As such, we accounted for the portion of the future cash receipts owed to Technicolor relating to patents existing as of the date of the acquisition as a contingent consideration liability, which was valued at $18.6 million as of the acquisition date. This revenue-sharing arrangement and associated contingent consideration liability were modified in conjunction with the R&I Acquisition, which closed during second quarter 2019. Refer to the discussion below. Additionally, as of the acquisition date, we estimated we would receive payments totaling 20.2 million relating to the transaction from Technicolor.

We allocated the fair value of consideration transferred to identifiable assets acquired and liabilities assumed based on their estimated fair values as of the acquisition date. We recorded the excess of the fair value of consideration transferred over the net values of these assets and liabilities as goodwill. We estimated the fair value of the intangible assets in this transaction through a combination of a discounted cash flow analysis (the income approach) and an analysis of comparable market transactions (the market approach). For the income approach, we based the inputs and assumptions used to develop these estimates on a market participant perspective and included estimates of projected revenues, discount rates, economic lives and income tax rates, among others, and all of these estimates require significant management judgment. For the market approach, we applied judgment to identify the most comparable market transactions to this transaction. Refer to Note 7 for discussion regarding the valuation methodologies used for the contingent consideration liability.

The following table summarizes the fair value of consideration transferred and our allocation of that consideration based on the fair values of the assets acquired and liabilities assumed as of the date of acquisition (in thousands):

		As of July 30, 2018
Cash	$	158,898
Contingent consideration liability		18,616
	$	177,514
Less: Transaction-related receivable		(20,200)
Net fair value of consideration transferred	**$**	**157,314**

Allocation:			**Estimated useful life (Years)**
Net tangible assets and liabilities:			
Restricted cash	$	15,913	
Other current assets		5,600	
Other non-current assets		3,116	
Current liabilities		(6,219)	
Long-term debt		(17,717)	
Other long-term liabilities		(3,767)	
Total net tangible assets and liabilities	$	(3,074)	
Identified intangible assets:			

Patents	$	154,000	9 - 10
Goodwill[1]		6,388	
Total identified intangible assets	$	160,388	
Total fair value of consideration transferred	**$**	**157,314**	

(1) Goodwill consists of expected synergies resulting from the combination of our and Technicolor's patent licensing businesses in the increasingly complementary areas of mobile and video technology. We expect almost all of the goodwill resulting from the Technicolor Patent Acquisition will be deductible for income tax purposes.

The amount of revenue and earnings that would have been included in the Company's consolidated statements of income for the years ended December 31, 2018 and 2017 had the acquisition date been January 1, 2017 are reflected in the

table below. These amounts have been calculated after applying the Company's accounting policies and adjusting the results to reflect additional interest expense as well as amortization that would have been charged assuming the fair value adjustments to amortizable intangible assets had been recorded as of January 1, 2017. In addition, pro forma adjustments have been made to reflect the impact of the transaction-related costs discussed below. These unaudited pro forma combined results of operations have been prepared for comparative purposes only, and they do not purport to be indicative of the results of operations that actually would have resulted had the acquisition occurred on the date indicated, or that may result in the future. The amounts in the table are unaudited (in thousands, except per share data):

	For the Year Ended December 31,	
	2018	2017
	(Unaudited)	
Actual revenue	$ 307,404	$ 532,938
Supplemental pro forma revenue	$ 314,096	$ 541,921
Actual earnings	$ 65,031	$ 176,220
Supplemental pro forma earnings	$ 52,754	$ 107,531
Actual diluted earnings per share	$ 1.84	$ 4.93
Supplemental pro forma diluted earnings per share	$ 1.49	$ 3.01

Acquisition of Technicolor's Research & Innovation Unit

On May 31, 2019, we completed the acquisition of the Research & Innovation unit of Technicolor SA, which we refer to as the R&I Acquisition. The R&I Acquisition expanded the Company's research capabilities in video coding, Internet of Things (IoT) and smart home, imaging sciences, augmented reality and virtual reality, and artificial intelligence and machine learning technologies. The Technicolor R&I unit was the driving creative force behind the patent portfolio that was acquired in the Technicolor Patent Acquisition discussed above.

The R&I Acquisition unit met the definition of an asset acquisition and was accounted for using the cost accumulation and allocation model. There was no cash consideration for the R&I Acquisition. As consideration for the R&I Acquisition, the jointly funded R&D collaboration that was entered into as part of the Technicolor Patent Acquisition was terminated. Technicolor will continue to fund research to be performed by the R&I unit for certain limited projects for a specified time period, subject to renewal. The Company also assumed certain employee-related liabilities, including obligations for certain defined benefit post-retirement plans for the acquired R&I unit employees, which are further discussed below. Additionally, Technicolor agreed to reduce its rights under the revenue-sharing arrangement entered into as part of the Technicolor Patent Acquisition, as further discussed below.

The R&I Acquisition resulted in a net gain of approximately $14.2 million, inclusive of the $20.5 million gain from the derecognition of the contingent consideration liability described below, all of which is included within "*Other Income (Expense), Net*" in the consolidated statement of income for the year ended December 31, 2019.

Contingent Consideration

As discussed above, in conjunction with the initial Technicolor Patent Acquisition, Technicolor was to receive 42.5% of all of InterDigital's future cash receipts (net of estimated operating expenses) from InterDigital's new licensing efforts in the consumer electronics field; there was no revenue sharing associated with InterDigital's mobile industry licensing efforts. The portion of the future cash receipts relating to patents existing as of the date of the acquisition was originally accounted for as a contingent consideration liability in accordance with ASC 805-30-25, *Business Combinations - Contingent Consideration.* There are no minimum or maximum payments under the revenue sharing arrangement, and, except in certain circumstances, the arrangement continues through December 31, 2038.

The estimated acquisition date fair value of the contingent consideration liability of $18.6 million was determined utilizing a Monte Carlo simulation model. This initial fair value measurement was based on the perspective of a market participant and included significant unobservable inputs that are classified as Level 3 inputs within the fair value hierarchy and are discussed further within Note 7.

As a result of the R&I Acquisition in second quarter 2019, under the amended revenue-sharing arrangement described above, Technicolor will now receive 42.5% of future cash receipts from new licensing efforts from the Madison Arrangement only, subject to certain conditions and hurdles, but will no longer receive revenue-sharing from other licensing efforts in the consumer electronics field outside of the Madison Arrangement. We determined that the initial contingent consideration liability from the Technicolor Patent Acquisition was significantly modified in conjunction with the R&I Acquisition, and, as

such, the contingent consideration liability will now be accounted for under *ASC 450 - Contingencies* under the asset acquisition framework when the liability is deemed probable and estimable. Since the contingent consideration liability arising from the amended revenue-sharing arrangement was not probable and estimable as of the R&I Acquisition date, the carrying value of the previous contingent consideration liability was derecognized, which resulted in a $20.5 million gain during the year ended December 31, 2019 and is included within "*Other Income (Expense), Net"* in the consolidated statement of income for the period. As of December 31, 2019, the contingent consideration liability from the amended revenue-sharing arrangement was deemed not probable and estimable and is therefore not reflected within the consolidated financial statements.

Defined Benefit Plans

In connection with the Technicolor Patent Acquisition and the R&I Acquisition, we assumed certain defined benefit plans which are accounted for in accordance with *ASC 715 - Compensation - Retirement Benefits*. These plans include a retirement lump sum indemnity plan and jubilee plan, both of which provide benefit payments to employees based upon years of service and compensation levels.

As of December 31, 2019, the combined accumulated projected benefit obligation related to these plans totaled $6.2 million. Service cost and interest cost for the combined plans totaled less than $0.1 million for the year ended December 31, 2019. The weighted average discount rate and assumed salary increase rate for these plans were 0.7% and 3.0%, respectively. These plans are not required to be funded and were not funded as of December 31, 2019. Expected future benefit payments under these plans as of December 31, 2019 were as follows (in thousands):

2020	$	94
2021		177
2022		239
2023		568
2024		248
2025 - 2029		1,935

Madison Arrangement

As discussed above, in conjunction with the Technicolor Patent Acquisition, effective July 30, 2018, we assumed Technicolor's rights and obligations under the Madison Arrangement, which commenced in 2015. The Madison Arrangement falls under the scope of ASC 808, *Collaborative Arrangements.*

Under the Madison Arrangement, Technicolor and Sony combined portions of their respective digital TV ("DTV") and computer display monitor ("CDM") patent portfolios and created a combined licensing opportunity to DTV and CDM manufacturers. Per an Agency and Management Services Agreement ("AMSA") entered into upon the creation of the Madison Arrangement, Technicolor was initially appointed as sole licensing agent of the arrangement, and InterDigital has now assumed that role. As licensing agent, we are responsible for making decisions regarding the prosecution and maintenance of the combined patent portfolio and the licensing and enforcement of the combined patent portfolio in the field of use of DTVs and CDMs on an exclusive basis during the term of the AMSA in exchange for an agent fee.

We were deemed to be the principal in this collaborative arrangement under ASC 808, and, as such, in accordance with ASC 606-10-55-36, *Revenue From Contracts with Customers - Principal Agent Considerations*, we record revenues generated on sales to third parties and costs incurred on a gross basis in the consolidated statements of income. Therefore, we recognize all royalties from customers as revenue and payments to Sony for its royalty share as operating expenses within the consolidated statements of income. Cost reimbursements for expenses incurred resulting from fulfilling the duties of the licensing agent are recorded as contra expenses. During the year ended December 31, 2019, gross revenues recorded related to the Madison Arrangement were $13.5 million and are reflected within "*Patent licensing royalties*" in the consolidated statement of income. Net operating expenses related to the Madison Arrangement during the year ended December 31, 2019 were approximately $12.0 million, including $6.3 million related to revenue sharing, and are reflected primarily within "*Patent administration and licensing*" expenses in the consolidated statement of income.

Long-term debt

An affiliate of CPPIB Credit Investments Inc. ("CPPIB Credit"), a wholly owned subsidiary of Canada Pension Plan Investment Board, is a third-party investor in the Madison Arrangement. CPPIB Credit has made certain payments to Technicolor and Sony and has agreed to contribute cash to fund certain capital reserve obligations under the arrangement in exchange for a percentage of future revenues, specifically through September 11, 2030 in regard to the Technicolor patents.

Upon our assumption of Technicolor's rights and obligations under the Madison Arrangement, our relationship with CPPIB Credit met the criteria in ASC 470-10-25, *Sales of Future Revenues or Various Other Measures of Income* ("ASC 470"), which relates to cash received from an investor in exchange for a specified percentage or amount of revenue or other measure of income of a particular product line, business segment, trademark, patent, or contractual right for a defined period. Under this guidance, we recognized the fair value of our contingent obligation to CPPIB Credit, as of the acquisition date, as long-term debt in our consolidated balance sheet. This initial fair value measurement was based on the perspective of a market participant and includes significant unobservable inputs which are classified as Level 3 inputs within the fair value hierarchy. The fair value of the long-term debt as of December 31, 2019 is disclosed within Note 7. Our repayment obligations are contingent upon future royalty revenues generated from the Madison Arrangement and there are no minimum or maximum payments under the arrangement.

Under ASC 470, amounts recorded as debt shall be amortized under the interest method. At each reporting period, we will review the discounted expected future cash flows over the life of the obligation. The Company made an accounting policy election to utilize the catch-up method when there is a change in the estimated future cash flows, whereby we will adjust the carrying amount of the debt to the present value of the revised estimated future cash flows, discounted at the original effective interest rate, with a corresponding adjustment recognized as interest expense within "*Interest expense*" in the consolidated statements of income. The effective interest rate as of the acquisition date was approximately 14.5%. This rate represents the discount rate that equates the estimated future cash flows with the fair value of the debt as of the acquisition date, and is used to compute the amount of interest to be recognized each period based on the estimated life of the future revenue streams. During the year ended December 31, 2019 and 2018, we recognized $2.7 million and $0.7 million of interest expense related to this debt which is included within "*Interest expense*" in the consolidated statements of income. Any future payments made to CPPIB Credit, or additional proceeds received from CPPIB Credit, will decrease or increase the long-term debt balance accordingly.

Restricted cash

Under the Madison Arrangement, the parties reserve cash in bank accounts to fund our activities to manage the portfolios. These accounts are custodial accounts for which the funds are restricted for this purpose. As of December 31, 2019 and 2018, the Company had $9.5 million and $13.7 million of restricted cash included within the consolidated balance sheet attributable to the Madison Arrangement. Refer to Note 6 for a reconciliation of cash, cash equivalents and restricted cash within the consolidated balance sheets.

Commitments

To receive consent from both Sony and CPPIB Credit to assume the rights and responsibilities of Technicolor under the Madison Arrangement, we committed to contributing cash to fund shortfalls in the Madison Arrangement, up to a maximum of $25.0 million, through 2020. A shortfall funding is only required in the scenario in which the restricted cash is not sufficient to fund current obligations. In the event that we fund a shortfall, any surplus cash resulting from subsequent royalty receipts would be used to repay our shortfall funding plus 25% interest in advance of distributions of royalties to either Sony or CPPIB Credit, assuming they have not participated in the funding of the shortfall. As of December 31, 2019, we have not contributed any shortfall funding.

Transaction Costs

Transaction and integration related costs related to the above transactions for the years ended December 31, 2019 and 2018 were $8.4 million and $17.8 million, respectively. The majority of these costs were recorded within "*Patent administration and licensing*" and "*Selling, general and administrative*" expenses in the consolidated statements of income.

Hillcrest Product Business

On December 20, 2016, we acquired Hillcrest Laboratories, Inc. ("Hillcrest"), a pioneer in sensor processing technology, for approximately $48.0 million in cash, net of $0.4 million cash acquired. The business combination transaction was accounted for using the acquisition method of accounting.

On July 19, 2019, we completed the sale of Hillcrest's product business to a subsidiary of CEVA, Inc. In connection with the sale, we received initial proceeds of $10.0 million, with a customary portion of the purchase price placed in escrow to secure potential indemnification claims. As part of the transaction, we retained substantially all of the Hillcrest patent assets that we acquired in 2016. As a result of this transaction, we recorded an $8.5 million gain on sale which is included within "*Other Income (Expense), Net*" in the consolidated statements of income for the year ended December 31, 2019.

6. CASH, CASH EQUIVALENTS, RESTRICTED CASH AND MARKETABLE SECURITIES

Cash, Cash Equivalents and Restricted Cash

Cash, cash equivalents and restricted cash as of December 31, 2019 and 2018 consisted of the following (in thousands):

	December 31,	
	2019	2018
Money market and demand accounts	$ 757,098	$ 488,733
Commercial paper	—	—
	$ 757,098	$ 488,733

The following table provides a reconciliation of total cash, cash equivalents and restricted cash as of December 31, 2019 and 2018 within the consolidated balance sheets (in thousands). The Company had no restricted cash prior to 2018.

	December 31,	
	2019	2018
Cash and cash equivalents	$ 745,491	$ 475,056
Restricted cash included within prepaid and other current assets	10,526	13,677
Restricted cash included within other non-current assets	1,081	—
Total cash, cash equivalents and restricted cash	$ 757,098	$ 488,733

Marketable Securities

As of December 31, 2019 and 2018, the majority of our marketable securities have been classified as available-for-sale and are carried at fair value, with unrealized gains and losses reported net-of-tax as a separate component of shareholders' equity. Substantially all of our investments are investment-grade government and corporate debt securities that have maturities of less than 2 years, and we have both the ability and intent to hold the investments until maturity. We recorded no other-than-temporary impairments recorded during 2019, 2018 or 2017. The gross realized gains and losses on sales of marketable securities were not significant during the years ended December 31, 2019, 2018 and 2017.

Marketable securities as of December 31, 2019 and 2018 consisted of the following (in thousands):

	December 31, 2019			
	Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value
Available-for-sale securities				
U.S. government securities	$ 105,453	$ 249	$ —	$ 105,702
Corporate bonds, asset backed and other securities	73,276	226	—	73,502
Total available-for-sale securities	$ 178,729	$ 475	$ —	$ 179,204
Reported in:				

Cash and cash equivalents	$	—
Short-term investments		179,204
Total marketable securities	$	179,204

	December 31, 2018			
	Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value
Available-for-sale securities				
Commercial paper	$ 14,548	$ —	$ —	$ 14,548
U.S. government securities	291,157	—	(1,581)	289,576
Corporate bonds, asset backed and other securities	167,579	5	(984)	166,600
Total available-for-sale securities	$ 473,284	$ 5	$ (2,565)	$ 470,724
Reported in:				
Cash and cash equivalents				$ —
Short-term investments				470,724
Total marketable securities				$ 470,724

As of December 31, 2019 and 2018, $163.1 million and $390.9 million, respectively, of our short-term investments had contractual maturities within one year. The remaining portions of our short-term investments had contractual maturities within one to two years.

7. CONCENTRATION OF CREDIT RISK AND FAIR VALUE OF FINANCIAL ASSETS AND FINANCIAL LIABILITIES

Concentration of Credit Risk and Fair Value of Financial Instruments

Financial instruments that potentially subject us to concentration of credit risk consist primarily of cash equivalents, short-term investments and accounts receivable. We primarily place our cash equivalents and short-term investments in highly rated financial instruments and in United States government instruments.

Our accounts receivable are derived principally from patent license and technology solutions agreements. As of December 31, 2019 and 2018, seven and five licensees comprised 73% and 76%, respectively, of our accounts receivable balance. We perform ongoing credit evaluations of our licensees, who generally include large, multinational, wireless telecommunications equipment manufacturers. We believe that the book values of our financial instruments approximate their fair values.

Fair Value Measurements

We use various valuation techniques and assumptions when measuring the fair value of our assets and liabilities. We utilize market data or assumptions that market participants would use in pricing the asset or liability, including assumptions about risk and the risks inherent in the inputs to the valuation technique. This guidance established a hierarchy that prioritizes fair value measurements based on the types of input used for the various valuation techniques (market approach, income approach and cost approach). The levels of the hierarchy are described below:

Level 1 Inputs — Level 1 includes financial instruments for which quoted market prices for identical instruments are available in active markets.

Level 2 Inputs — Level 2 includes financial instruments for which there are inputs other than quoted prices included within Level 1 that are observable for the instrument such as quoted prices for similar instruments in active markets, quoted prices for identical or similar instruments in markets with insufficient volume or infrequent transactions (less active markets) or model-driven valuations in which significant inputs are observable or can be derived principally from, or corroborated by, observable market data, including market interest rate curves, referenced credit spreads and pre-payment rates.

Level 3 Inputs — Level 3 includes financial instruments for which fair value is derived from valuation techniques including pricing models and discounted cash flow models in which one or more significant inputs are unobservable, including the company's own assumptions. The pricing models incorporate transaction details such as contractual terms, maturity and, in certain instances, timing and amount of future cash flows, as well as assumptions related to liquidity and credit valuation adjustments of marketplace participants.

Our assessment of the significance of a particular input to the fair value measurement requires judgment and may affect the valuation of financial assets and financial liabilities and their placement within the fair value hierarchy. We use quoted market prices for similar assets to estimate the fair value of our Level 2 investments.

Recurring Fair Value Measurements

Our financial assets are included within short-term investments on our consolidated balance sheets, unless otherwise indicated. Our financial assets and liabilities that are accounted for at fair value on a recurring basis are presented in the tables below as of December 31, 2019 and December 31, 2018 (in thousands):

	Fair Value as of December 31, 2019			
	Level 1	**Level 2**	**Level 3**	**Total**
Assets:				
Money market and demand accounts (a)	$ 757,098	$ —	$ —	$ 757,098
U.S. government securities	—	105,702	—	105,702
Corporate bonds, asset backed and other securities	—	73,502	—	73,502
	$ 757,098	$ 179,204	$ —	$ 936,302

	Fair Value as of December 31, 2018			
	Level 1	**Level 2**	**Level 3**	**Total**
Assets:				
Money market and demand accounts (a)	$ 488,733	$ —	$ —	$ 488,733
Commercial paper	—	14,548	—	14,548
U.S. government securities	—	289,576	—	289,576
Corporate bonds and asset backed securities	—	166,600	—	166,600
	$ 488,733	$ 470,724	$ —	$ 959,457
Liabilities:				
Contingent consideration resulting from the Technicolor Patent Acquisition	$ —	$ —	$ 19,800	$ 19,800
	$ —	$ —	$ 19,800	$ 19,800

(a) Included within cash and cash equivalents.

Level 3 Fair Value Measurements

Contingent Consideration

As discussed further in Note 5, we completed the Technicolor Patent Acquisition during third quarter 2018. In conjunction with the Technicolor Patent Acquisition, we initially recognized a contingent consideration liability which was measured at fair value on a recurring basis using significant unobservable inputs classified as Level 3

measurements within the fair value hierarchy. We utilized a Monte Carlo simulation model to determine the estimated fair value of the contingent consideration liability through first quarter 2019. A Monte Carlo simulation uses random numbers together with volatility assumptions to generate individual paths, or trials, for variables of interest governed by a Geometric Brownian Motion in a risk-neutral framework.

During second quarter 2019, we completed the R&I Acquisition. The transaction met the definition of an asset acquisition and was accounted for using the cost accumulation and allocation model. As discussed in Note 5, "*Business Combinations and Other Transactions*," as part of this acquisition, Technicolor reduced its rights to the revenue-sharing arrangement that created the initial contingent consideration liability from the Technicolor Patent Acquisition. We determined that the initial contingent consideration liability from the Technicolor Patent Acquisition was significantly modified in conjunction with the R&I Acquisition, and, as such, the contingent consideration liability will now be accounted for under *ASC 450 - Contingencies* under the asset acquisition framework when the liability is deemed probable and estimable. Since the contingent consideration liability arising from the amended revenue-sharing arrangement was not probable and estimable as of the acquisition date, the carrying value of the previous contingent consideration liability was derecognized, which resulted in a $20.5 million gain which is included within "*Other Income (Expense), Net*" in the consolidated statement of income for the

year ended December 31, 2019. Therefore, effective as of the acquisition date of May 31, 2019, the contingent consideration liability was no longer a Level 3 fair value recurring measurement. As of December 31, 2019, the contingent consideration liability from the amended revenue-sharing arrangement was deemed not probable and estimable and is therefore not reflected within the consolidated financial statements.

Level 3 significant unobservable inputs used in the December 31, 2018 valuation of the contingent consideration liability included the following:

Significant Unobservable Input	Ranges	Weighted Average
Risk-adjusted discount rate for revenue	13.5% - 14.2%	13.9%
Credit risk discount rate	6.2% - 8.0%	7.1%
Revenue volatility	35.0%	35.0%
Projected years of earn out	2019 - 2030	N/A

Significant increases or decreases in any of those inputs in isolation could have resulted in a significantly lower or higher fair value measurement. Adjustments to the fair value of contingent consideration were reflected in operating expenses within our consolidated statements of income through first quarter 2019.

The following table provides a reconciliation of the beginning and ending balances of our Level 3 fair value measurements from December 31, 2018 to December 31, 2019, which includes the contingent consideration liability resulting from the Technicolor Patent Acquisition discussed further above and within Note 5 (in thousands). The Level 3 contingent consideration liability was historically included within "*Other long-term liabilities*" in the consolidated balance sheet prior to its derecognition in second quarter 2019.

Level 3 Fair Value Measurements	**Contingent Consideration Liability**
Balance as of December 31, 2018	$ 19,800
Changes in fair value recognized in the consolidated statements of income	710
Derecognition of contingent consideration liability as a Level 3 fair value measurement	(20,510)
Balance as of December 31, 2019	$ —

Fair Value of Long-Term Debt

2024 and 2020 Senior Convertible Notes

The principal amount, carrying value and related estimated fair value of the Company's senior convertible debt reported in the consolidated balance sheets as of December 31, 2019 and December 31, 2018 was as follows (in thousands). The aggregate fair value of the principal amount of the senior convertible long-term debt is a Level 2 fair value measurement.

	December 31, 2019	**December 31, 2018**

	Principal Amount	Carrying Value	Fair Value	Principal Amount	Carrying Value	Fair Value
Senior Convertible Long-Term Debt	$ 494,909	$ 423,657	$ 492,969	$ 316,000	$ 298,951	$ 331,595

Technicolor Patent Acquisition Long-term Debt

As more fully disclosed in Note 5, we recognized long-term debt in conjunction with the Technicolor Patent Acquisition. The carrying value and related estimated fair value of the Technicolor Patent Acquisition long-term debt reported in the consolidated balance sheet as of December 31, 2019 and December 31, 2018 was as follows (in thousands). The aggregate fair value of the Technicolor Patent Acquisition long-term debt is a Level 3 fair value measurement.

	December 31, 2019		December 31, 2018	
	Carrying Value	Fair Value	Carrying Value	Fair Value
Technicolor Patent Acquisition Long-Term Debt	$ 21,101	$ 23,305	$ 18,428	$ 19,100

Non-Recurring Fair Value Measurements

Investments in Other Entities

As disclosed in Note 2, we made an accounting policy election to utilize a measurement alternative for equity investments that do not have readily determinable fair values, which applies to our long-term strategic investments in other entities. Under the alternative, our long-term strategic investments in other entities that do not have readily determinable fair values are measured at cost, less any impairment, plus or minus changes resulting from observable price changes in orderly transactions for an identical or similar investment of the same issuer. Any adjustments to the carrying value of those investments are considered non-recurring fair value measurements.

During the year ended December 31, 2019, we recognized a net loss of $2.6 million resulting from the partial impairment of one of our strategic investments partially offset by a gain on sale of a separate strategic investment, which is included within "*Other Income (Expense), Net*" in the consolidated statement of income. During the year ended December 31, 2018, we recognized an aggregate $8.4 million loss resulting from the sale of our entire ownership interest in one of our strategic investments and the impairment of a separate strategic investment. Certain of our investments in other entities may be seeking additional financing in the next twelve months or potential exit strategies. We will continue to review and monitor our investments in other entities for any indications of an increase in fair value or impairment.

Patents

During fourth quarter 2019, we entered into a multi-year, worldwide, non-exclusive, royalty-bearing patent license and settlement agreement with ZTE Corporation. A portion of the future consideration for the agreement was in the form of patents. We have yet to record these patents on our balance sheet as of December 31, 2019 as they have not yet been transferred. However, we have determined the estimated fair value of the patents for determining the transaction price for revenue recognition purposes, which was estimated to be $14.0 million utilizing the market approach. The value will be amortized as a non-cash expense over the patents' estimated useful lives once transferred. Additionally, as previously disclosed, during 2018 and 2017, we entered into patent license agreements with Sony and LG, respectively, for which a portion of the consideration was patents. The estimated fair value of the Sony patents was $22.5 million, and the estimated fair value of the LG patents was $19.7 million, which are being amortized as a non-cash expense over their estimated useful lives. We estimated the fair value of the patents in the Sony and LG transactions through a combination of the income approach, the market approach, and the cost approach.

As noted above, we estimated the fair value of the patents in these transactions using one of, or a combination of, an analysis of comparable market transactions (the market approach), a discounted cash flow analysis (the income approach) and/or by quantifying the amount of money required to replace the future service capability of the assets (the cost approach). For the market approach, judgment was applied as to which market transactions were most comparable to the transaction. For the income approach, the inputs and assumptions used to develop these estimates were based on a market participant perspective and included estimates of projected royalties, discount rates, economic lives and income tax rates, among others. For the cost approach, we utilized the historical cost of assets of similar technologies to determine the estimated replacement cost, including research, development, testing and patent application fees.

8. PROPERTY AND EQUIPMENT

Property and equipment, net is comprised of the following (in thousands):

	December 31,	
	2019	2018
Computer equipment and software	$ 11,320	$ 20,876
Engineering and test equipment	1,333	4,168
Building and improvements	3,702	3,711
Leasehold improvements	11,315	11,364
Furniture and fixtures	1,121	1,549
Property and equipment, gross	28,791	41,668
Less: accumulated depreciation	(18,574)	(31,617)
Property and equipment, net	$ 10,217	$ 10,051

Depreciation expense was $3.9 million, $3.7 million and $3.9 million in 2019, 2018 and 2017, respectively. Depreciation expense included depreciation of computer software costs of $0.2 million, $0.3 million and $0.5 million in 2019, 2018 and 2017, respectively. Accumulated depreciation related to computer software costs was $1.1 million and $9.2 million as of December 31, 2019 and 2018, respectively. The net book value of our computer software was $0.2 million and $0.3 million as of December 31, 2019 and 2018, respectively.

9. PATENTS, GOODWILL AND OTHER INTANGIBLE ASSETS

Patents

As of December 31, 2019 and 2018, patents consisted of the following (in thousands, except for useful life data):

	December 31,	
	2019	2018
Weighted average estimated useful life (years)	9.9	10
Gross patents	$ 905,814	$ 851,846
Accumulated amortization	(469,475)	(397,279)
Patents, net	$ 436,339	$ 454,567

Amortization expense related to capitalized patent costs was $72.3 million, $61.8 million and $52.9 million in 2019, 2018 and 2017, respectively. These amounts are recorded within the "*Patent administration and licensing"* line of our Consolidated Statements of Income.

The estimated aggregate amortization expense for the next five years related to our patents balance as of December 31, 2019 is as follows (in thousands):

2020	$ 71,572
2021	66,962

2022	62,673
2023	56,748
2024	48,135

Goodwill

The following table shows the change in the carrying amount of our goodwill balance from December 31, 2017 to December 31, 2019, all of which is allocated to our one reportable segment (in thousands):

Goodwill balance as of December 31, 2017	$	16,033
Technicolor Patent Acquisition		6,388
Goodwill balance as of December 31, 2018	$	22,421
Activity		—
Goodwill balance as of December 31, 2019	$	22,421

Other Intangible Assets

During the year ended December 31, 2019, our other intangible assets were sold in conjunction with the sale of our Hillcrest product business which is discussed further in Note 5, "*Business Combinations and Other Transactions*". As of December 31, 2018, intangible assets excluding patents were included in "*Other Non-Current Assets*" on the consolidated balance sheet and consisted of the following (in thousands):

		December 31, 2018		
	Average Life (Years)	**Gross Assets**	**Accumulated Amortization**	**Net**
Trade Names	9	$ 600	$ (133)	$ 467
Customer Relationships	10	1,700	(340)	1,360
		$ 2,300	$ (473)	$ 1,827

10. OBLIGATIONS

Refer to Note 5, "*Business Combinations and Other Transactions,*" and Note 7, "*Concentration of Credit Risk and Fair Value of Financial Assets and Financial Liabilities*," for information regarding the long-term debt initially recognized during 2018 resulting from the Technicolor Patent Acquisition.

Long-term debt obligations, excluding the long-term debt resulting from the Technicolor Patent Acquisition, are comprised of the following (in thousands):

	December 31,	
	2019	**2018**
2.00% Senior Convertible Notes due 2024	$ 400,000	$ —
1.50% Senior Convertible Notes due 2020	94,909	316,000
Less:		
Unamortized interest discount	(65,393)	(15,428)
Deferred financing costs	(5,859)	(1,621)
Total net carrying amount of Senior Convertible Notes	423,657	298,951
Less: Current portion of long-term debt	94,170	—
Long-term net carrying amount of Senior Convertible Notes	$ 329,487	$ 298,951

There were no finance leases as of December 31, 2019 or December 31, 2018.

Maturities of principal of the long-term debt obligations of the Company as of December 31, 2019, excluding the long-term debt resulting from the Technicolor Patent Acquisition, are as follows (in thousands):

2020	$	94,909
2021		—
2022		—
2023		—
2024		400,000
Thereafter		—
	$	494,909

2024 Senior Convertible Notes, and Related Note Hedge and Warrant Transactions

On June 3, 2019 we issued $400.0 million in aggregate principal amount of 2.00% Senior Convertible Notes due 2024 (the "2024 Notes"). The net proceeds from the issuance of the 2024 Notes, after deducting the initial purchasers' transaction fees and offering expenses, were approximately $391.6 million. The 2024 Notes bear interest at a rate of 2.00% per year, payable in cash on June 1 and December 1 of each year, commencing on December 1, 2019, and mature on June 1, 2024, unless earlier converted or repurchased.

The 2024 Notes will be convertible into cash, shares of our common stock or a combination thereof, at our election, at an initial conversion rate of 12.3018 shares of common stock per $1,000 principal amount of 2024 Notes (which is equivalent to an initial conversion price of approximately $81.29 per share), as adjusted pursuant to the terms of the indenture governing the 2024 Notes (the "Indenture"). The conversion rate of the 2024 Notes, and thus the conversion price, may be adjusted in certain circumstances, including in connection with a conversion of the 2024 Notes made following certain fundamental changes and under other circumstances set forth in the Indenture. It is our current intent and policy to settle all conversions of the 2024 Notes through combination settlements of cash and shares of common stock, with a specified dollar amount of $1,000 per $1,000 principal amount of 2024 Notes and any remaining amounts in shares of common stock.

Prior to 5:00 p.m., New York City time, on the business day immediately preceding March 1, 2024, the 2024 Notes will be convertible only under certain circumstances as set forth in the Indenture, including on any date during any calendar quarter (and only during such calendar quarter) beginning after September 30, 2019 if the closing sale price of the common stock was more than 130% of the applicable conversion price (approximately $105.68 based on the current conversion price of the 2024 Notes) on each applicable trading day for at least 20 trading days (whether or not consecutive) in the period of the 30 consecutive trading days ending on the last trading day of the immediately preceding calendar quarter.

Commencing on March 1, 2024, the 2024 Notes will be convertible at any time prior to 5:00 p.m., New York City time, on the second scheduled trading day immediately preceding the maturity date of the 2024 Notes.

The Company may not redeem the 2024 Notes prior to their maturity date.

If a fundamental change (as defined in the Indenture) occurs, holders may require the Company to purchase all or a portion of their 2024 Notes for cash at a repurchase price equal to 100% of the principal amount of the 2024 Notes to be repurchased, plus any accrued and unpaid interest to, but excluding, the fundamental change repurchase date. The 2024 Notes are our senior unsecured obligations and rank equally in right of payment with any of our current and any future senior unsecured indebtedness, including our 1.50% senior convertible notes due 2020 (the “2020 Notes”) discussed below. The 2024 Notes are effectively subordinated to all of our future secured indebtedness to the extent of the value of the related collateral, and the 2024 Notes are structurally subordinated to indebtedness and other liabilities, including trade payables, of our subsidiaries.

On May 29 and May 31, 2019, in connection with the offering of the 2024 Notes, we entered into convertible note hedge transactions (collectively, the “2024 Note Hedge Transactions”) that cover, subject to customary anti-dilution adjustments, approximately 4.9 million shares of common stock, in the aggregate, at a strike price that initially corresponds to the initial conversion price of the 2024 Notes, subject to adjustment, and are exercisable upon any conversion of the 2024 Notes. The aggregate cost of the 2024 Note Hedge Transactions was $72.0 million.

On May 29 and May 31, 2019, we also entered into privately negotiated warrant transactions (collectively, the “2024 Warrant Transactions” and, together with the 2024 Note Hedge Transactions, the “2024 Call Spread Transactions”), whereby we sold warrants to acquire, subject to customary anti-dilution adjustments, approximately 4.9 million shares of common stock at an initial strike price of approximately $109.43 per share, subject to adjustment. As consideration for the 2024 Warrant Transactions, we received aggregate proceeds of $47.6 million. The net cost of the 2024 Call Spread Transactions was $24.4 million.

The net proceeds from the issuance of the 2024 Notes, after deducting fees and offering expenses, were used for the following: (i) $232.7 million was used to repurchase $221.1 million in aggregate principal amount of the 2020 Notes (as defined below) in privately negotiated transactions concurrently with the offering of the 2024 Notes (ii) $19.6 million was used to repurchase shares of common stock at $62.53 per share, the closing price of the stock on May 29, 2019; and (iii) $24.4 million, in addition to the proceeds from the 2024 Warrant Transactions discussed above, was used to fund the cost of the 2024 Call Spread Transactions.

Accounting Treatment of the 2024 Notes and Related Convertible Note Hedge and Warrant Transactions

The 2024 Call Spread Transactions were classified as equity. The Company bifurcated the proceeds from the offering of the 2024 Notes between liability and equity components. On the date of issuance, the liability and equity components were calculated to be approximately $328.0 million and $72.0 million, respectively. The initial $328.0 million liability component was determined based on the fair value of similar debt instruments excluding the conversion feature. The initial $72.0 million ($56.9 million net of tax)

equity component represents the difference between the fair value of the initial $328.0 million in debt and the $400.0 million gross proceeds. The related initial debt discount of $72.0 million is being amortized over the life of the 2024 Notes using the effective interest method. An effective interest rate of 6.25% was used to calculate the debt discount on the 2024 Notes.

In connection with the above-noted transactions, the Company incurred approximately $8.4 million of directly related costs. The initial purchasers' transaction fees and related offering expenses were allocated to the liability and equity

components in proportion to the allocation of proceeds and accounted for as debt and equity issuance costs, respectively. We allocated $6.4 million of debt issuance costs to the liability component, which were capitalized as deferred financing costs. These costs are being amortized as interest expense over the term of the debt using the effective interest method. The remaining $1.9 million of costs ($1.7 million net of tax) allocated to the equity component were recorded as a reduction of the equity component.

2020 Senior Convertible Notes, and Related Note Hedge and Warrant Transactions

On March 11, 2015, we issued $316.0 million in aggregate principal amount of 1.50% Senior Convertible Notes due 2020 (the “2020 Notes”). The 2020 Notes bear interest at a rate of 1.50% per year, payable in cash on March 1 and September 1 of each year, commencing September 1, 2015, and mature on March 1, 2020, unless earlier converted or repurchased. In connection with the initial offering of the 2020 Notes, on March 5 and March 9, 2015, we entered into convertible note hedge transactions (the “2020 Note Hedge Transactions”) that initially covered approximately 4.4 million shares of common stock at a strike price that initially corresponded to the initial conversion price of the 2020 Notes and are exercisable upon any conversion of the 2020 Notes. On March 5 and March 9, 2015, we also entered into warrant transactions (collectively, the "2020 Warrant Transactions" and, together with the 2020 Note Hedge Transactions, the "2020 Call Spread Transactions") to initially acquire, subject to customary anti-dilution adjustments, approximately 4.4 million shares of common stock. The warrants become exercisable and expire in daily tranches over a three and a half month period starting in June 2020.

As noted above, during second quarter 2019, the Company used $232.7 million from the offering of the 2024 Notes to repurchase $221.1 million in aggregate principal amount of the 2020 Notes in privately negotiated transactions concurrently with the offering of the 2024 Notes. As a result of the partial repurchase of the 2020 Notes, $94.9 million in aggregate principal amount of the 2020 Notes remain outstanding as of December 31, 2019. Additionally, on May 29, 2019, in connection with the partial repurchase of the 2020 Notes, the Company entered into partial unwind agreements that amend the terms of the 2020 Note Hedge Transactions to reduce the number of options corresponding to the principal amount of the repurchased 2020 Notes. The unwind agreements also reduce the number of warrants exercisable under the 2020 Warrant Transactions. As a result of the partial unwind transactions, approximately 1.3 million shares of common stock in the aggregate were covered under each of the 2020 Note Hedge Transactions and the 2020 Warrant Transactions as of December 31, 2019. As of December 31, 2019, the warrants under the 2020 Warrant Transactions had a strike price of approximately $86.34 per share, as adjusted. Proceeds received from the unwind of the 2020 Note Hedge Transactions were $9.0 million, and consideration paid for the unwind of the 2020 Warrant Transactions was $4.2 million, resulting in net proceeds received of $4.9 million for the combined unwind transactions which was recorded to equity during the year ended December 31, 2019.

We recognized a $5.5 million loss on extinguishment of debt during the year ended December 31, 2019 in connection with this repurchase, which was included within "*Other Income (Expense), Net*" in the consolidated statement of income for the period. The loss on extinguishment represents the difference between the calculated fair value of the debt immediately prior to its derecognition and the carrying amount of the debt component, including any unamortized debt discount and issuance costs. The remaining consideration paid for the partial repurchase of the 2020 Notes was allocated to the reacquisition of the equity component, which equaled $13.0 million ($10.6 million net of tax) and was recorded as a reduction of equity during the year ended December 31, 2019. The remaining unamortized debt discount and issuance costs of $3.3 million will continue to be amortized throughout the remaining life of the 2020 Notes, which are set to mature in March 2020.

The remaining 2020 Notes will be convertible into cash, shares of our common stock or a combination thereof, at our election, at a current conversion rate of 14.1559 shares of common stock per $1,000 principal amount of 2020 Notes as of December 31, 2019 (which is equivalent to a conversion price of approximately $70.64 per share), as adjusted pursuant to the terms of the indenture governing the 2020 Notes (the "2020 Notes Indenture"). The conversion rate of the 2020 Notes, and thus the conversion price, may be adjusted in certain circumstances, including in connection with a conversion of the 2020 Notes made following certain fundamental changes and under other circumstances set forth in the 2020 Notes Indenture. It is our current intent and policy to settle all conversions of the 2020 Notes through combination settlements of cash and shares of common stock, with a specified dollar amount of $1,000 per $1,000 principal amount of the 2020 Notes and any remaining amounts in shares of common stock.

Prior to 5:00 p.m., New York City time, on the business day immediately preceding December 1, 2019, the 2020 Notes will be convertible only under certain circumstances as set forth in the 2020 Notes Indenture, including on any date during any calendar quarter (and only during such calendar quarter) if the closing sale price of our common stock was more than 130% of the applicable conversion price (approximately $91.83 based on the current conversion price of the 2020 Notes) on each applicable trading day for at least 20 trading days in the period of the 30 consecutive trading days ending on the last trading day of the immediately preceding calendar quarter.

Commencing on December 1, 2019, the 2020 Notes will be convertible at any time prior to 5:00 p.m., New York City time, on the second scheduled trading day immediately preceding the maturity date of the 2020 Notes.

The Company may not redeem the 2020 Notes prior to their maturity date.

On April 3, 2018, in connection with the reorganization of the Company's holding company structure, the predecessor company (now known as InterDigital Wireless, Inc., the "Predecessor Company") and the successor company (now known as InterDigital, Inc., the "Successor Company") entered into a First Supplemental Indenture (the "2020 Notes Supplemental Indenture") to the 2020 Notes Indenture with the trustee. The 2020 Notes Supplemental Indenture effected certain amendments to the 2020 Notes Indenture in connection with the Reorganization, which, among other things, amended the conversion right of the 2020 Notes so that at the effective time of the Reorganization, the holder of each Note outstanding as of the effective time of the Reorganization will have the right to convert, subject to the terms of the 2020 Notes Indenture, each $1,000 principal amount of such 2020 Note into the number of shares of the Successor Company's common stock that a holder of a number of shares of the Predecessor Company's common stock equal to the conversion rate immediately prior to the effective time of the Reorganization would have been entitled to receive upon the Reorganization. In addition, pursuant to the 2020 Notes Supplemental Indenture, the Successor Company guaranteed the Predecessor Company's obligations under the 2020 Notes and the 2020 Notes Indenture.

Accounting Treatment of the 2020 Notes and Related Convertible Note Hedge and Warrant Transactions

The offering of the 2020 Notes on March 5, 2015 was for $275.0 million and included an overallotment option that allowed the initial purchasers to purchase up to an additional $41.0 million aggregate principal amount of 2020 Notes. The initial purchasers exercised their overallotment option on March 9, 2015, bringing the total amount of 2020 Notes issued on March 11, 2015 to $316.0 million. The Company bifurcated the proceeds from the offering of the 2020 Notes between liability and equity components. On the date of issuance, the liability and equity components were calculated to be approximately $256.7 million and $59.3 million ($38.6 million net of tax), respectively. The related initial debt discount of $59.3 million is being amortized using the effective interest method over the life of the 2020 Notes. An effective interest rate of 5.89% was used to calculate the debt discount on the 2020 Notes. Directly related costs are being amortized to interest expense over the term of the debt using the effective interest method.

The following table presents the amount of interest cost recognized for the years ended December 31, 2019, 2018 and 2017 related to the contractual interest coupon, accretion of the debt discount and the amortization of financing costs (in thousands).

	For the Year Ended December 31,				
	2019			2018	2017
	2024 Notes	2020 Notes	Total	2020 Notes	2020 Notes
Contractual coupon interest	$ 4,600	$ 2,824	$ 7,424	$ 4,740	$ 4,740
Accretion of debt discount	7,322	7,743	15,065	12,434	11,715
Amortization of financing costs	654	821	1,475	1,390	1,390
Total	$ 12,576	$ 11,388	$ 23,964	$ 18,564	$ 17,845

11. COMMITMENTS

Minimum future payments for accounts payable and other purchase commitments, excluding long-term operating leases for office space, as of December 31, 2019 were as follows (in thousands):

2020	$ 16,664
2021	—
2022	—
2023	—

2024	—
Thereafter	—

As part of the Technicolor Patent Acquisition, we committed to contributing cash, subject to certain requirements, of up to a maximum of $25.0 million to fund a collaborative arrangement related to the transaction. Additionally, we are subject to a revenue-sharing arrangement with Technicolor resulting from the Technicolor Patent Acquisition and the R&I Acquisition. We also assumed certain defined benefit plan liabilities in conjunction with these transactions. Refer to Note 5, "*Business Combinations and Other Transactions*," for further information.

Refer to Note 10, "*Obligations*," for details of the Company's long-term debt obligations. Refer to Note 17, "*Leases*," for maturities of the Company's operating lease liabilities as of December 31, 2019.

12. LITIGATION AND LEGAL PROCEEDINGS

COURT PROCEEDINGS

Huawei

China Proceedings

On January 3, 2019, InterDigital was notified that a civil complaint was filed on January 2, 2019 by Huawei Technologies Co., Ltd. and certain of its subsidiaries against InterDigital, Inc. and certain of its subsidiaries in the Shenzhen Intermediate People's Court (the "Shenzhen Court"). The complaint seeks a ruling that the InterDigital defendants have violated an obligation to license their patents that are essential to 3G, 4G and 5G wireless telecommunication standards on fair, reasonable and non-discriminatory ("FRAND") terms and conditions. The complaint also seeks a determination of the terms for licensing all of the InterDigital defendants' Chinese patents that are essential to 3G, 4G and 5G wireless telecommunication standards to the Huawei plaintiffs for the plaintiffs' wireless terminal unit products made and/or sold in China from 2019 to 2023. On September 17, 2019, InterDigital filed a petition challenging the jurisdiction of the Shenzhen Court to hear the action. On December 25, 2019, InterDigital was notified that the Shenzhen Court rejected InterDigital's jurisdictional challenge. On January 23, 2020, InterDigital filed an appeal of the Shenzhen Court's decision to deny InterDigital's jurisdictional challenge with the IP Tribunal of the China Supreme People's Court. InterDigital's appeal is pending.

U.K. Proceedings

On December 3, 2019, InterDigital, Inc. and certain of its subsidiaries filed a claim in the High Court of Justice, Business and Property Courts of England and Wales, Intellectual Property List (Chancery Division), Patents Court (the "High Court") against Huawei Technologies Co., Ltd. and Huawei Technologies (UK) Co., Ltd. ("Huawei UK"). The claim alleges infringement of five of InterDigital's patents relating to 3G, 4G/LTE and/or 5G standards: European Patent (U.K.) Nos. 2,363,008; 2,421,318; 2,485,558; 2,557,714; and 3,355,537.

In these proceedings, InterDigital is seeking a declaration that the terms offered by InterDigital to Huawei for a worldwide license are consistent with InterDigital's FRAND commitments, or, alternatively, a determination of FRAND terms for a license to the litigated patents. InterDigital is also seeking a 'FRAND injunction' of the type previously awarded by the High Court in Unwired Planet v. Huawei (such injunction, a "FRAND Injunction"), preventing further infringement of the litigated patents where the court has settled the terms of a worldwide FRAND license and the defendant does not enter into a license on those terms, along with other relief concerning declarations, damages and costs.

On December 20, 2019, Huawei UK filed an application seeking an extension of time to challenge the jurisdiction of the High Court to hear the action against it until the later of January 17, 2020 or fourteen days following the Supreme Court of the United Kingdom's (the "U.K. Supreme Court") decision in *Unwired Planet v. Huawei* and *Conversant v. Huawei* and ZTE (together, the "Unwired Planet and Conversant Cases"). On January 17, 2020, the parties filed a consent order directing that Huawei UK's challenge to the jurisdiction of the High Court be heard before July 31, 2020, and setting the deadline for Huawei UK to file its application challenging jurisdiction to be fourteen days following the Supreme Court's decision in the Unwired Planet and Conversant Cases, which the court entered into with minor amendments. On January 24, 2020, the High Court listed Huawei UK's application challenging jurisdiction to be heard between July 1 and July 3, 2020.

Lenovo

U.K. Proceedings

On August 27, 2019, InterDigital, Inc. and certain of its subsidiaries filed a claim in the High Court against Lenovo Group Limited and certain of its subsidiaries. The claim, as amended, alleges infringement of five of InterDigital's patents relating to 3G and/or 4G/LTE standards: European Patent (U.K.) Nos. 2,363,008 (the "'008 Patent"); 2,421,318; 2,485,558; 2,557,714; and 3,355,537.

In these proceedings, InterDigital is seeking a FRAND Injunction, preventing further infringement of the litigated patents where the court has settled the terms of a worldwide FRAND license and the defendant does not enter into a license on those terms, along with other relief concerning declarations, damages and costs.

On October 3, 2019, Lenovo filed an application challenging the jurisdiction of the High Court to hear the action, as well as the order which permitted service outside of the United Kingdom with respect to the U.S. and Hong Kong defendants (the "Lenovo Jurisdiction Challenge"). The High Court listed the Lenovo Jurisdiction Challenge to be heard over two days between February 24 and February 27, 2020. On February 11, 2020, Lenovo filed an application seeking to adjourn the Lenovo Jurisdiction Challenge to allow time for the U.K. Supreme Court to issue its ruling in the Unwired Planet and Conversant Cases. On February 17, 2020, the parties filed a consent order adjourning the hearing of the Lenovo Jurisdiction Challenge until between May 5, 2020 and July 30, 2020.

Also on February 11, 2020, the High Court listed a five-day trial in relation to the '008 Patent to begin between March 1, 2021 and March 5, 2021. On February 17, 2020, the High Court listed a second five-day trial to begin on June 21, 2021. The patent in suit to be addressed at such trial, if not previously agreed, will be determined at a case management conference scheduled to take place in late May, 2020, with the exact date to be determined. Also at the case management conference, the parties will ask the High Court to determine directions for the remaining trials in the proceedings if they cannot be agreed.

District of Delaware Proceedings

On August 28, 2019, InterDigital, Inc. and certain of its subsidiaries filed a complaint in the United States District Court for the District of Delaware (the "Delaware District Court") against Lenovo Holding Company, Inc. and certain of its subsidiaries alleging that Lenovo infringes eight of InterDigital's U.S. patents—U.S. Patent Nos. 8,085,665 (the "'665 Patent"); 8,199,726 (the "'726 Patent"); 8,427,954 (the "'954 Patent"); 8,619,747; 8,675,612 (the "'612 Patent"); 8,797,873 (the "'873 Patent"); 9,203,580; and 9,456,449 (the "'449 Patent")—by making, using, offering for sale, and/or selling Lenovo wireless devices with 3G and/or 4G LTE capabilities. As relief, InterDigital is seeking: (a) a declaration that InterDigital is not in breach of its relevant FRAND commitments with respect to Lenovo; (b) to the extent Lenovo does not agree to negotiate a worldwide patent license, does not agree to enter into binding international arbitration to set the terms of a FRAND license, and does not agree to be bound by the FRAND terms to be set by the High Court in the separately filed U.K. Proceedings described above, an injunction prohibiting Lenovo from continued infringement; (c) damages, including enhanced damages for willful infringement and supplemental damages; and (d) attorneys' fees and costs.

On November 4, 2019, Lenovo filed a motion to dismiss InterDigital's patent infringement claims for six of the eight litigated patents—the '873, '665, '954, '726, '449 and '612 Patents—on the basis that such patents allegedly are not directed to patent-eligible subject matter. On December 9, 2019, InterDigital amended its complaint and on January 10, 2020, Lenovo filed a renewed motion to dismiss the same claims from InterDigital's amended complaint as they moved to dismiss from the original complaint. On February 7, 2020, InterDigital filed its opposition to Lenovo's renewed motion to dismiss InterDigital's amended complaint. Lenovo's response to InterDigital's opposition is due on February 21, 2020.

Asustek

Information regarding the legal proceeding that Asustek Computer Incorporated ("Asus") filed against InterDigital, Inc. and certain of its subsidiaries in the U.S. District Court for the Northern District of California (the "CA Northern District Court") on April 15, 2015 can be found in the description of legal proceedings contained in InterDigital's Annual Report on Form 10-K for the fiscal year ended December 31, 2018 as filed with the SEC on February 21, 2019 (the "2018 Form 10-K").

On March 11, 2019, as a result of the CA Northern District Court's ruling on December 20, 2018 that Asus was judicially estopped from arguing that the parties' April 2008 patent license agreement was not entered into on FRAND terms and conditions, Asus revised its damages calculations downward, and updated the calculations to include sales through 2018. Asus was seeking damages for what it called "4G capable products" in the amount of $58.3 million for sales through 2018. Any damages attributable to a violation Section 2 of the Sherman Act would have been subject to mandatory trebling, as well as an award of reasonable attorneys' fees.

On April 4, 2019, Asus informed the court that it would not be proceeding to trial on its waiver and Delaware Consumer Fraud Act claims. A jury trial on Asus' remaining claims—violation of Section 2 of the Sherman Act and breach of contract resulting from ongoing negotiations—was scheduled to commence on May 6, 2019 in the CA Northern

District Court.

On April 9, 2019, the parties participated in another court-mandated settlement conference. On April 12, 2019, certain subsidiaries of InterDigital entered into a Settlement Agreement and First Amendment to the Patent License Agreement with Asus (the "Asus Settlement Agreement"), pursuant to which, among other things, the parties agreed to a multi-year amendment to the 2008 Asus PLA that added coverage for 4G technologies and amended certain other terms. The parties also agreed to dismiss all outstanding litigation and other proceedings among the parties. Accordingly, the action in the CA Northern District Court described herein was dismissed on April 15, 2019, and there are no further proceedings in this matter.

ZTE USITC Proceedings and Related Delaware District Court Proceedings

Information regarding legal proceedings that InterDigital filed against ZTE Corporation and ZTE (USA) Inc. (collectively, "ZTE") with the United States International Trade Commission ("USITC") and the Delaware District Court can be found in the description of legal proceedings contained in InterDigital's 2018 Form 10-K. With respect to the Delaware District Court proceeding related to the 2013 USITC Proceeding (337-TA-868), on January 23, 2019, InterDigital and ZTE filed a joint status report that informed the Delaware District Court of the Federal Circuit's decision regarding the '966 and '847 patents and that the PTAB proceedings regarding the '244 patent remained pending. The parties jointly requested that the case remain stayed so that the portion of the case related to damages potentially owed by ZTE as to the three patents-in-suit could be coordinated. The court granted that request on January 25, 2019.

On October 18, 2019, InterDigital and ZTE entered into a Patent License Agreement (the "ZTE PLA") pursuant to which the parties agreed that, upon the performance of certain obligations by ZTE, the parties would end all legal proceedings initiated by either party or otherwise pending between them. Pursuant to the ZTE PLA, on October 25, 2019, ZTE filed an unopposed motion with the Federal Circuit to withdraw from the appeal of the PTAB's remand ruling that claim 8 of the '244 patent is invalid. On November 22, 2019, the Federal Circuit reversed and vacated the PTAB's remand decision. The court's mandate issued on December 30, 2019.

Also on December 30, 2019, InterDigital and ZTE filed a stipulation and proposed order to dismiss the Delaware District Court proceedings related to the 2011 USITC Proceeding (337-TA-800) and 2013 USITC Proceeding (337-TA-868), which was granted by the court on January 2, 2020. There are no further proceedings in either of these matters.

REGULATORY PROCEEDING

Investigation by National Development and Reform Commission of China

On September 23, 2013, counsel for InterDigital was informed by China's National Development and Reform Commission ("NDRC") that the NDRC had initiated a formal investigation into whether InterDigital has violated China's Anti-Monopoly Law ("AML") with respect to practices related to the licensing of InterDigital's standards-essential patents to Chinese companies. Companies found to violate the AML may be subject to a cease and desist order, fines and disgorgement of any illegal gains. On March 3, 2014, the Company submitted to NDRC, pursuant to a procedure set out in the AML, a formal application for suspension of the investigation that included proposed commitments by the Company. On May 22, 2014, NDRC formally suspended its investigation of the Company based on the commitments proposed by the Company. The Company's commitments with respect to the licensing of its patent portfolio for wireless mobile standards to Chinese manufacturers of cellular terminal units ("Chinese Manufacturers") are as follows:

1. Whenever InterDigital engages with a Chinese Manufacturer to license InterDigital's patent portfolio for 2G, 3G and 4G wireless mobile standards, InterDigital will offer such Chinese Manufacturer the option of taking a worldwide portfolio license of only its standards-essential wireless patents, and comply with F/RAND principles when negotiating and entering into such licensing agreements with Chinese Manufacturers.
2. As part of its licensing offer, InterDigital will not require that a Chinese Manufacturer agree to a royalty-free, reciprocal cross-license of such Chinese Manufacturer's similarly categorized standards-essential wireless patents.
3. Prior to commencing any action against a Chinese Manufacturer in which InterDigital may seek exclusionary or injunctive relief for the infringement of any of its wireless standards-essential patents, InterDigital will offer such Chinese Manufacturer the option to enter into expedited binding arbitration under fair and reasonable procedures to resolve the royalty rate and other terms of a worldwide license under InterDigital's wireless standards-essential patents. If the Chinese Manufacturer accepts InterDigital's binding arbitration offer or otherwise enters into an agreement with InterDigital on a binding arbitration mechanism, InterDigital will, in accordance with the terms of the arbitration agreement and patent license agreement, refrain from seeking exclusionary or injunctive relief against such company.

The commitments contained in item 3 above expired on May 22, 2019. With the consolidation of China's antimonopoly enforcement authorities into the State Administration for Market Regulation ("SAMR") in April 2018, SAMR is now responsible for overseeing InterDigital's commitments.

OTHER

We are party to certain other disputes and legal actions in the ordinary course of business, including arbitrations and legal proceedings with licensees regarding the terms of their agreements and the negotiation thereof. We do not currently believe that these matters, even if adversely adjudicated or settled, would have a material adverse effect on our financial

condition, results of operations or cash flows. None of the preceding matters have met the requirements for accrual or disclosure of a potential range as of December 31, 2019.

13. COMPENSATION PLANS AND PROGRAMS

Compensation Programs

We use a variety of compensation programs to attract, retain and motivate our employees, and to more closely align employee compensation with company performance. These programs include, but are not limited to, short-term incentive awards tied to performance goals, cash awards to inventors for filed patent applications and patent issuances, and long-term incentives in the form of stock option awards, time-based RSU awards, performance-based awards and cash awards.

Our long-term incentives typically include annual time-based RSU grants or cash awards with a three-year vesting period, as well as annual performance-based RSU grants or cash awards with a three to five-year performance period; as a result, in any one year, we are typically accounting for at least three active cycles. We issue new shares of our common stock to satisfy our obligations under the share-based components of these programs. However, our Board of Directors has the right to authorize the issuance of treasury shares to satisfy such obligations in the future.

Equity Incentive Plans

On June 14, 2017, our shareholders adopted and approved the 2017 Equity Incentive Plan (the "2017 Plan"), under which officers, employees, non-employee directors and consultants can receive share-based awards such as RSUs, restricted stock and stock options as well as other stock or cash awards. From June 2009 through June 14, 2017, we granted such awards pursuant to our 2009 Stock Incentive Plan (the “2009 Plan," and, together with the 2017 Plan, the "Equity Plans"), which was adopted and approved by our shareholders on June 4, 2009, and the material terms of which were re-approved on June 12, 2014. Upon the adoption of the 2017 Plan in June 2017, the 2009 Plan was terminated and all shares remaining available for grant under the 2009 Plan were canceled. The number of shares available for issuance under the 2017 Plan is equal to 2,400,000 shares plus any shares subject to awards granted under the 2009 Plan that, on or after June 14, 2017, expire or otherwise terminate without having been exercised in full, or that are forfeited to or repurchased by us.

RSUs and Restricted Stock

We may issue RSUs and/or shares of restricted stock to officers, employees, non-employee directors and consultants. Any cancellations of unvested RSUs granted under the Equity Plans will increase the number of shares remaining available for grant under the 2017 Plan. Time-based RSUs vest over periods generally ranging from 1 to 3 years from the date of the grant. Performance-based RSUs generally have a vesting period of between 3 and 5 years.

As of December 31, 2019, we had unrecognized compensation cost related to share-based awards of $10.2 million, at current performance accrual rates. For grants made that cliff vest, we expect to amortize the associated unrecognized compensation cost as of December 31, 2019, on a straight-line basis generally over a three to five-year period.

Vesting of performance-based RSU awards is subject to attainment of specific goals established by the Compensation Committee of the Board of Directors. Depending upon performance achievement against these goals, the number of shares that vest can be anywhere from 0 to 2 times the target number of shares.

Information with respect to current RSU activity is summarized as follows (in thousands, except per share amounts):

Number of Unvested RSUs	Weighted Average Per Share Grant Date Fair Value

Balance at December 31, 2018	915	$ 63.70
Granted*	512	66.19
Forfeited	(274)	76.44
Vested	(198)	58.84
Balance at December 31, 2019	955	$ 62.40

** These numbers include less than 0.1 million RSUs credited on unvested RSU awards as dividend equivalents. Dividend equivalents accrue with respect to unvested RSUs when and as cash dividends are paid on the Company's common stock, and vest if and when the underlying RSUs vest. Granted amounts include performance-based RSU awards at their maximum potential payout level of 200%.*

During 2019, 2018 and 2017, we granted approximately 0.3 million, 0.3 million and 0.2 million RSUs under the Equity Plans, respectively, with weighted-average grant date fair values of $66.19, $73.75 and $58.63, respectively. The total vest date fair value of the RSUs that vested in 2019, 2018 and 2017 was $12.7 million, $25.2 million and $56.0 million, respectively. The weighted average per share grant date fair value of the awards that vested in 2019, 2018 and 2017 was $58.84, $54.75 and $35.14, respectively.

Other Equity Grants

We may also grant equity awards to non-management Board members and certain consultants.

Stock Options

The 2009 Plan allowed, and the 2017 Plan allows, for the granting of incentive and non-qualified stock options, as well as other securities. The administrator of the Equity Plans, the Compensation Committee of the Board of Directors, determines the number of options to be granted, subject to certain limitations set forth in the 2017 Plan. Annually, since 2013, both incentive and non-qualified stock options have been granted as part of our long-term incentive programs, which have generally vested over three years. During the year ended December 31, 2018, performance-based options were granted for the first time. The number of options which cliff vest, if at all, is anywhere from 0 to 2 times the target number of options subject to the attainment of performance goals measured at the end of the performance period. These performance-based options have a vesting period between three and five years.

Under the terms of the Equity Plans, the exercise price per share of each option, other than in the event of options granted in connection with a merger or other acquisition, cannot be less than 100% of the fair market value of a share of common stock on the date of grant. Options granted under the Equity Plans are generally exercisable for a period of between 7 to 10 years from the date of grant and may vest on the grant date, another specified date, over a period of time and/or dependent upon the attainment of specified performance goals. We also have approximately 0.1 million options outstanding under a prior stock plan that have an indefinite contractual life.

The fair value for option awards is computed using the Black-Scholes pricing model, whose inputs and assumptions are determined as of the date of grant and which require considerable judgment. Expected volatility was based upon a combination of implied and historic volatilities. The weighted-average grant date fair value per option award granted during the years ended December 31, 2019, 2018 and 2017 was $13.68, $24.56, and $19.90, respectively, based upon the assumptions included in the table below:

	For the Year Ended December 31,		
	2019	**2018**	**2017**
Expected term (in years)	4.5	7.7	4.5
Expected volatility	25.8%	30.1%	28.5%
Risk-free interest rate	2.4%	3.0%	1.9%
Dividend yield	2.0%	1.8%	1.4%

Information with respect to current year stock option activity is summarized as follows (in thousands, except per share amounts):

	Outstanding Options	**Weighted Average Exercise Price**
Balance at December 31, 2018	695	$ 57.21
Granted*	130	67.61
Forfeited	—	—
Exercised	—	8.25

Balance at December 31, 2019	825	$ 58.83

The weighted average remaining contractual life of our outstanding options was 8.3 years as of December 31, 2019. These options were granted between 1983 and 1986 under a prior stock plan. For purposes of calculating the weighted average remaining contractual life, these options were assigned an original life in excess of 50 years. The majority of these options have an exercise price between $9.00 and $11.63.

The total intrinsic value of our outstanding options as of December 31, 2019 was $7.2 million. Of the 0.8 million outstanding options as of December 31, 2019, 0.4 million were exercisable with a weighted-average exercise price of $35.81. Options exercisable as of December 31, 2019 had total intrinsic value of $7.2 million and a weighted average remaining contractual life of 8.6 years. The total intrinsic value of stock options exercised during the years ended December 31, 2019, 2018 and 2017 was less than $0.1 million, $5.6 million and $0.3 million, respectively. In 2019, we recorded cash received from the exercise of options of less than $0.1 million. Upon option exercise, we issued new shares of stock.

As of December 31, 2019, we had unrecognized compensation cost on our unvested stock options of $0.9 million, at current performance accrual rates. As of December 31, 2019 and 2018, we had approximately 0.3 million and 0.3 million options outstanding, respectively, that had exercise prices less than the fair market value of our stock at the respective balance sheet date. These options would have generated cash proceeds to the Company of $7.6 million and $11.2 million, respectively, if they had been fully exercised on those dates.

Defined Contribution Plans

We have a 401(k) plan ("Savings Plan") wherein employees can elect to defer compensation within federal limits. We match a portion of employee contributions. Our 401(k) contribution expense was approximately $1.1 million, $1.3 million and $1.4 million for 2019, 2018 and 2017, respectively. At our discretion, we may also make a profit-sharing contribution to our employees' 401(k) accounts. Additionally, the company contributed $0.2 million, $0.2 million and $0.3 million in 2019, 2018 and 2017, respectively, to other defined contribution plans.

14. TAXES

Our income tax provision (benefit) consists of the following components for 2019, 2018 and 2017 (in thousands):

	2019	2018	2017
Current			
Federal	$ (11,436)	$ (3,148)	$ 3,656
State	207	239	(1)
Foreign source withholding tax	19,850	25,187	47,592
	8,621	22,278	51,247
Deferred			
Federal	(21,735)	(63,030)	21,671
State	2,457	(1,554)	(1,074)
Foreign source withholding tax	21,648	14,889	49,832
	2,370	(49,695)	70,429
Total	$ 10,991	$ (27,417)	$ 121,676

The deferred tax assets and liabilities were comprised of the following components at December 31, 2019 and 2018 (in thousands):

	2019	2018
	Total	Total
Net operating losses	$ 131,501	$ 126,946
Deferred revenue, net	33,131	39,711
Tax credit carryforward	11,744	—
Stock compensation	3,307	5,037
Patent amortization	18,522	18,520
Depreciation	443	246
Goodwill	(1,933)	—
Other-than-temporary impairment	1,138	490
Other accrued liabilities	785	2,981
Other employee benefits	7,520	6,405
Right of use asset	(4,913)	—
Lease liability	5,760	—
	207,005	200,336
Less: valuation allowance	(133,797)	(125,158)
Net deferred tax asset	$ 73,208	$ 75,178

Note: Included within the balance sheet, but not reflected in the tables are deferred tax assets primarily related to foreign withholding taxes that are expected to be paid within the next twelve months of $0.1 million and $1.5 million as of December 31, 2019 and December 31, 2018, respectively.

The following is a reconciliation of income taxes at the federal statutory rate with income taxes recorded by the Company for the years ended December 31, 2019, 2018 and 2017:

	2019	2018	2017
Tax at U.S. statutory rate	21.0 %	21.0 %	35.0 %
State tax provision (a)	10.2 %	(8.9)%	— %
Effects of rates different than statutory	(2.8)%	(1.4)%	— %
Change in valuation allowance	23.3 %	8.5 %	0.5 %
Research and development tax credits	(4.5)%	(4.3)%	(0.8)%
Uncertain tax positions	(0.8)%	3.9 %	(2.4)%
Permanent differences	2.3 %	4.9 %	1.0 %
Domestic production activities deduction	— %	— %	(2.0)%
Stock compensation	(0.6)%	(5.0)%	(4.0)%

Rate change (b)	— %	— %	14.6 %
Foreign derived intangible income deduction (c)	— %	(56.3)%	— %
Amended return benefit	(8.4)%	(49.4)%	— %
Other	2.7 %	1.5 %	(0.3)%
Total tax provision (benefit)	42.4 %	(85.5)%	41.6 %

(a) In 2019, we determined that we would not be able to utilize our state deferred tax assets for our parent company in Delaware and Pennsylvania, therefore we put a full valuation allowance on these assets.
(b) In 2017, the inclusion of the revaluation of the deferred tax assets attributable to the TCJA signed into law in December 2017 increased the tax provision by 14.6%.
(c) In 2018, the new Foreign Derived Intangible Income ("FDII") deduction that was enacted as part of the TCJA decreased the tax provision by 56.3%.

Valuation Allowances and Net Operating Losses

We establish a valuation allowance for any portion of our deferred tax assets for which management believes it is more likely than not that we will be unable to utilize the assets to offset future taxes. We believe it is more likely than not that the majority of our state net operating losses and net operating losses in certain subsidiaries in France and the United Kingdom will not be utilized; therefore we have maintained a near full valuation allowance against our state, French and United Kingdom net operating losses as of December 31, 2019. All other deferred tax assets are fully benefited.

Uncertain Income Tax Positions

As of December 31, 2019, 2018 and 2017, we had 4.5 million, 4.4 million and 3.3 million, respectively, of unrecognized tax benefits that, if recognized, would impact the Company's effective tax rate. The total amount of unrecognized tax benefits could change within the next twelve months for a number of reasons including audit settlements, tax examination activities and the recognition and measurement considerations under this guidance.

During 2019, we established a reserve of $0.3 million related to an additional deduction related to the issuance cost of the convertible debt that is recorded through equity.

During 2018, we established a reserve of 1.1 million related to the recognition of the 2006 to 2010 research and development credits and manufacturing deduction credits.

During 2017, we released a reserve of $6.5 million as a result of the IRS Joint Committee issuing a letter ruling in acceptance of the refund claims associated with the domestic production activities deduction and research and development credit. Additionally, we reduced the previously established reserve for the 2016 domestic production activities deduction and research and development credit by 1.6 million. These reductions in reserves were partially offset by the establishment of a 1.0 million reserve related to the 2017 research and development and manufacturing deduction credit, as well an increase for interest and penalty on previously recognized reserves.

The following is a roll forward of our total gross unrecognized tax benefits, which if reversed would impact the effective tax rate, for the fiscal years 2017 through 2019 (in thousands):

	2019	2018	2017
Balance as of January 1	$ 4,352	$ 3,252	$ 10,397
Tax positions related to current year:			
Additions	402	73	1,009
Reductions	—	—	—
Tax positions related to prior years:			
Additions	34	1,054	—
Reductions	—	(27)	(1,610)
Settlements	—	—	(6,544)
Lapses in statues of limitations	(332)	—	—
Balance as of December 31	$ 4,456	$ 4,352	$ 3,252

Our policy is to recognize interest and/or penalties related to income tax matters in income tax expense. For certain positions that related to years prior to 2019, we have recorded approximately $0.1 million of accrued interest during 2019 and 2018.

The Company and its subsidiaries are subject to United States federal income tax, foreign income and withholding taxes and income taxes from multiple state jurisdictions. Our federal income tax returns for 2006 to the present, with the exception of 2011 and 2012, are currently open and will not close until the respective statutes of limitations have

expired. The statutes of limitations generally expire three years following the filing of the return or in some cases three years following the utilization or expiration of net operating loss carry forwards. The statute of limitations applicable to our open federal returns will expire at the end of 2021. Excluding the Korea Competent Authority Proceeding and the Finland Competent Authority Proceeding described in the section below, specific tax treaty procedures remain open for certain jurisdictions for 2014 to the present. Many of our subsidiaries have filed state income tax returns on a separate company basis. To the extent these subsidiaries have unexpired net operating losses, their related state income tax returns remain open. These returns have been open for varying periods, some exceeding ten years. The total amount of state net operating losses is $1.6 billion. In November 2018, the Company received notice that its 2016 U.S. Federal income tax return will be subject to audit. In February 2020, the Company received a no change letter from the IRS indicating the audit is closed. In December 2018, the

Company received a notice of proposed assessment related to an ongoing audit of its California tax returns for 2013 through 2015. The Company filed a protest to the California assessment in February 2019.

Foreign Taxes

We pay foreign source withholding taxes on patent license royalties when applicable. We apply foreign source withholding tax payments against our United States federal income tax obligations to the extent we have foreign source income to support these credits. In 2019, 2018 and 2017, we paid $18.8 million, $25.1 million and $46.7 million in foreign source withholding taxes, respectively, and applied these payments as credits against our United States federal tax obligation.

Between 2006 and 2019, we paid approximately $177.4 million in foreign taxes to foreign governments that have tax treaties with the U.S., for which we have claimed foreign tax credits against our U.S. tax obligations, and for which the tax treaty procedures are still open. It is possible that as a result of tax treaty procedures, the U.S. government may reach an agreement with the related foreign governments that will result in a partial refund of foreign taxes paid with a related reduction in our foreign tax credits. Due to foreign currency fluctuations, any such agreement could result in foreign currency gain or loss.

On November 8, 2019, the Company received notification that its request for competent authority pertaining to Article 25 (Mutual Agreement Procedure) of the United States-Republic of Finland Income Tax Convention had been reviewed by the IRS and an agreement has been reached (the “Finland Competent Authority Proceeding”). As a result of this agreement, the Company does not anticipate any tax consequences.

On July 24, 2018, the Company received notification that its request for competent authority pertaining to Article 27 (Mutual Agreement 14 Table of Contents Procedure) of the United States-Republic of Korea Income Tax Convention had been reviewed by the IRS and an agreement had been reached (the "Korea Competent Authority Proceeding"). As a result of this agreement, the Company received refunds of $97.4 million, inclusive of interest. In addition, we have recorded a net tax benefit of $14.7 million in our full year 2018. In September 2019 the amended tax returns for tax years covered by this agreement were filed and an additional benefit of $2.2 million was recorded related to the final refund the Company expects to receive.

15. NET INCOME PER SHARE

Basic Earnings Per Share ("EPS") is calculated by dividing net income available to common shareholders by the weighted-average number of common shares outstanding for the period. Diluted EPS reflects the potential dilution that could occur if options or other securities with features that could result in the issuance of common stock were exercised or converted to common stock. The following table reconciles the numerator and the denominator of the basic and diluted net income per share computation (in thousands, except for per share data):

	For the Year Ended December 31,					
	2019		2018		2017	
	Basic	Diluted	Basic	Diluted	Basic	Diluted
Numerator:						
Net income applicable to common shareholders	$ 20,928	$ 20,928	$ 65,031	$ 65,031	$ 176,220	$ 176,220
Denominator:						
Weighted-average shares outstanding: Basic	31,546	31,546	34,491	34,491	34,605	34,605

Dilutive effect of stock options, RSUs and convertible securities		239		816		1,174
Weighted-average shares outstanding: Diluted		31,785		35,307		35,779
Earnings Per Share:						
Net income: Basic	$ 0.66	0.66	$ 1.89	1.89	$ 5.09	5.09
Dilutive effect of stock options, RSUs and convertible securities		—		(0.05)		(0.16)
Net income: Diluted		$ 0.66		$ 1.84		$ 4.93

Certain shares of common stock issuable upon the exercise or conversion of certain securities have been excluded from our computation of earnings per share because the strike price or conversion rate, as applicable, of such securities was greater than the average market price of our common stock for the years ended December 31, 2019, 2018 and 2017, as

applicable, and, as a result, the effect of such exercise or conversion would have been anti-dilutive. Set forth below are the securities and the weighted average number of shares of common stock underlying such securities that were excluded from our computation of earnings per share for the periods presented (in thousands):

	For the Year Ended December 31,		
	2019	**2018**	**2017**
Restricted stock units and stock options	128	25	19
Convertible securities	5,495	—	—
Warrants	5,495	4,404	—
Total	11,118	4,429	19

16. EQUITY TRANSACTIONS

Repurchase of Common Stock

In June 2014, our Board of Directors authorized a $300 million share repurchase program (the "2014 Repurchase Program"). In June 2015, September 2017, December 2018, and May 2019, our Board of Directors authorized four $100 million increases to the program, respectively, bringing the total amount of the 2014 Repurchase Program to $700 million. The Company may repurchase shares under the 2014 Repurchase Program through open market purchases, pre-arranged trading plans or privately negotiated purchases.

The table below sets forth the total number of shares repurchased and the dollar value of shares repurchased under the 2014 Repurchase Program (in thousands). As of December 31, 2019, there was approximately $71.8 million remaining under the stock repurchase authorization.

	2014 Repurchase Program	
	# of Shares	**Value**
2019	2,962	$ 196,269
2018	1,478	110,505
2017	107	7,693
2016	1,304	64,685
2015	1,836	96,410
2014	3,554	152,625
Total	11,241	$ 628,187

Dividends

Cash dividends on outstanding common stock declared in 2019 and 2018 were as follows (in thousands, except per share data):

2019	Per Share	Total	Cumulative by Fiscal Year
First quarter	$ 0.35	$ 11,180	$ 11,180
Second quarter	0.35	10,895	22,075
Third quarter	0.35	10,897	32,972
Fourth quarter	0.35	10,746	43,718
	$ 1.40	$ 43,718	
2018			
First quarter	$ 0.35	$ 12,124	$ 12,124
Second quarter	0.35	12,192	24,316
Third quarter	0.35	11,996	36,312
Fourth quarter	0.35	11,610	47,922
	$ 1.40	$ 47,922	

In September 2017, we announced that our Board of Directors had approved an increase in the Company's quarterly cash dividend to $0.35 per share. We currently expect to continue to pay dividends comparable to our quarterly $0.35 per share cash dividend in the future; however, continued payment of cash dividends and changes in the Company's dividend policy will depend on the Company's earnings, financial condition, capital resources and capital requirements, alternative uses of capital, restrictions imposed by any existing debt, economic conditions and other factors considered relevant by our Board of Directors.

17. LEASES

In February 2016, the FASB issued ASC 842, which outlines a comprehensive change to the lease accounting model and supersedes prior lease guidance ("ASC 840"). The new guidance requires lessees to recognize lease liabilities and corresponding right-of-use assets for all leases with lease terms of greater than 12 months, and also changes the definition of a lease and expands the disclosure requirements of lease arrangements.

The Company adopted this guidance on January 1, 2019 using the modified retrospective transition effective date method. As part of that adoption, we have elected the package of three practical expedients, which includes the following: an entity may elect not to reassess whether expired or existing contracts contain a lease under the revised definition of a lease; an entity may elect not to reassess the lease classification for expired or existing leases; and an entity may elect not to reassess whether previously capitalized initial direct costs would qualify for capitalization. The Company has elected not to utilize the hindsight expedient in determining the lease term, and to not record leases with an initial term of 12 months or less on our balance sheet. Additionally, the Company has elected to account for lease components and non-lease components as a single lease component for all asset classes. Lease expense is recognized over the expected term on a straight-line basis. The adoption did not have a material impact on the Company's condensed consolidated statements of income or cash flows.

The Company enters into operating leases primarily for real estate to support research and development ("R&D") sites and general office space in North America, with additional locations in Europe and Asia. The Company does not currently have any finance leases. Certain of our leases include options to extend the lease at our discretion at the end of the lease term, or terminate the lease early subject to certain conditions and penalties. We do not include any renewal options in our lease terms for calculating our lease liabilities, as the renewal options allow us to maintain operational flexibility and we are not reasonably certain we will exercise these options.

At the inception of an arrangement, the Company determines whether the arrangement is or contains a lease based on the specific facts and circumstances present. Operating lease liabilities and their corresponding right-of-use assets are recorded based on the present value of lease payments over the expected lease term. The interest rate implicit in lease contracts is typically not readily determinable, and, as such, the Company utilizes its incremental borrowing rate as the discount rate based on information available on the lease commencement date. Our incremental borrowing rate represents the rate we would incur to borrow on a collateralized basis over a similar term for an amount equal to the lease payments in a similar economic environment. We utilized the incremental borrowing rate as of January 1, 2019, our adoption date, for operating leases that commenced prior to that date. Upon our adoption of ASU 2016-02, the Company recorded the following operating lease right-of-use assets and operating lease liabilities as of January 1, 2019. Additionally, the table below includes the balances of operating lease right-of-use assets and operating lease liabilities as of December 31, 2019 (in thousands):

	Balance Sheet Classification	January 1, 2019	December 31, 2019
Assets			
Operating lease right-of-use assets, net	Other Non-current Assets	$ 13,634	$ 24,513
Total Lease Assets		$ 13,634	$ 24,513
Liabilities			
Operating lease liabilities - Current	Other Accrued Expenses	$ 3,519	$ 3,437
Operating lease liabilities - Noncurrent	Other Long-Term Liabilities	13,652	24,142
Total Lease Liabilities		$ 17,171	$ 27,579

The components of lease costs which were included within operating expenses in our consolidated statement of income were as follows (in thousands):

	For the Year Ended December 31,
	2019
Operating lease cost	$ 4,776
Short-term lease cost	925
Variable lease cost	1,502

For the year ended December 31, 2019, sublease income was insignificant. Cash paid for amounts included in the measurement of operating lease liabilities for the year ended December 31, 2019 was $5.2 million and was included in net cash provided by operating activities in our consolidated statement of cash flows. Operating lease right-of-use assets obtained in exchange for operating lease obligations totaled $14.4 million during the year ended December 31, 2019. As of December 31, 2019, the weighted average remaining operating lease term was 7.1 years and the weighted average discount rate used to determine the operating lease liabilities was 5.8%.

The maturities of our operating lease liabilities as of December 31, 2019 under ASC 842, excluding short-term leases with terms less than 12 months, were as follows (in thousands):

Maturity of Operating Lease Liabilities	December 31, 2019
2020	$ 3,296
2021	5,311
2022	5,341
2023	4,605
2024	4,409
Thereafter	11,355
Total lease payments	$ 34,317
Less: Imputed interest	(6,738)
Present value of lease liabilities	$ 27,579

The undiscounted maturities of our operating leases as of December 31, 2018 under ASC 840, including short-term leases with terms less than 12 months, were as follows (in thousands):

Maturity of Operating Leases	December 31, 2018
2019	$ 5,362
2020	3,386
2021	2,883
2022	2,920
2023	2,184
Thereafter	5,582

18. OTHER INCOME (EXPENSE), NET

The amounts included in "*Other Income (Expense), Net*" in the consolidated statements of income for the year ended December 31, 2019, 2018 and 2017 were as follows (in thousands):

	For the Year Ended December 31,		
	2019	**2018**	**2017**
Interest and investment income	14,991	14,590	8,488
Gain on asset acquisition and sale of business	22,690	—	—
Loss on extinguishment of long-term debt	(5,488)	—	—
Other	(3,131)	(9,171)	252
Other income (expense), net	$ 29,062	$ 5,419	$ 8,740

Refer to Note 5, "*Business Combinations and Other Transactions,*" for further information regarding the $14.2 million gain resulting from the R&I Acquisition and the $8.5 million gain on sale of our Hillcrest product business. Refer to Note 10, "*Obligations,*" for further information on the $5.5 million loss on extinguishment of long-term debt recognized during the year ended December 31, 2019.

During the year ended December 31, 2019, we recognized a net loss of $2.6 million resulting from the partial impairment of one of our strategic investments partially offset by a gain on sale of a separate strategic investment. During the year ended December 31, 2018, we recognized an aggregate $8.4 million loss resulting from the sale of our entire ownership interest in one of our strategic investments and the impairment of a separate strategic investment. These items are included in the "*Other*" caption in the table above.

19. SELECTED QUARTERLY RESULTS (UNAUDITED)

The table below presents quarterly data for the years ended December 31, 2019 and 2018.

	(In thousands, except per share amounts, unaudited)			
	First	**Second**	**Third**	**Fourth**
2019				
Revenues (a)	$ 68,631	$ 75,609	$ 72,523	$ 102,161
Net income applicable to InterDigital, Inc.'s common shareholders	$ (2,803)	$ 7,743	$ 2,234	$ 13,754
Net income per common share — basic	$ (0.09)	$ 0.25	$ 0.07	$ 0.44
Net income per common share — diluted	$ (0.09)	$ 0.24	$ 0.07	$ 0.44
2018	**First**	**Second**	**Third**	**Fourth[c]**
Revenues (b)	$ 87,444	$ 69,555	$ 75,079	$ 75,326
Net income applicable to InterDigital, Inc.'s common shareholders	$ 30,230	$ 10,966	$ 21,752	$ 2,083
Net income per common share — basic	$ 0.87	$ 0.32	$ 0.63	$ 0.06

Net income per common share — diluted	$ 0.85	$ 0.31	$ 0.61	$ 0.06

(a) In 2019, we recognized $19.8 million of non-current patent royalties primarily attributable to the Funai, ZTE Corporation, and Innovius LLC patent license agreements, all of which were signed in fourth quarter 2019.

(b) In 2018, we recognized $26.3 million of non-current patent royalties primarily attributable to the Kyocera and Signal Trust for Wireless Innovation patent license agreements, both signed in first quarter 2018.

(c) Fourth quarter 2018 amounts have been revised due to the revision to noncontrolling interest that is discussed further in Note 21, "*Revision to Noncontrolling Interest*." As reported amounts for net income applicable to InterDigital, Inc's common shareholders, net income per common share - basic, and net income per common share - diluted for fourth quarter 2018 were $1,830, $0.05, and $0.05, respectively.

20. VARIABLE INTEREST ENTITIES

As further discussed below, we are the primary beneficiary of three variable interest entities. As of December 31, 2019, the combined book values of the assets and liabilities associated with these variable interest entities included in our consolidated balance sheet were $60.6 million and $5.4 million, respectively. Assets included $18.5 million of cash and cash equivalents, $1.7 million of accounts receivable and prepaid assets, $39.3 million of patents, net, and $1.3 million of other non-current assets. As of December 31, 2018, the combined book values of the assets and liabilities associated with these variable interest entities included in our consolidated balance sheet were $29.9 million and $6.1 million, respectively. Assets included $11.7 million of cash and cash equivalents, $1.3 million of accounts receivable, $14.4 million of patents, net, and $2.5 million of other non-current assets.

Chordant

On January 31, 2019, we launched the Company's Chordant™ business as a standalone company. The spinout of the unit, which now includes an affiliate of Sony as an investor along with the Company, gives Chordant added independence and flexibility in driving into its core operator and smart city markets. Chordant is a variable interest entity and we have determined that we are the primary beneficiary for accounting purposes and will consolidate Chordant. For the year ended December 31, 2019, we have allocated approximately $1.5 million of Chordant's net loss to noncontrolling interests held by other parties.

Convida Wireless

Convida Wireless was launched in 2013 and most recently renewed in 2018 to combine Sony's consumer electronics expertise with our pioneering IoT expertise to drive IoT communications and connectivity. Based on the terms of the agreement, the parties will contribute funding and resources for additional research and platform development, which we will perform. SCP IP Investment LLC, an affiliate of Stephens Inc., is a minority investor in Convida Wireless.

Convida Wireless is a variable interest entity. Based on our provision of research and platform development services to Convida Wireless, we have determined that we are the primary beneficiary for accounting purposes and will continue to consolidate Convida Wireless. For the years ended December 31, 2019, 2018 and 2017, we have allocated approximately $4.5 million, $5.6 million and $5.5 million, respectively, of Convida Wireless' net loss to noncontrolling interests held by other parties.

Signal Trust for Wireless Innovation

During 2013, we announced the establishment of the Signal Trust for Wireless Innovation (the "Signal Trust"), the goal of which is to monetize a large InterDigital patent portfolio related to cellular infrastructure.

The more than 500 patents and patent applications transferred from InterDigital to the Signal Trust focus primarily on 3G and LTE technologies, and were developed by InterDigital's engineers and researchers over more than a decade, with a number of the innovations contributing to the worldwide standards process.

InterDigital is the primary beneficiary of the Signal Trust. The distributions from the Signal Trust will support continued research related to cellular wireless technologies. A small portion of the proceeds from the Signal Trust will be used to fund, through the Signal Foundation for Wireless Innovation, scholarly analysis of intellectual property rights and the technological, commercial and creative innovations they facilitate.

The Signal Trust is a variable interest entity. Based on the terms of the trust agreement, we have determined that we are the primary beneficiary for accounting purposes and must consolidate the Signal Trust.

21. REVISION TO NONCONTROLLING INTEREST

As discussed in Note 1, "*Background and Basis of Presentation*," we revised our prior period presentation of noncontrolling interest. The following tables present the effect of the revision on the consolidated statements of income, statements of comprehensive income, balance sheets and statements of shareholders' equity (in thousands, except per share data). The correction of this error has no impact to the previously reported consolidated statements of cash flows for any periods.

	Statements of Income and Statements of Comprehensive Income Impact	
	Year Ended	
	December 31, 2017	December 31, 2018
Net loss attributable to noncontrolling interest - As Reported	$3,579	$4,393
Net loss attributable to noncontrolling interest - As Revised	$5,506	$5,556
Net income attributable to InterDigital, Inc. - As Reported	$174,293	$63,868
Net income attributable to InterDigital, Inc. - As Revised	$176,220	$65,031
Net income per common share, Basic - As Reported	$5.04	$1.85
Net income per common share, Basic - As Revised	$5.09	$1.89
Net income per common share, Diluted - As Reported	$4.87	$1.81
Net income per common share, Diluted - As Revised	$4.93	$1.84
Total comprehensive income attributable to InterDigital, Inc. - As Reported	$172,724	$63,929
Total comprehensive income attributable to InterDigital, Inc. - As Revised	$174,651	$65,092

	Balance Sheets and Statements of Shareholders' Equity Impact		
	December 31, 2016	December 31, 2017	December 31, 2018
Retained earnings - As Reported	$1,120,766	$1,249,091	$1,426,266
Retained earnings - As Revised	$1,127,380	$1,257,632	$1,435,970
Total InterDigital, Inc. shareholders' equity - As Reported	$739,709	$855,267	$927,025
Total InterDigital, Inc. shareholders' equity - As Revised	$746,323	$863,808	$936,729
Noncontrolling interest - As Reported	$14,659	$17,881	$10,988
Noncontrolling interest - As Revised	$8,045	$9,340	$1,284
Total equity - As Reported	$754,368	$873,148	$938,013
Total equity - As Revised	$754,368	$873,148	$938,013

Item 9. *CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE.*

None.

Item 9A. *CONTROLS AND PROCEDURES.*

Evaluation of Disclosure Controls and Procedures

The Company's Chief Executive Officer and its Chief Financial Officer, with the assistance of other members of management, have evaluated the effectiveness of our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) under the Securities Exchange Act of 1934) as of December 31, 2019. Based on that evaluation, the Chief Executive Officer and Chief Financial Officer have concluded that our disclosure controls and procedures were effective to ensure that the information required to be disclosed by us in the reports that we file or submit under the Securities Exchange Act of 1934 is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms and to ensure that the information required to be disclosed by us in the reports that we file or submit under the Securities Exchange Act of 1934 is accumulated and communicated to our management, including our Chief Executive Officer and Chief Financial Officer, as appropriate to allow timely decisions regarding required disclosure.

Management's Annual Report on Internal Control Over Financial Reporting

Management is responsible for establishing and maintaining adequate internal control over financial reporting as defined in Rules 13a-15(f) and 15d-15(f) under the Securities Exchange Act of 1934. The Company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with accounting principles generally accepted in the United States of America. Internal control over financial reporting includes those policies and procedures that:

- Pertain to the maintenance of records that in reasonable detail accurately and fairly reflect the transactions and dispositions of the assets of the company;
- Provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with accounting principles generally accepted in the United States of America, and that receipts and expenditures of the company are being made only in accordance with authorization of management and directors of the company; and
- Provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of the company's assets that could have a material effect on the consolidated financial statements.

Management, including the Chief Executive Officer and Chief Financial Officer, assessed the effectiveness of internal control over financial reporting as of December 31, 2019. Management based this assessment on criteria for effective internal control over financial reporting described in "*Internal Control — Integrated Framework"* issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013). Based on this assessment, management determined that, as of December 31, 2019, the Company maintained effective internal control over financial reporting at a reasonable assurance level.

The effectiveness of the Company's internal control over financial reporting as of December 31, 2019 has been audited by PricewaterhouseCoopers LLP, an independent registered public accounting firm, as stated in their report that appears under Part II, Item 8, of this Form 10-K.

Changes in Internal Control over Financial Reporting

There were no changes in our internal control over financial reporting during fourth quarter 2019 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Item 9B. *OTHER INFORMATION.*

None.

PART III

Item 10. *DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE.*

The information required by this item is incorporated by reference to the information following the captions "Election of Directors," "EXECUTIVE OFFICERS," "Section 16(a) Beneficial Ownership Reporting Compliance," "Code of Ethics," "Nominating and Corporate Governance Committee" and "Audit Committee" in the definitive proxy statement to be filed pursuant to Regulation 14A in connection with our 2020 annual meeting of shareholders not later than 120 days after the end of the fiscal year covered by this Annual Report on Form 10-K (the "Proxy Statement").

Item 11. ***EXECUTIVE COMPENSATION.***

The information required by this item is incorporated by reference to the information following the captions "EXECUTIVE COMPENSATION" and "DIRECTOR COMPENSATION" in the Proxy Statement.

Item 12. ***SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS.***

The information required by this item is incorporated by reference to the information following the captions "EQUITY COMPENSATION PLAN INFORMATION" and "SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT" in the Proxy Statement.

Item 13. ***CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE.***

The information required by this item is incorporated by reference to the information following the captions "CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS" and "Director Independence" in the Proxy Statement.

Item 14. ***PRINCIPAL ACCOUNTANT FEES AND SERVICES.***

The information required by this item is incorporated by reference to the information following the captions "Fees Paid to Independent Registered Public Accounting Firm" and "Audit Committee Pre-Approval Policy for Audit and Non-Audit Services of Independent Registered Public Accounting Firm" in the Proxy Statement.

PART IV

Item 15. *EXHIBITS AND FINANCIAL STATEMENT SCHEDULES.*

(a) The following documents are filed as a part of this Form 10-K:

(1) Financial Statements.

The information required by this item begins on Page 61.

(2) Financial Statement Schedules.

The following financial statement schedule of InterDigital is included herewith and should be read in conjunction with the Financial Statements included in this Item 15.

Valuation and Qualifying Accounts

	Balance Beginning of Period	Increase/ (Decrease)		Reversal of Valuation Allowance	Balance End of Period
2019 valuation allowance for deferred tax assets	$ 125,158	$ 8,639	(a)	$ —	$ 133,797
2018 valuation allowance for deferred tax assets	$ 123,916	$ 1,568	(a)	$ (326)	$ 125,158
2017 valuation allowance for deferred tax assets	$ 89,815	$ 34,430	(b)	$ (329)	$ 123,916
2019 reserve for uncollectible accounts	$ 693	$ (156)	(c)	$ —	$ 537
2018 reserve for uncollectible accounts	$ 456	$ 237		$ —	$ 693
2017 reserve for uncollectible accounts	$ —	$ 456		$ —	$ 456

—

(a) The increase was primarily necessary to maintain a full, or near full, valuation allowance against our state deferred tax assets and deferred tax assets for certain subsidiaries in France as well as a non-wholly owned subsidiary in the United States and the United Kingdom.

(b) The increase was primarily a result of the Tax Cut and Jobs Act signed into law in December of 2017. There was also a release of a state VA during the year that was recorded through tax expense. The remainder of the increase was necessary to maintain a full, or near full, valuation allowance against our state deferred tax assets and did not result in additional tax expense.

(c) The decrease relates to the write-off of a previously recorded reserve during 2019.

(3) Exhibits.

See Item 15(b) below.

(b)

Exhibit Number	Exhibit Description
*3.1	Amended and Restated Articles of Incorporation of InterDigital, Inc. ("InterDigital") (Exhibit 3.1 to InterDigital's Current Report on Form 8-K dated June 7, 2011).

*3.2	Amended and Restated Bylaws of InterDigital (Exhibit 3.1 to InterDigital's Current Report on Form 8-K dated January 30, 2015).
*4.1	Specimen Stock Certificate of InterDigital (Exhibit 4.3 to InterDigital's Quarterly Report on Form 10-Q dated April 28, 2011).
4.2	Description of InterDigital's Securities.
*4.3	Indenture, dated March 11, 2015, between InterDigital and the Bank of New York Mellon Trust Company, N.A., as trustee (Exhibit 4.1 to InterDigital's Current Report on Form 8-K dated March 11, 2015).
*4.4	Form of 1.50% Senior Convertible Note due 2020 (Exhibit 4.2 to InterDigital's Current Report on Form 8-K dated March 11, 2015).
*4.5	Indenture, dated June 3, 2019, between InterDigital and The Bank of New York Mellon Trust Company, N.A., as trustee (Exhibit 4.1 to InterDigital's Current Report on Form 8-K dated May 29, 2019).

*4.6	Form of 2.00% Senior Convertible Note due 2024 (included in Exhibit 4.1 to InterDigital's Current Report on Form 8-K dated May 29, 2019).
	Real Estate Leases
*10.1	Lease Agreement effective March 1, 2012 by and between InterDigital and Musref Bellevue Parkway, LP (Exhibit 10.5 to InterDigital's Annual Report on Form 10-K for the year ended December 31, 2012).
	Benefit Plans
†*10.2	Non-Qualified Stock Option Plan, as amended (Exhibit 10.4 to InterDigital's Annual Report on Form 10-K for the year ended December 31, 1991). (P)
†*10.3	Amendment to Non-Qualified Stock Option Plan (Exhibit 10.31 to InterDigital's Quarterly Report on Form 10-Q dated August 14, 2000).
†*10.4	Amendment to Non-Qualified Stock Option Plan, effective October 24, 2001 (Exhibit 10.6 to InterDigital's Annual Report on Form 10-K for the year ended December 31, 2001).
†*10.5	2009 Stock Incentive Plan (Exhibit 99.1 to InterDigital's Registration Statement on Form S-8 filed with the Securities and Exchange Commission ("SEC") on June 4, 2009 (File No. 333-159743)).
†*10.6	Amendment to 2009 Stock Incentive Plan, effective as of June 12, 2013 (Exhibit 10.1 to InterDigital's Quarterly Report on Form 10-Q dated July 26, 2013).
†*10.7	2015 Amendment to 2009 Stock Incentive Plan, effective as of June 11, 2015 (Exhibit 10.1 to InterDigital's Quarterly Report on Form 10-Q dated July 30, 2015).
†*10.8	2009 Stock Incentive Plan, Term Sheet and Standard Terms and Conditions for Stock Options (Exhibit 10.5 to InterDigital's Current Report on Form 8-K dated January 28, 2013).
†*10.9	2009 Stock Incentive Plan, Term Sheet and Standard Terms and Conditions for Time-Based Restricted Stock Units (Exhibit 10.3 to InterDigital's Quarterly Report on Form 10-Q dated April 29, 2015).
†*10.10	2009 Stock Incentive Plan, Term Sheet and Standard Terms and Conditions for Performance-Based Restricted Stock Units (Exhibit 10.4 to InterDigital's Quarterly Report on Form 10-Q dated April 29, 2015).
†*10.11	2009 Stock Incentive Plan, Term Sheet and Standard Terms and Conditions for Stock Options (Exhibit 10.5 to InterDigital's Quarterly Report on Form 10-Q dated April 29, 2015).
†*10.12	2009 Stock Incentive Plan, Term Sheet for Restricted Stock Units (Non-Employee Directors) (Exhibit 10.3 to InterDigital's Quarterly Report on Form 10-Q dated July 26, 2013).
†*10.13	2009 Stock Incentive Plan, Standard Terms and Conditions for Restricted Stock Units (Non-Employee Directors) (Exhibit 10.4 to InterDigital's Quarterly Report on Form 10-Q dated July 26, 2013).
†*10.14	2017 Equity Incentive Plan (Exhibit 10.1 to InterDigital's Registration Statement on Form S-8 filed with the SEC on June 15, 2017 (File No. 333-218755)).
†*10.15	2017 Equity Incentive Plan, Form of Agreement for Time-Based Restricted Stock Unit Awards (Exhibit 10.2 to InterDigital's Current Report on Form 8-K dated June 16, 2017).
†*10.16	2017 Equity Incentive Plan, Form of Agreement for Performance-Based Restricted Stock Unit Awards (Exhibit 10.3 to InterDigital's Current Report on Form 8-K dated June 16, 2017).
†*10.17	2017 Equity Incentive Plan, Form of Agreement for Option Awards (Exhibit 10.4 to InterDigital's Current Report on Form 8-K dated June 16, 2017).

†*10.18	2017 Equity Incentive Plan, Form of Agreement for Restricted Stock Unit Awards to Non-Employee Directors (Exhibit 10.18 to InterDigital's Annual Report on Form 10-K for the year ended December 31, 2017 dated February 22, 2018).
†*10.19	Compensation Program for Non-Management Directors (as amended March 2017) (Exhibit 10.1 to InterDigital's Current Report on Form 8-K dated April 3, 2017).
†10.20	Compensation Program for Non-Management Directors (as amended June 2019).
†*10.21	Deferred Compensation Plan (Exhibit 10.1 to InterDigital's Current Report on Form 8-K dated June 18, 2013).

†*10.22	2017 Equity Incentive Plan, Form of Term Sheet for 2018 Performance-Based Restricted Stock Unit Awards (Exhibit 10.1 to InterDigital, Inc.'s Current Report on Form 8-K dated July 9, 2018).
†*10.23	2017 Equity Incentive Plan, Form of Term Sheet for 2018 Performance-Based Stock Option Awards (Exhibit 10.2 to InterDigital, Inc.'s Current Report on Form 8-K dated July 9, 2018).
†*10.24	2017 Equity Incentive Plan, Form of Agreement for Time-Based Restricted Stock Unit Awards (revised October 2018) (Exhibit 10.3 to InterDigital's Quarterly Report on Form 10-Q dated November 1, 2018).
†*10.25	2017 Equity Incentive Plan, Form of Agreement for Performance-Based Restricted Stock Unit Awards (revised October 2018) (Exhibit 10.4 to InterDigital's Quarterly Report on Form 10-Q dated November 1, 2018).
†*10.26	2017 Equity Incentive Plan, Form of Agreement for Stock Option Awards (revised October 2018) (Exhibit 10.5 to InterDigital's Quarterly Report on Form 10-Q dated November 1, 2018).
†*10.27	InterDigital Inc. Executive Severance and Change in Control Policy (Exhibit 10.6 to InterDigital's Quarterly Report on Form 10-Q dated November 1, 2018).
	Employment-Related Agreements
†*10.28	Indemnity Agreement dated as of March 19, 2003 by and between InterDigital and Howard E. Goldberg (pursuant to Instruction 2 to Item 601 of Regulation S-K, the Indemnity Agreements, which are substantially identical in all material respects, except as to the parties thereto and the dates, between the Company and the following individuals, were not filed: Jeffrey K. Belk, Richard J. Brezski, Joan H. Gillman, S. Douglas Hutcheson, John A. Kritzmacher, Jannie K. Lau, John D. Markley, Jr., Scott A. McQuilkin, William J. Merritt, James J. Nolan, Kai O. Oistamo, Jean F. Rankin, Lawrence F. Shay, Philip P. Trahanas and Richard L.Gulino)(Exhibit 10.47 to InterDigital's Quarterly Report on Form 10-Q dated May 15, 2003).
†*10.29	Assignment and Assumption of Indemnity Agreement dated as of July 2, 2007, by and between InterDigital Communications Corporation, InterDigital and Bruce G. Bernstein (pursuant to Instruction 2 to Item 601 of Regulation S-K, the Indemnity Agreements, which are substantially identical in all material respects, except as to the parties thereto, between InterDigital Communications Corporation, InterDigital, Inc. and the following individuals, were not filed: Richard J. Brezski, William J. Merritt, James J. Nolan and Lawrence F. Shay) (Exhibit 10.90 to InterDigital's Quarterly Report on Form 10-Q dated August 9, 2007).
†*10.30	Offer Letter Between InterDigital and Kai Oistamo dated October 10, 2018 (Exhibit 10.7 to InterDigital's Quarterly Report on Form 10-Q dated November 1, 2018).
†10.31	Retirement & Transition Agreement and Release, dated December 9, 2019, by and between InterDigital and Jannie K. Lau.
	Other Material Contracts
*10.32	Form of Convertible Note Hedge Transaction Confirmation (Exhibit 10.1 to InterDigital's Current Report on Form 8-K dated March 11, 2015).
*10.33	Form of Warrant Transaction Confirmation (Exhibit 10.2 to InterDigital's Current Report on Form 8-K dated March 11, 2015).
*10.34	Purchase Agreement, dated May 29, 2019, between InterDigital and Barclays Capital Inc., as representative of the several initial purchasers named in Schedule I attached thereto (Exhibit 10.1 to InterDigital's Current Report on Form 8-K dated

	May 29, 2019).
*10.35	Form of Convertible Note Hedge Transaction Confirmation (Exhibit 10.2 to InterDigital's Current Report on Form 8-K dated May 29, 2019).
*10.36	Form of Warrant Transaction Confirmation (Exhibit 10.3 to InterDigital's Current Report on Form 8-K dated May 29, 2019).
*10.37	Form of Unwind Agreement (Exhibit 10.4 to InterDigital's Current Report on Form 8-K dated May 29, 2019).
21	Subsidiaries of InterDigital.
23.1	Consent of PricewaterhouseCoopers LLP.
31.1	Certification of Principal Executive Officer pursuant to Rule 13a-14(a) of the Securities Exchange Act of 1934, as amended.

31.2	Certification of Principal Financial Officer pursuant to Rule 13a-14(a) of the Securities Exchange Act of 1934, as amended.
32.1	Certification of Principal Executive Officer pursuant to 18 U.S.C. Section 1350. +
32.2	Certification of Principal Financial Officer pursuant to 18 U.S.C. Section 1350. +
101.INS	XBRL Instance Document - The instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.
101.SCH	Inline XBRL Taxonomy Extension Schema Document.
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document.
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document.
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document.
104	Cover Page Interactive Data File (formatted as Inline XBRL and contained in Exhibit 101).

* Incorporated by reference to the previous filing indicated.

† Management contract or compensatory plan or arrangement.

\+ This exhibit will not be deemed “filed” for purposes of Section 18 of the Securities Exchange Act of 1934, as amended (15 U.S.C. 78r), or otherwise subject to the liability of that section. Such exhibit will not be deemed to be incorporated by reference into any filing under the Securities Act or Securities Exchange Act, except to the extent that InterDigital, Inc. specifically incorporates it by reference.

Table of Contents

Item 16. ***FORM 10-K SUMMARY.***

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

INTERDIGITAL, INC.

Date: February 20, 2020

By: /s/ William J. Merritt
William J. Merritt
President and Chief Executive Officer

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Date: February 20, 2020

/s/ S. Douglas Hutcheson
S. Douglas Hutcheson, Chairman of the Board of Directors

Date: February 20, 2020

/s/ Joan H. Gillman
Joan H. Gillman, Director

Date: February 20, 2020

/s/ John A. Kritzmacher
John A. Kritzmacher, Director

Date: February 20, 2020

/s/ John D. Markley, Jr.
John D. Markley, Jr., Director

Date: February 20, 2020

/s/ Jean F. Rankin
Jean F. Rankin, Director

Date: February 20, 2020

/s/ Philip P. Trahanas
Philip P. Trahanas, Director

Date: February 20, 2020	/s/ William J. Merritt
	William J. Merritt, Director, President and Chief Executive Officer (Principal Executive Officer)
Date: February 20, 2020	/s/ Richard J. Brezski
	Richard J. Brezski, Chief Financial Officer (Principal Financial Officer and Principal Accounting Officer)